



US012419112B2

(12) **United States Patent**
Tanaka

(10) **Patent No.:** **US 12,419,112 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **ISOLATION TRANSFORMER**

(56)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 174 days.

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(21) Appl. No.: **18/475,261**

(22) Filed: **Sep. 27, 2023**

Machine translation of Akita Japanese Patent Document JP 2016-6816 A Jan. 2016 (Year: 2016).*

(65) **Prior Publication Data**

US 2024/0021599 A1 Jan. 18, 2024

(Continued)

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2022/015019, filed on Mar. 28, 2022.

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(30) **Foreign Application Priority Data**

Mar. 29, 2021 (JP) 2021-055721

(51) **Int. Cl.**

H10D 86/85 (2025.01)
H01L 23/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H10D 86/85** (2025.01); **H01L 23/49575** (2013.01); **H01L 24/05** (2013.01);
(Continued)

(58) **Field of Classification Search**

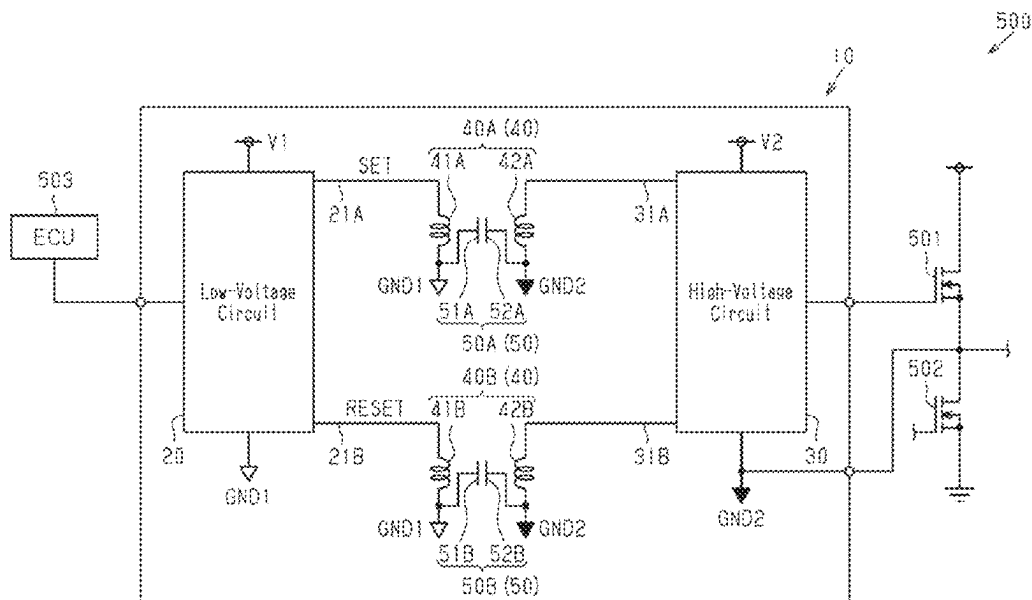
CPC ... H01F 19/04; H01F 27/00; H01L 23/49562; H01L 23/49575; H01L 24/05;
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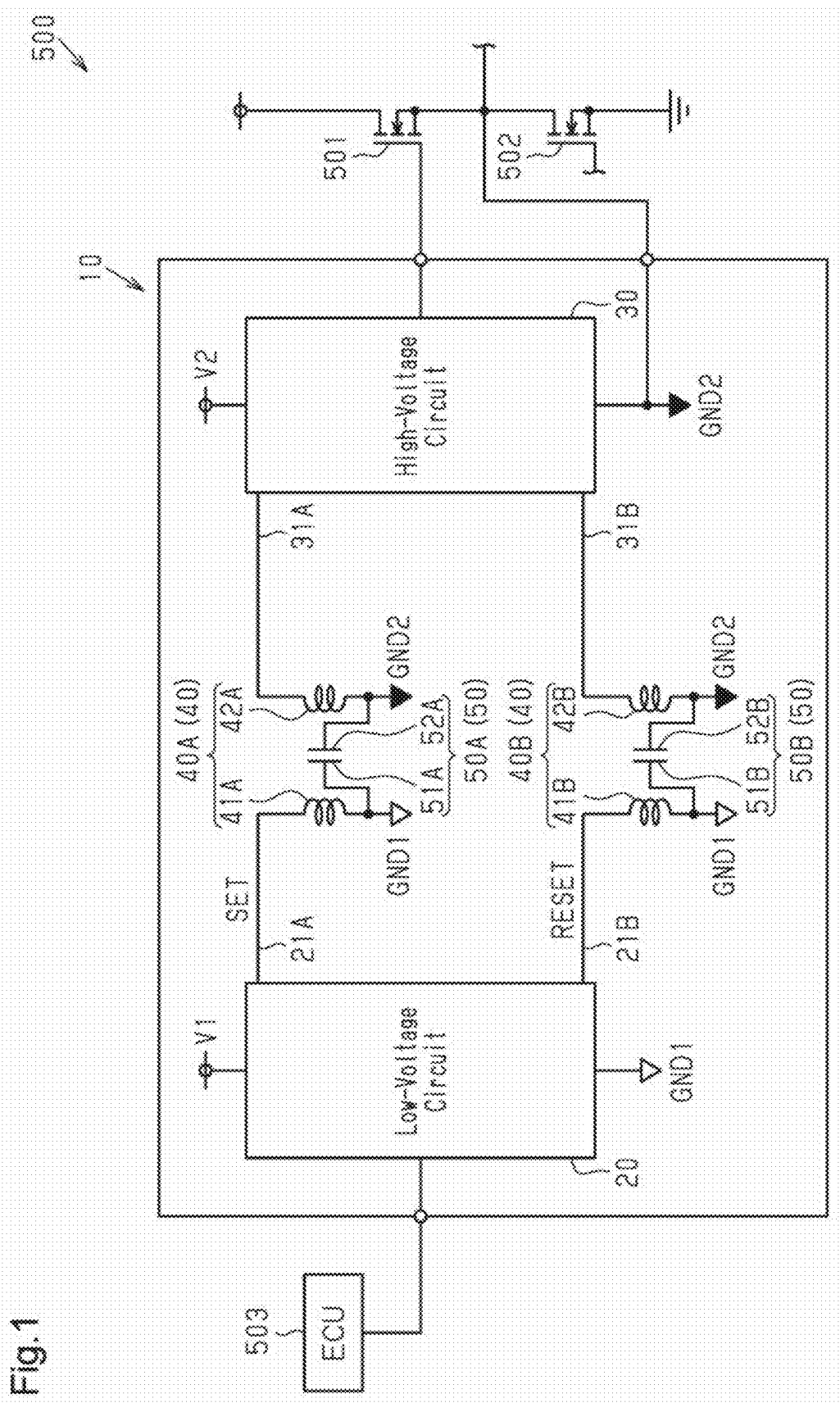
ABSTRACT

An isolation transformer includes an insulation layer, a transformer, and a capacitor. The transformer includes first and second coils separated from each other in a thickness-wise direction of the insulation layer. The capacitor includes a first capacitor electrode and a second capacitor electrode. The insulation layer includes thin films and interlayer insulation films alternately stacked in the direction. The thin films include first and second thin films separated from each other in the direction. The interlayer insulation films include a first interlayer insulation film located next to the first thin film in the direction and a second interlayer insulation film located next to the second thin film in the direction. The first capacitor electrode is formed between the first thin film and the first interlayer insulation film. The second capacitor electrode is formed between the second thin film and the second interlayer insulation film.

16 Claims, 30 Drawing Sheets



- (51) **Int. Cl.** 2224/0616; H01L 2224/45124; H01L 2224/45144; H01L 2224/45147; H01L 2224/48137; H01L 2224/49113; H01L 2224/49175; H01L 2225/06531; H02M 1/08; H03K 17/691; H10D 1/20; H10D 1/68; H10D 86/85
- H01L 23/495 (2006.01)
- H01L 25/18 (2023.01)
- (52) **U.S. Cl.**
- CPC H01L 24/06 (2013.01); H01L 24/45 (2013.01); H01L 24/48 (2013.01); H01L 24/49 (2013.01); H01L 25/18 (2013.01); H01L 2224/05567 (2013.01); H01L 2224/06152 (2013.01); H01L 2224/0616 (2013.01); H01L 2224/45124 (2013.01); H01L 2224/45144 (2013.01); H01L 2224/45147 (2013.01); H01L 2224/48137 (2013.01); H01L 2224/49113 (2013.01); H01L 2224/49175 (2013.01)
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- (58) **Field of Classification Search**
- CPC H01L 24/06; H01L 24/45; H01L 24/48; H01L 24/49; H01L 25/18; H01L 2224/05567; H01L 2224/06152; H01L



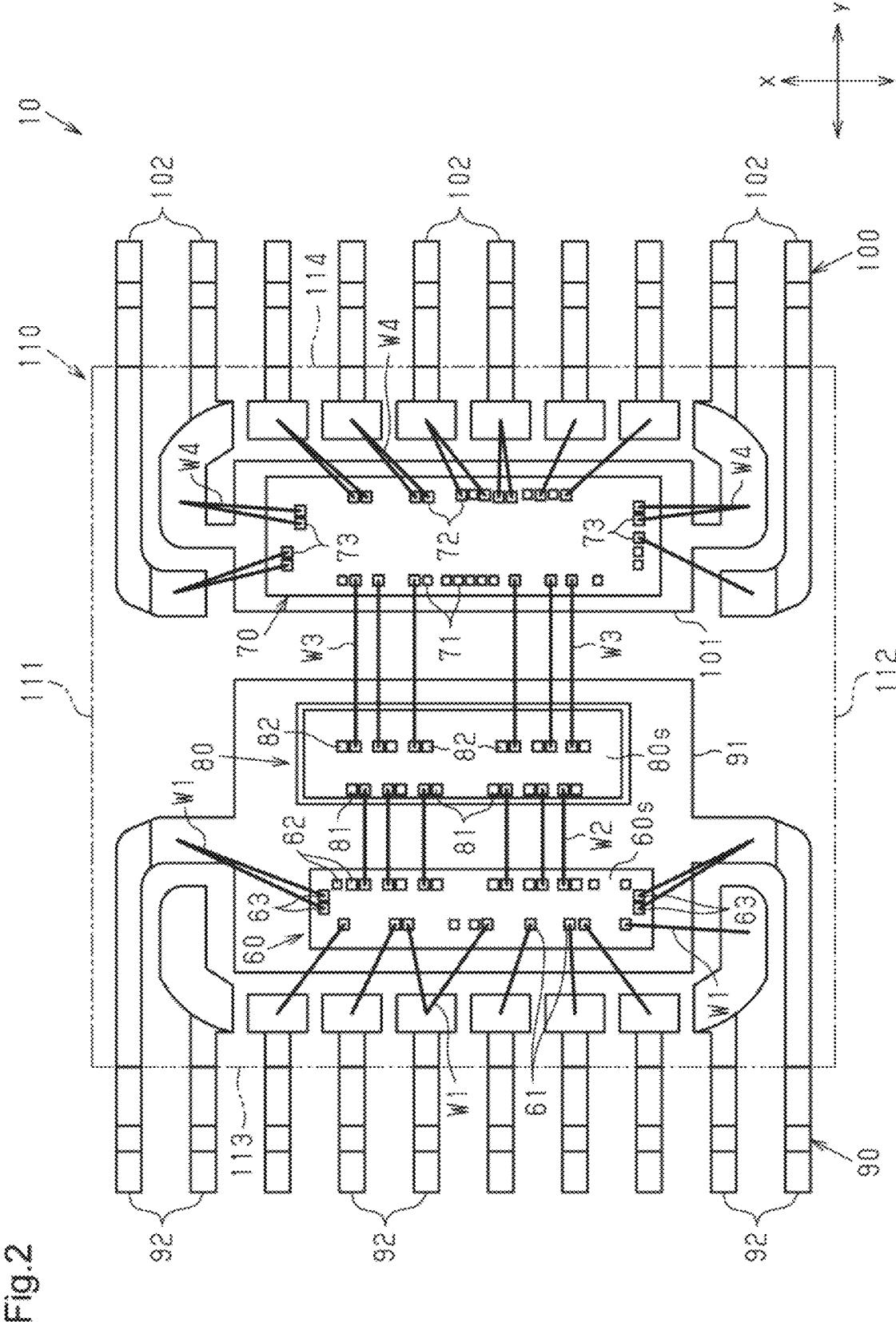


Fig.3

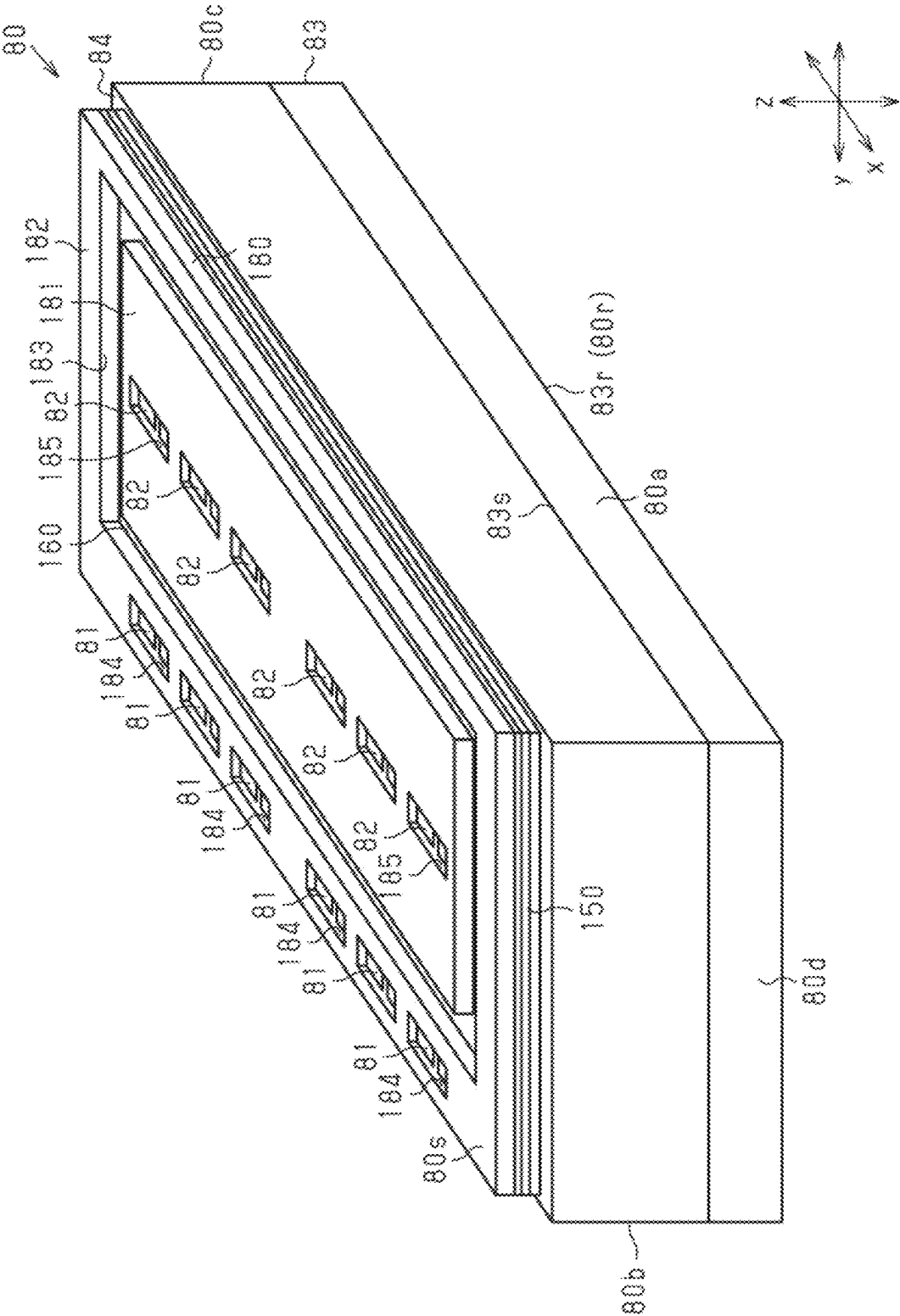
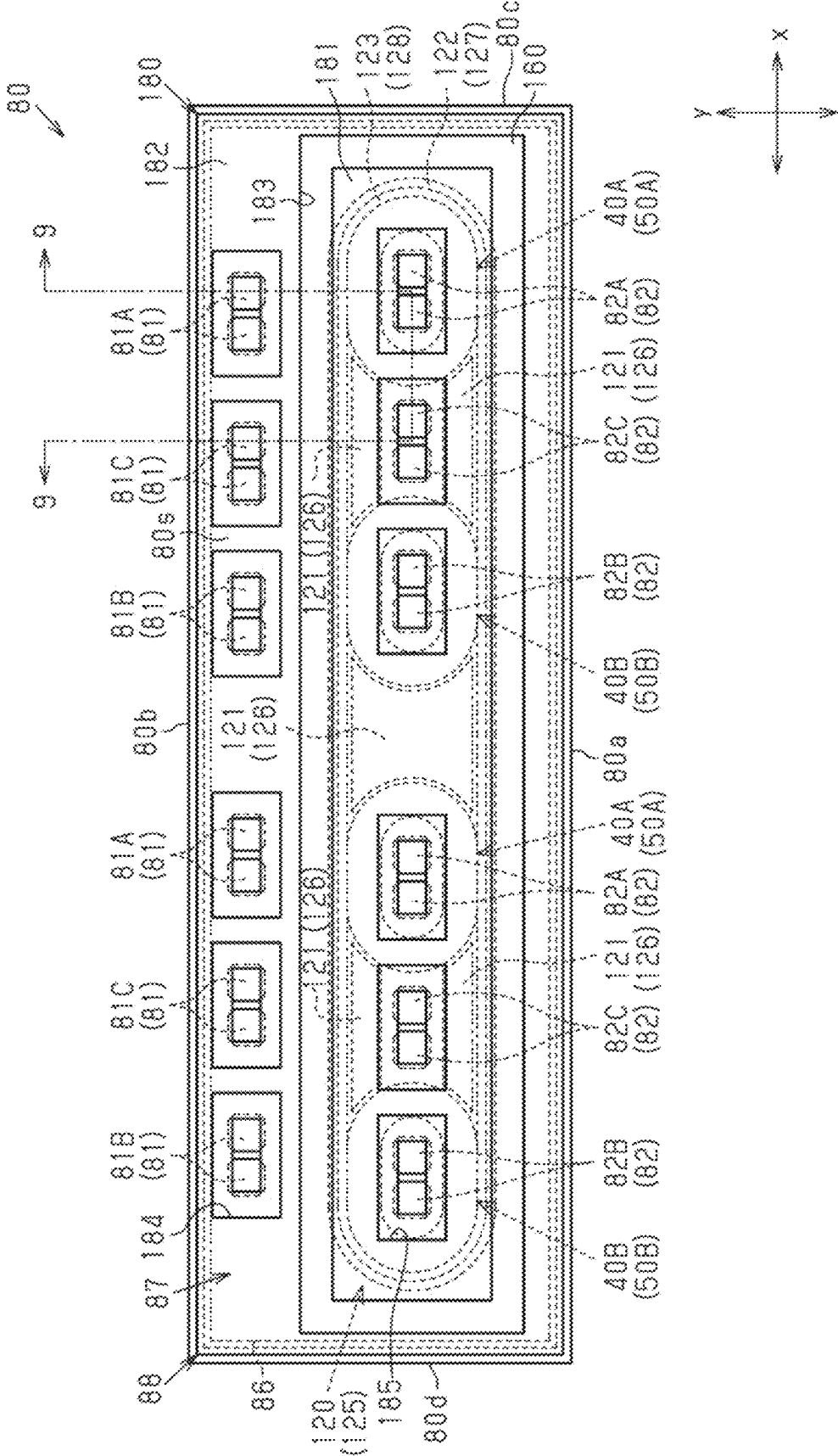


Fig.4



50

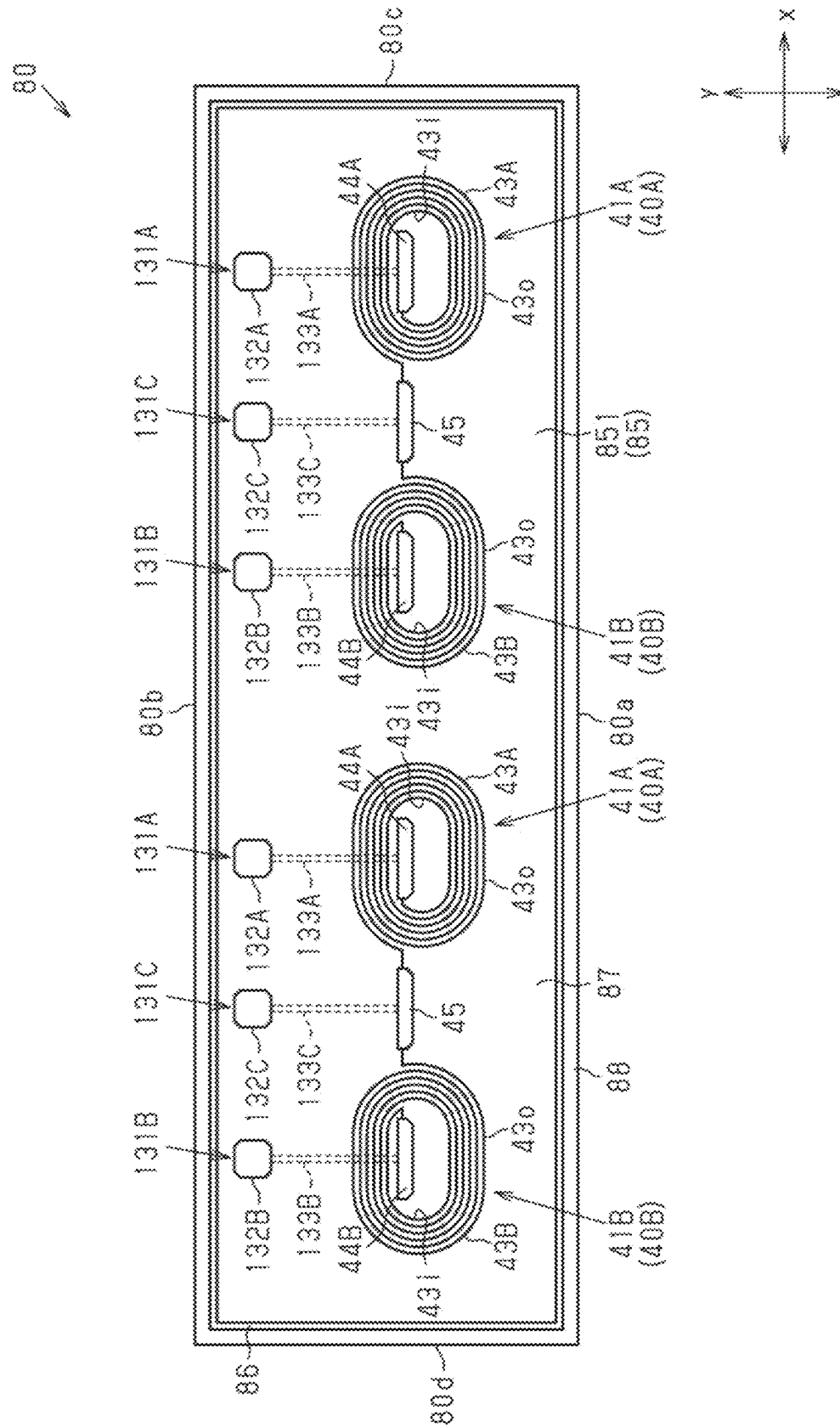


Fig.6

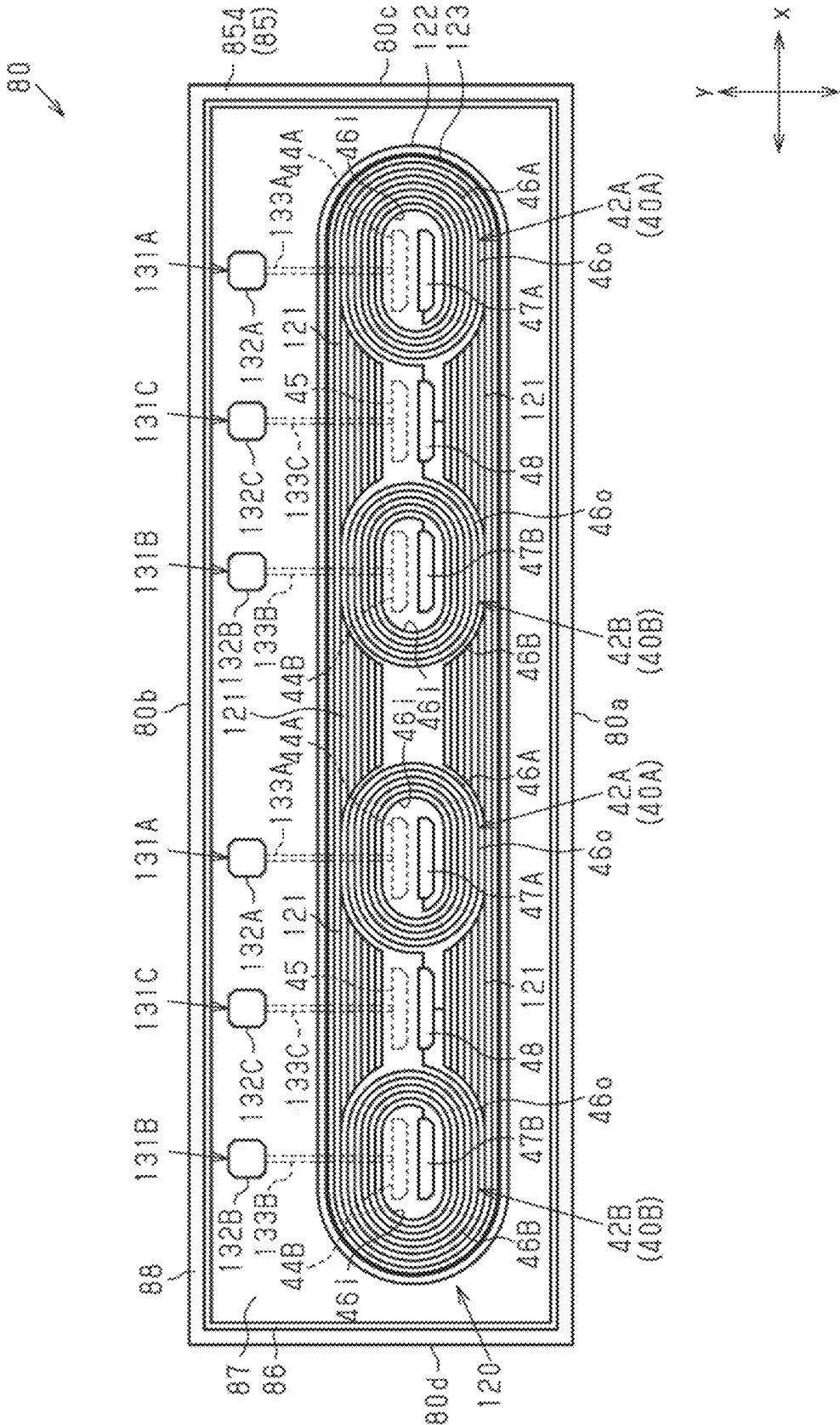


Fig.7

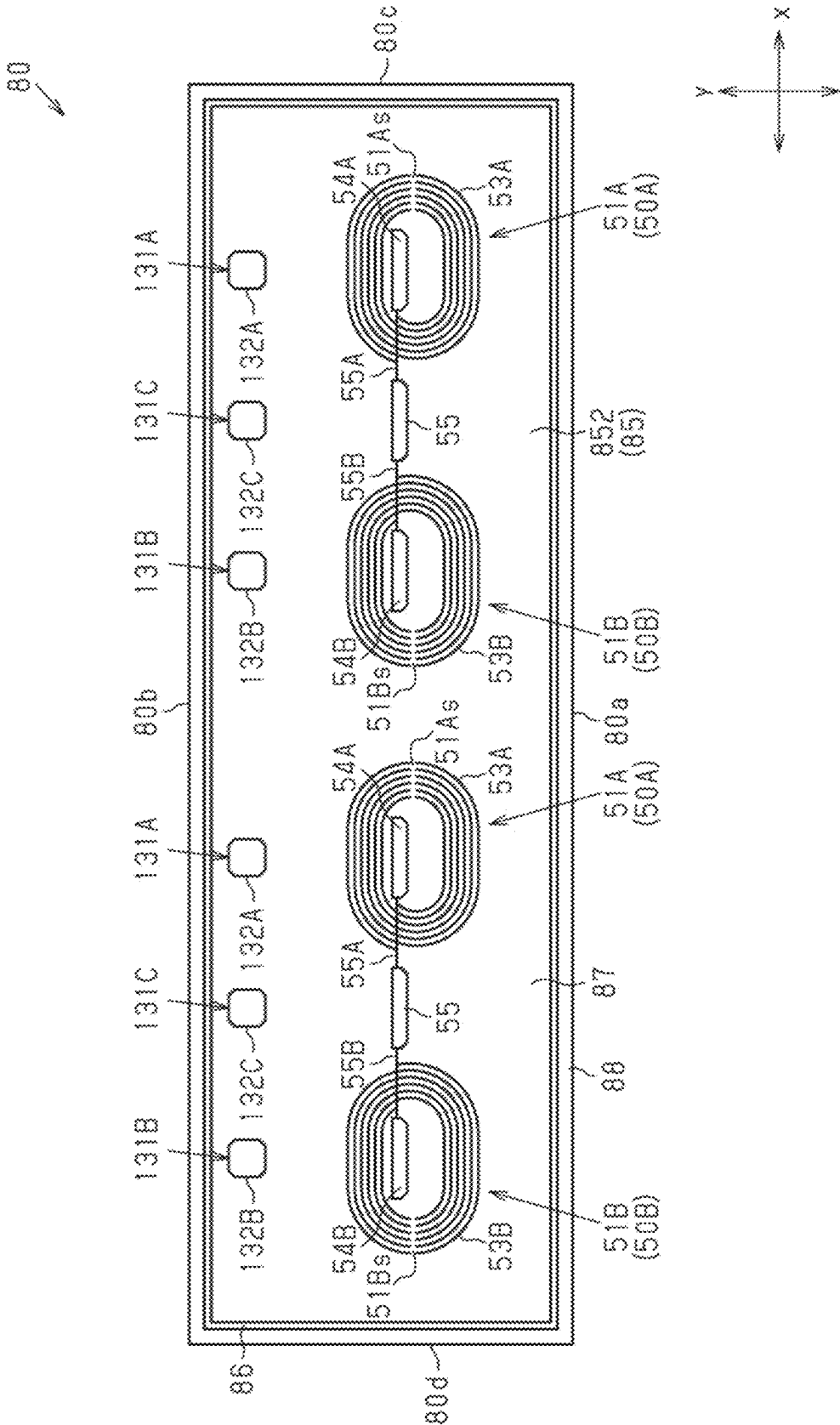
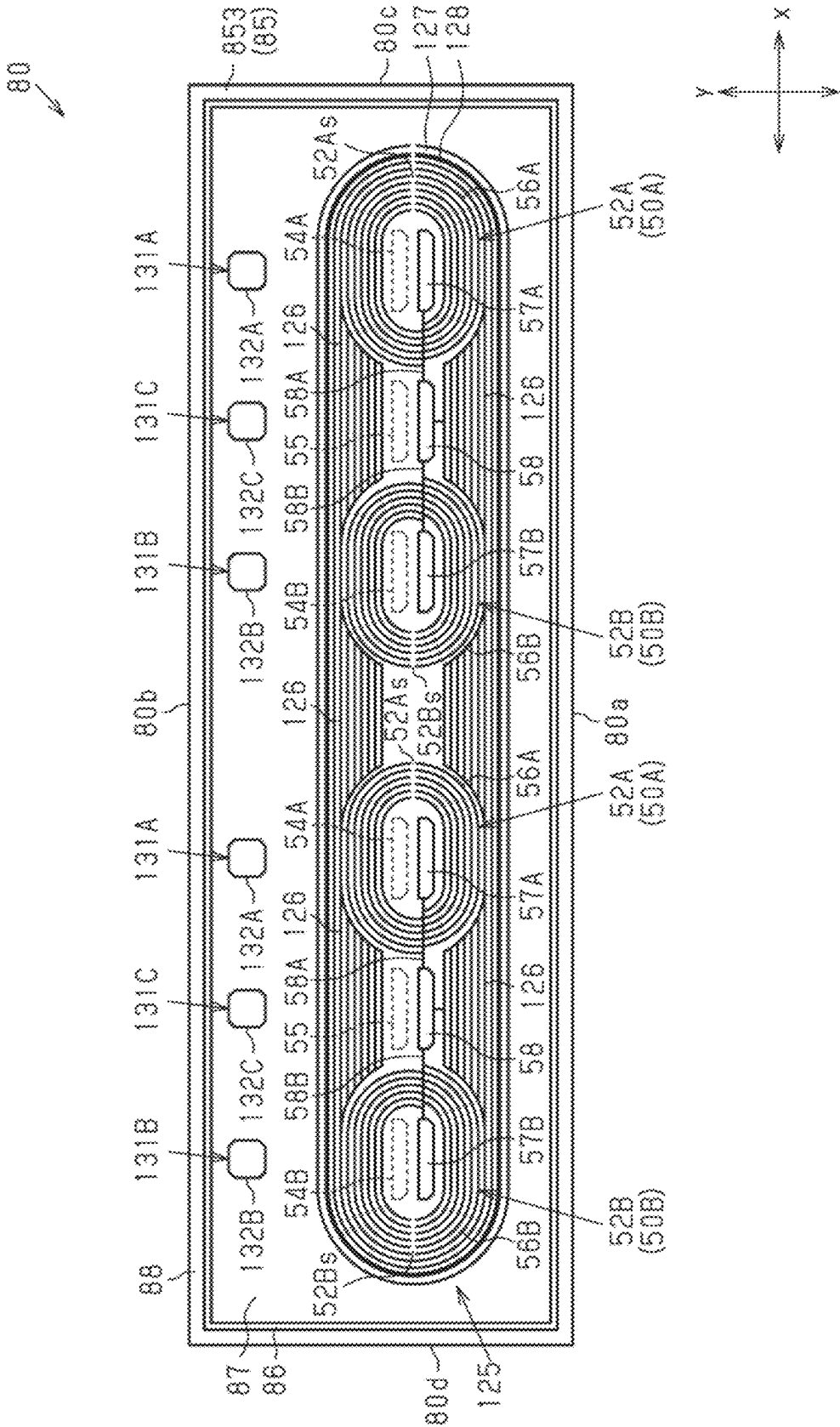


Fig. 8



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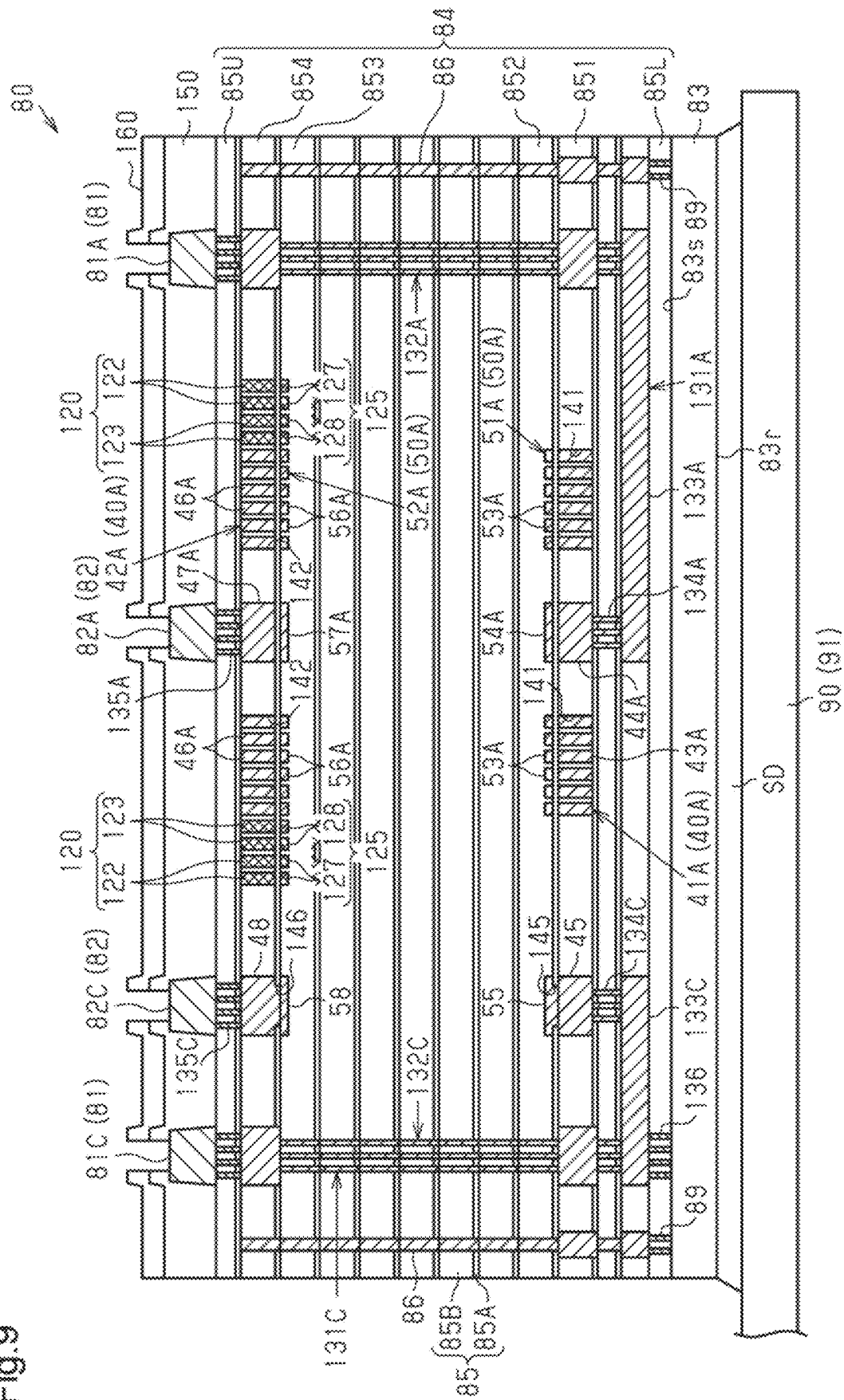


Fig.10

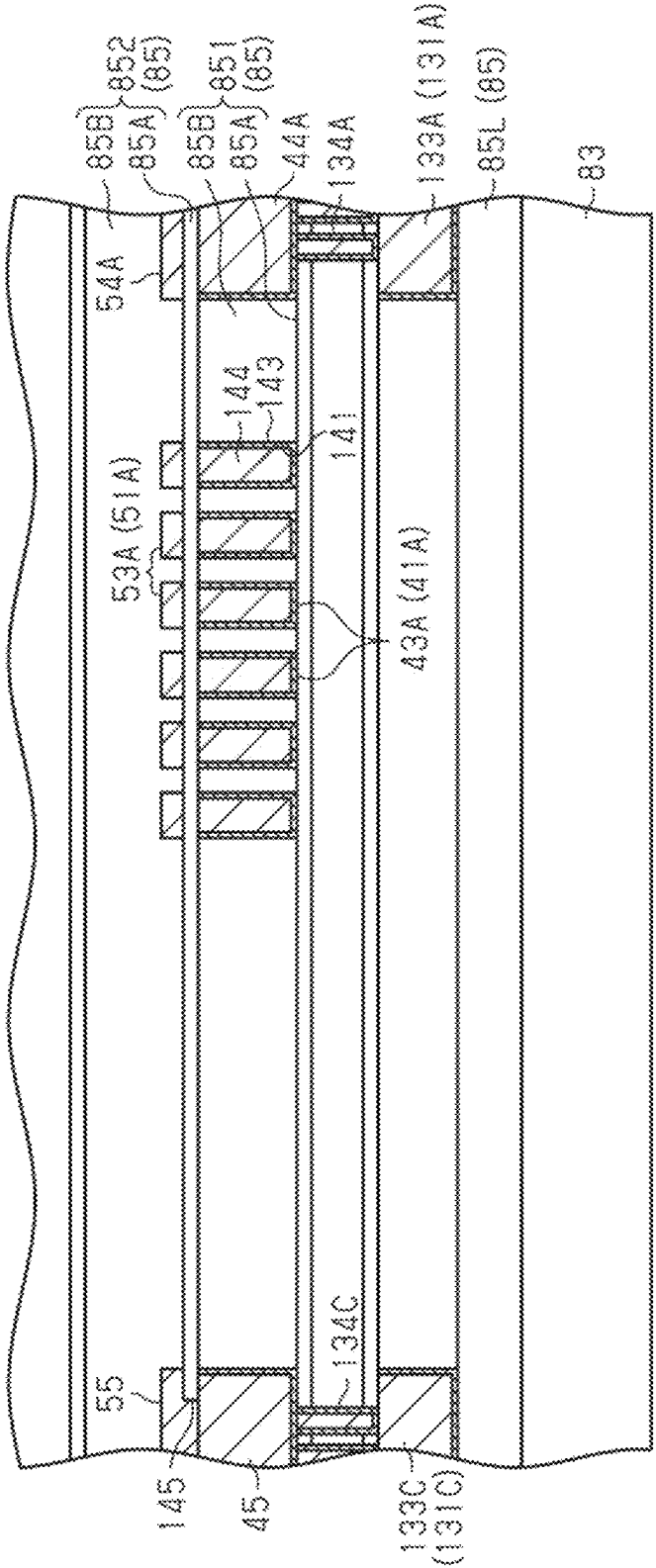


Fig.11

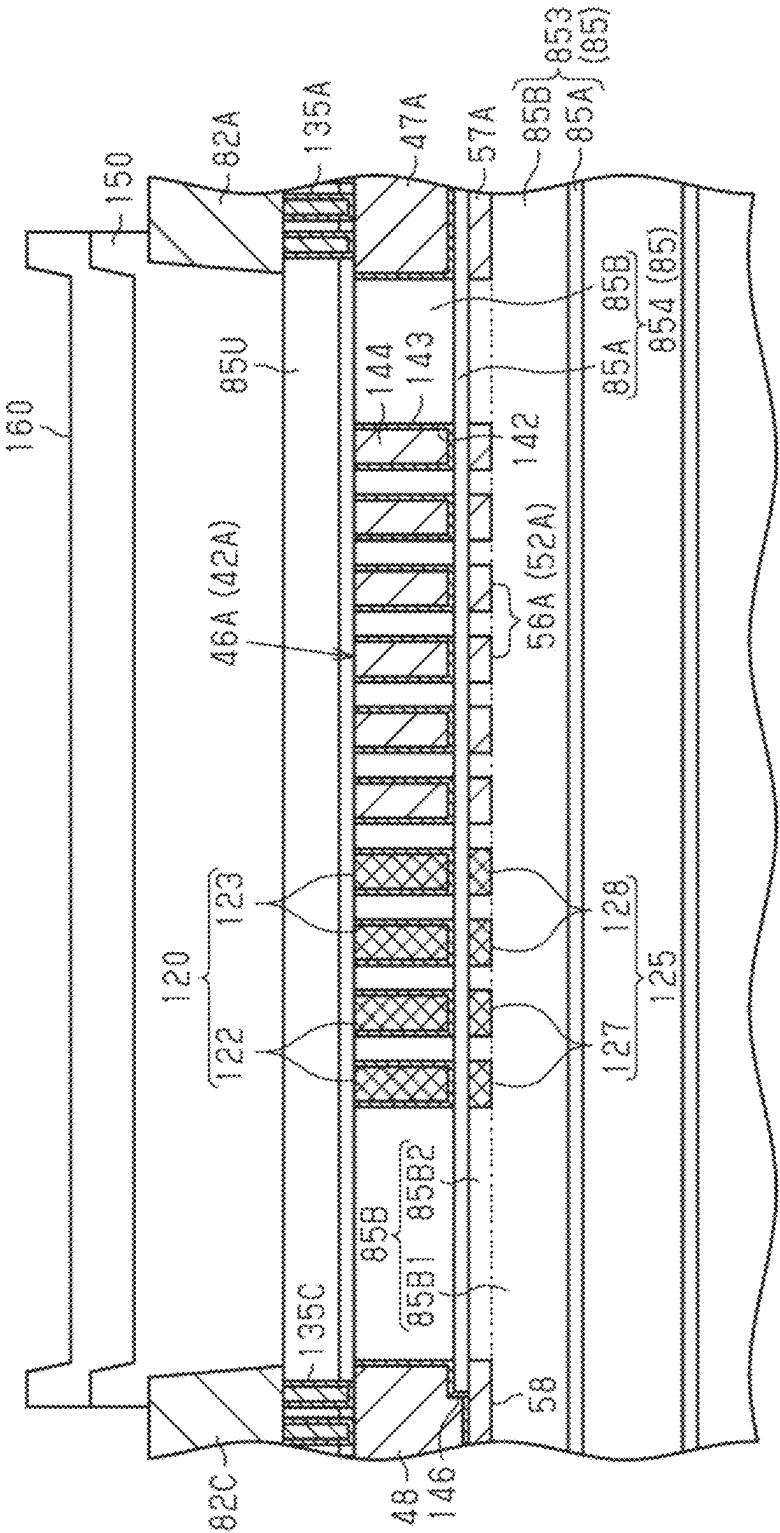


Fig.12

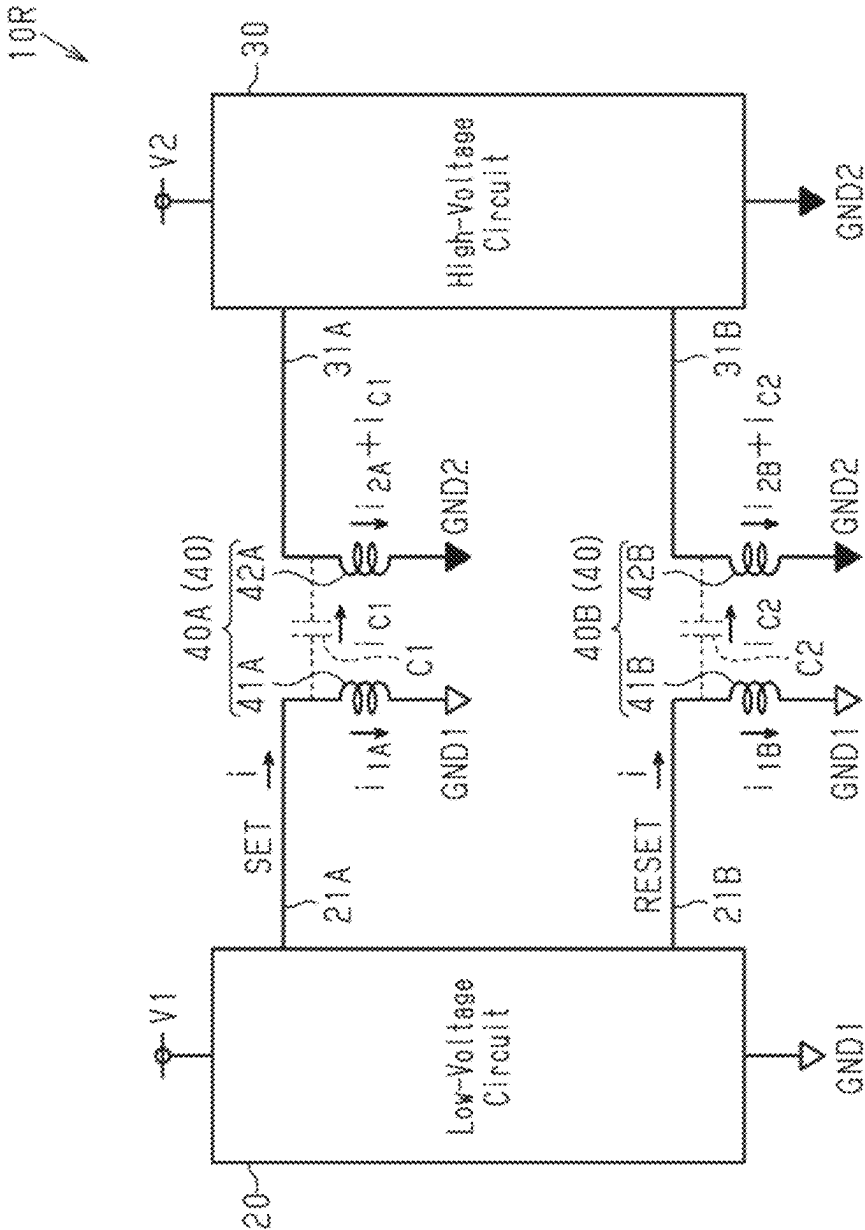
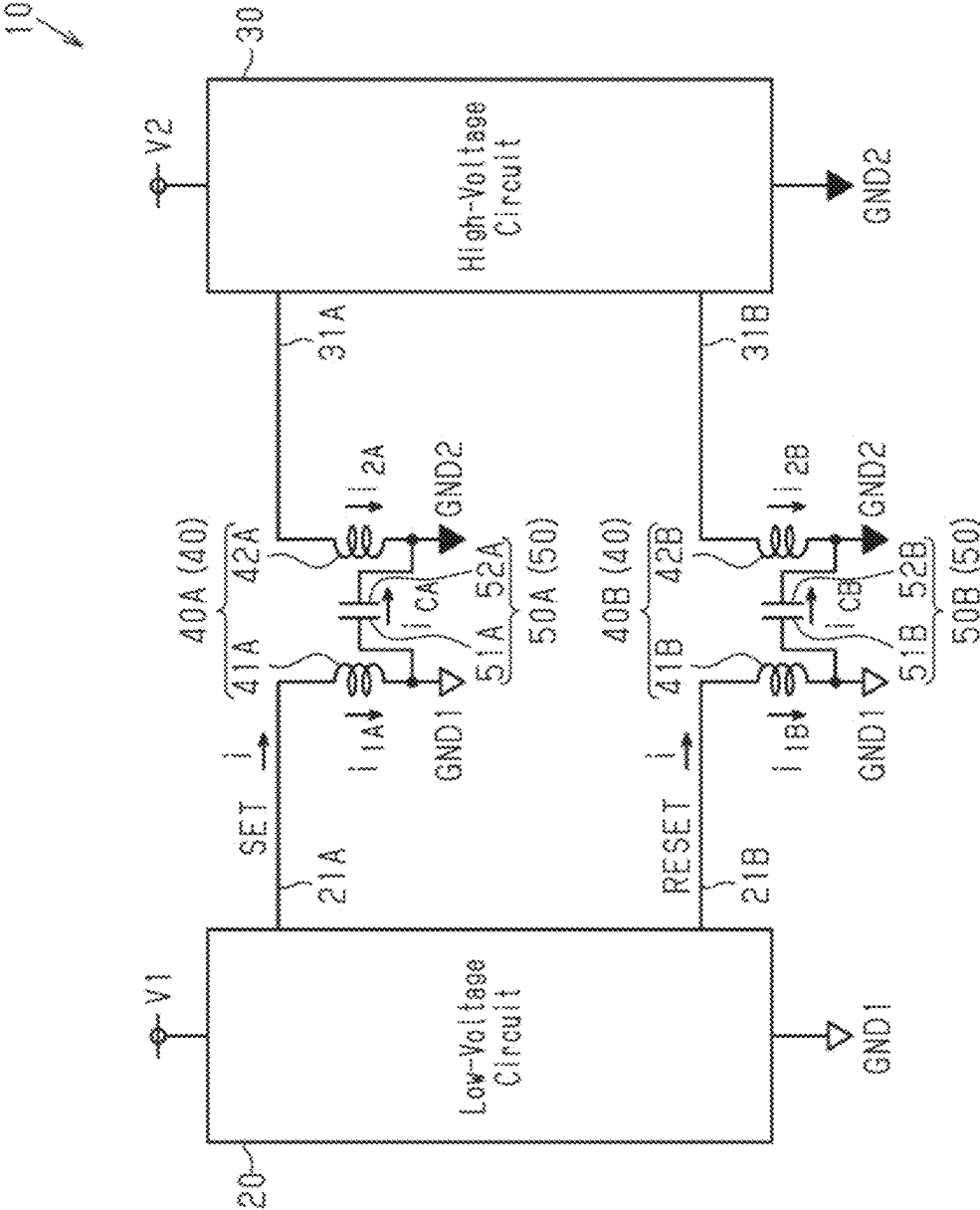


Fig. 13



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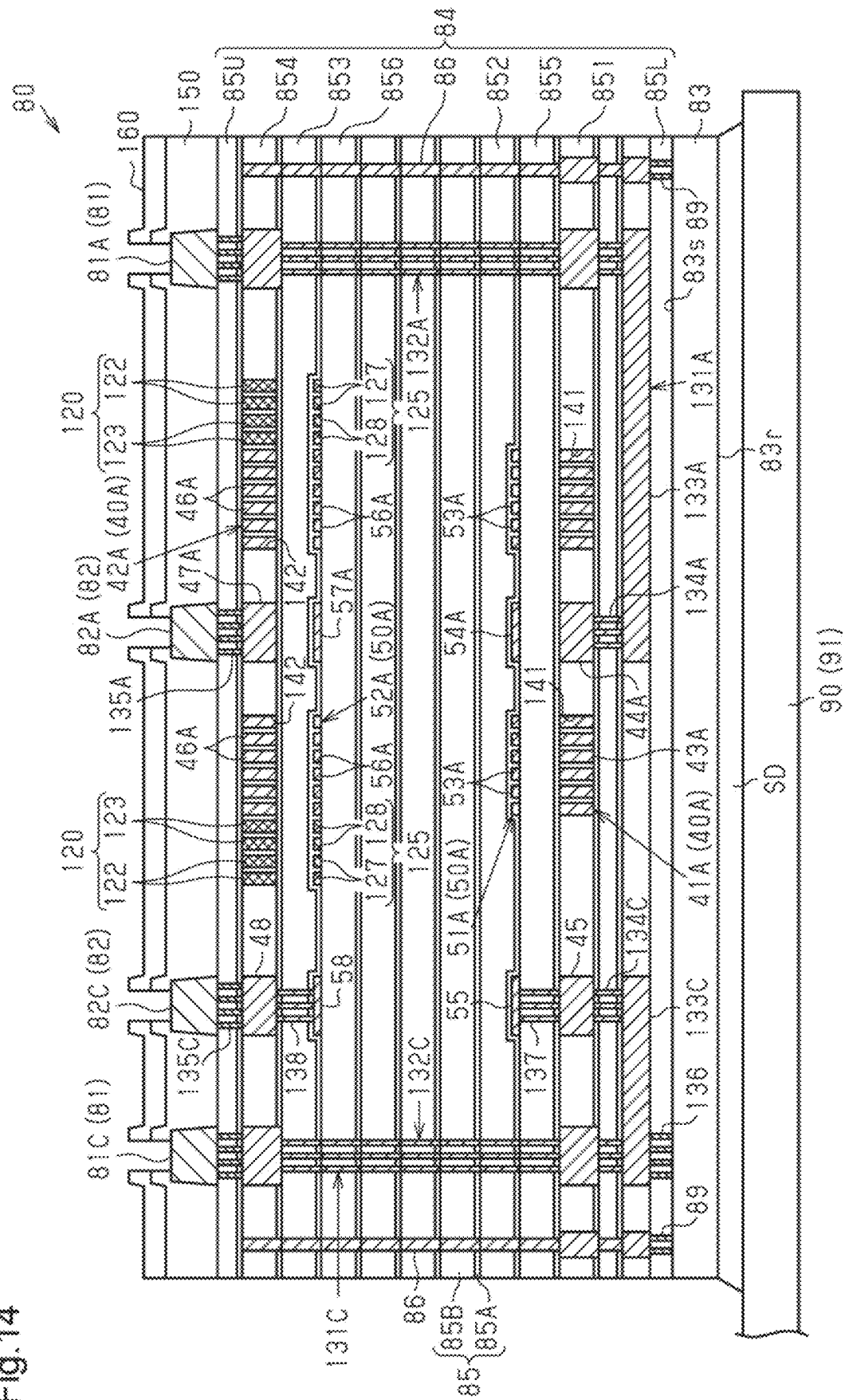


Fig.15A

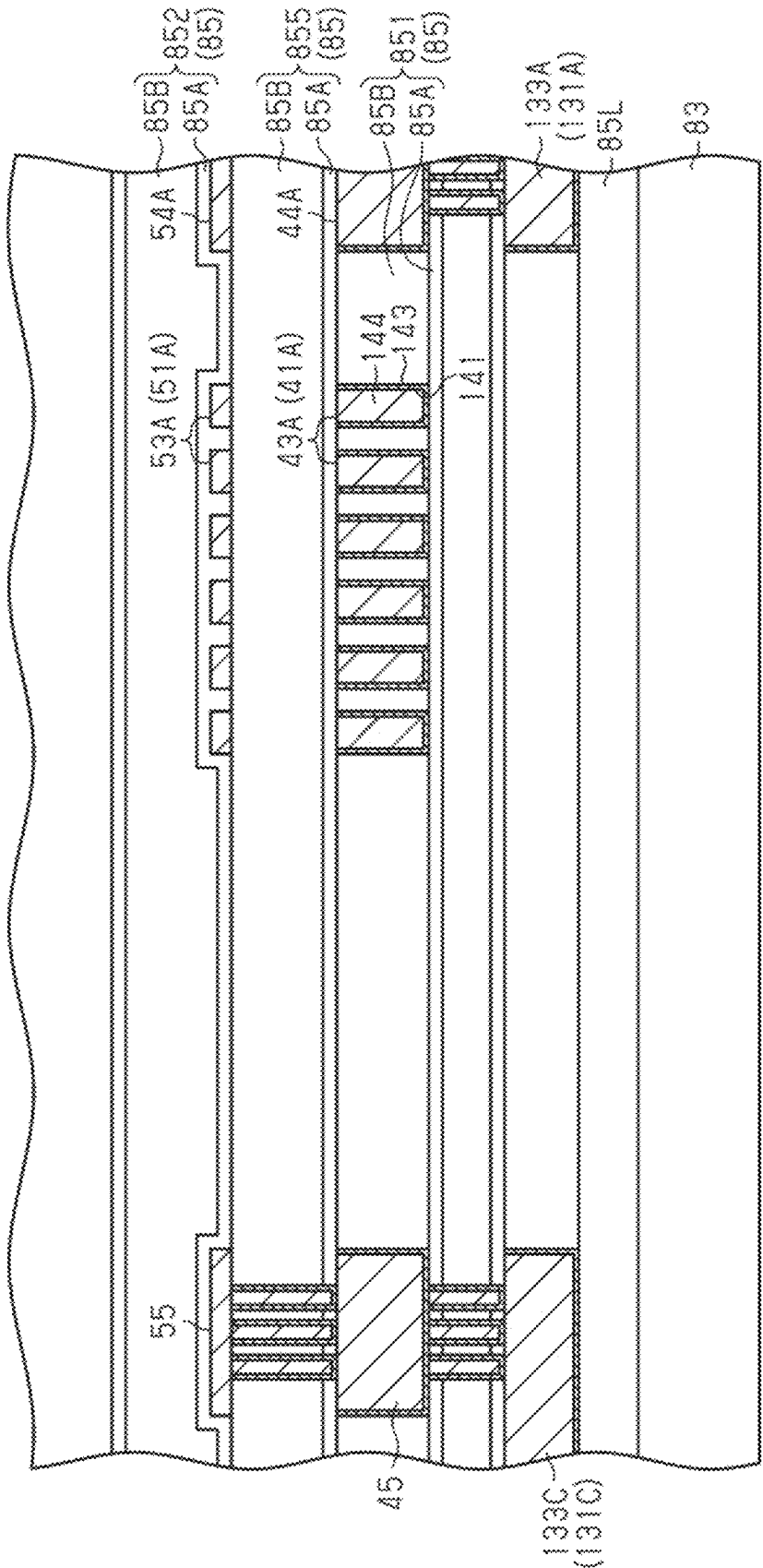


Fig.15B

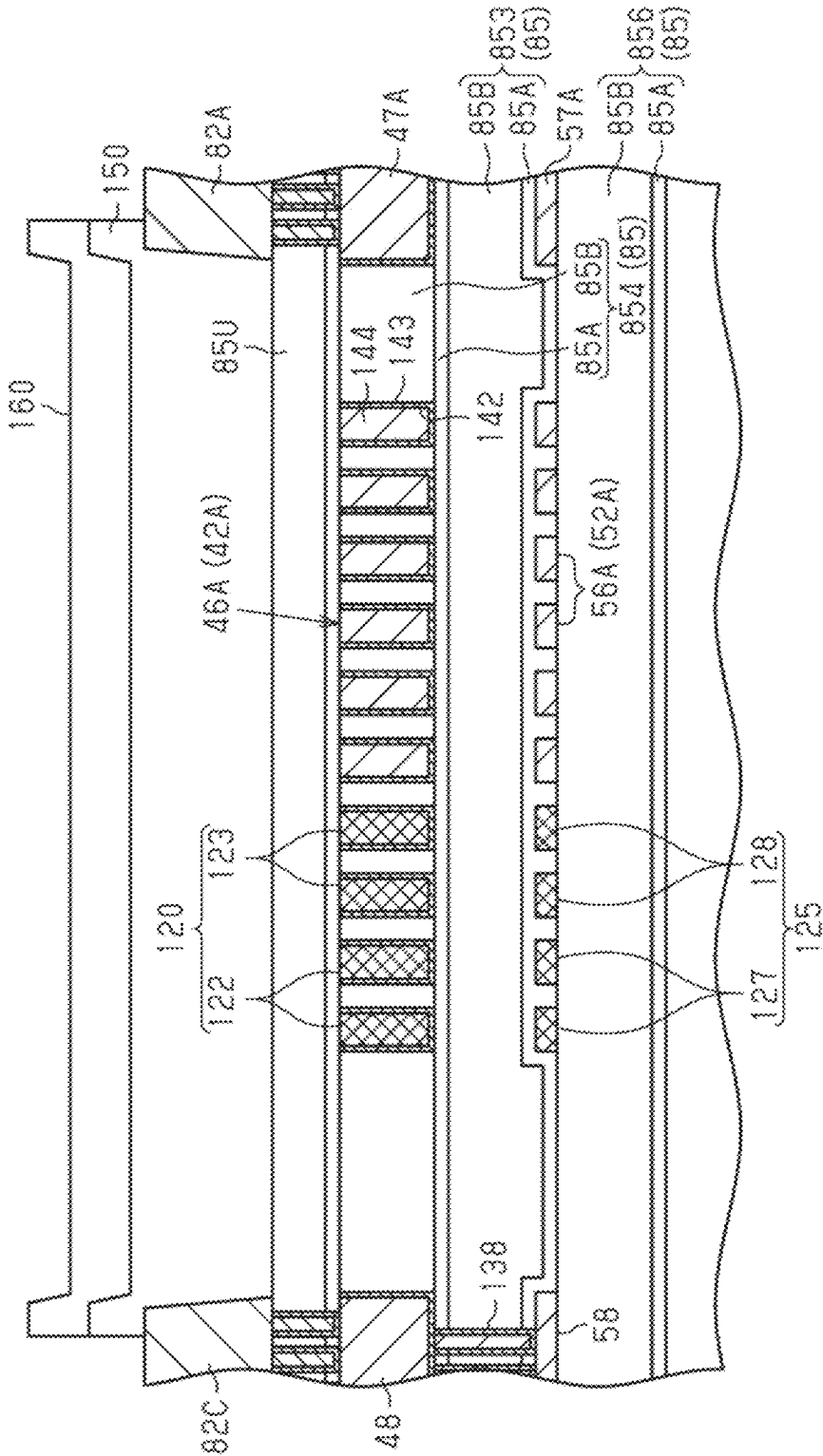


Fig. 16A

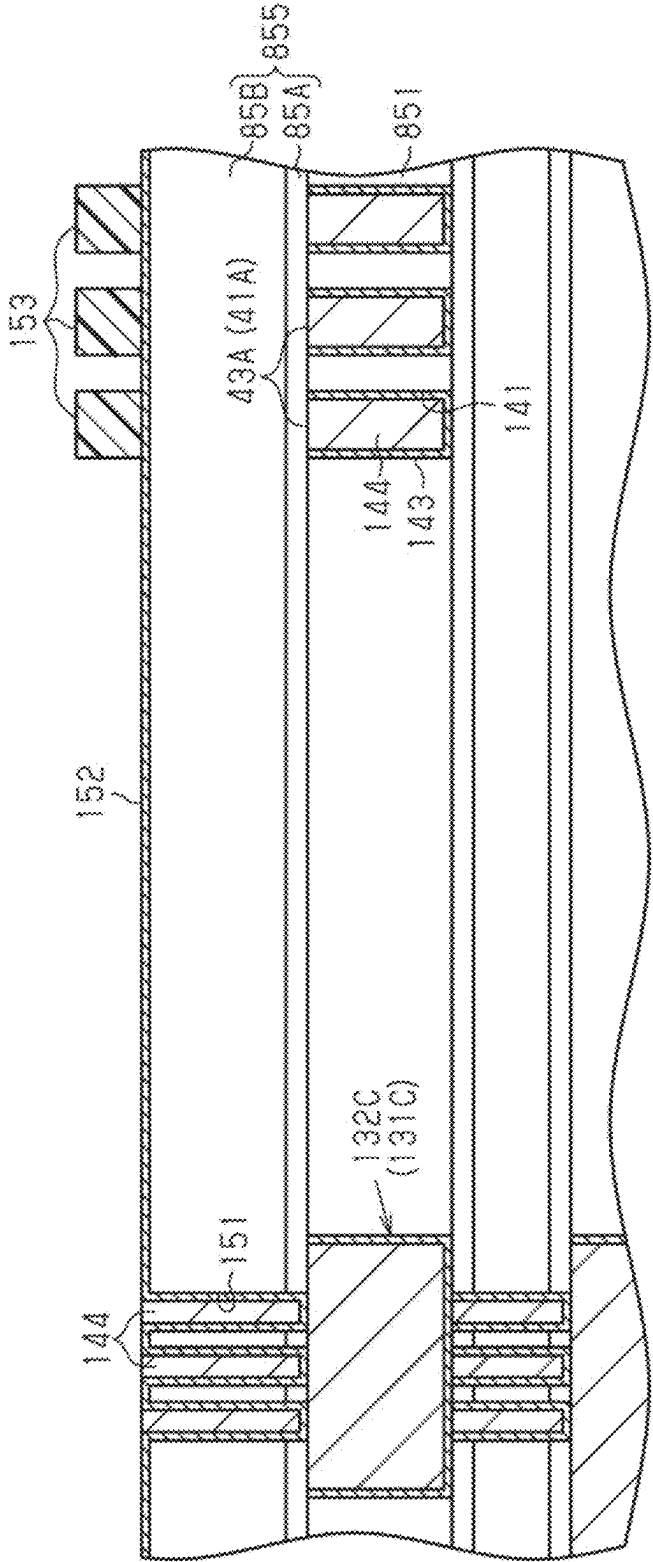
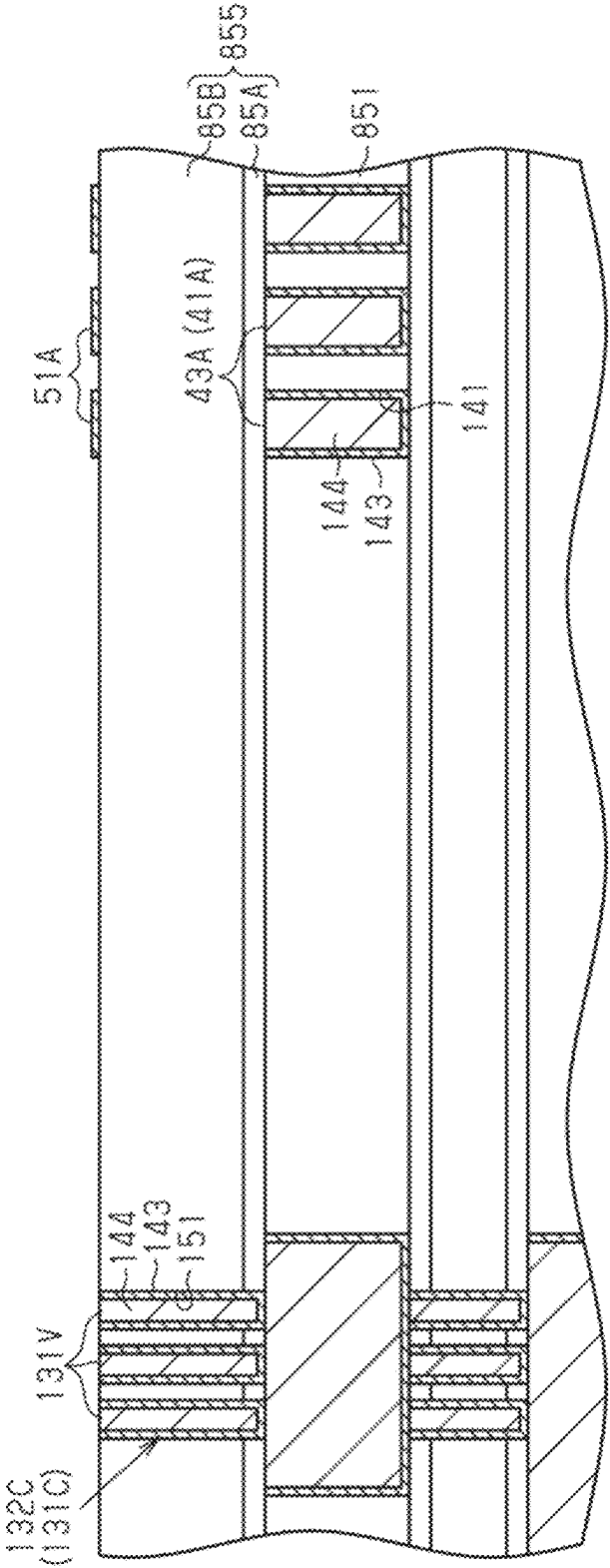


Fig. 16B



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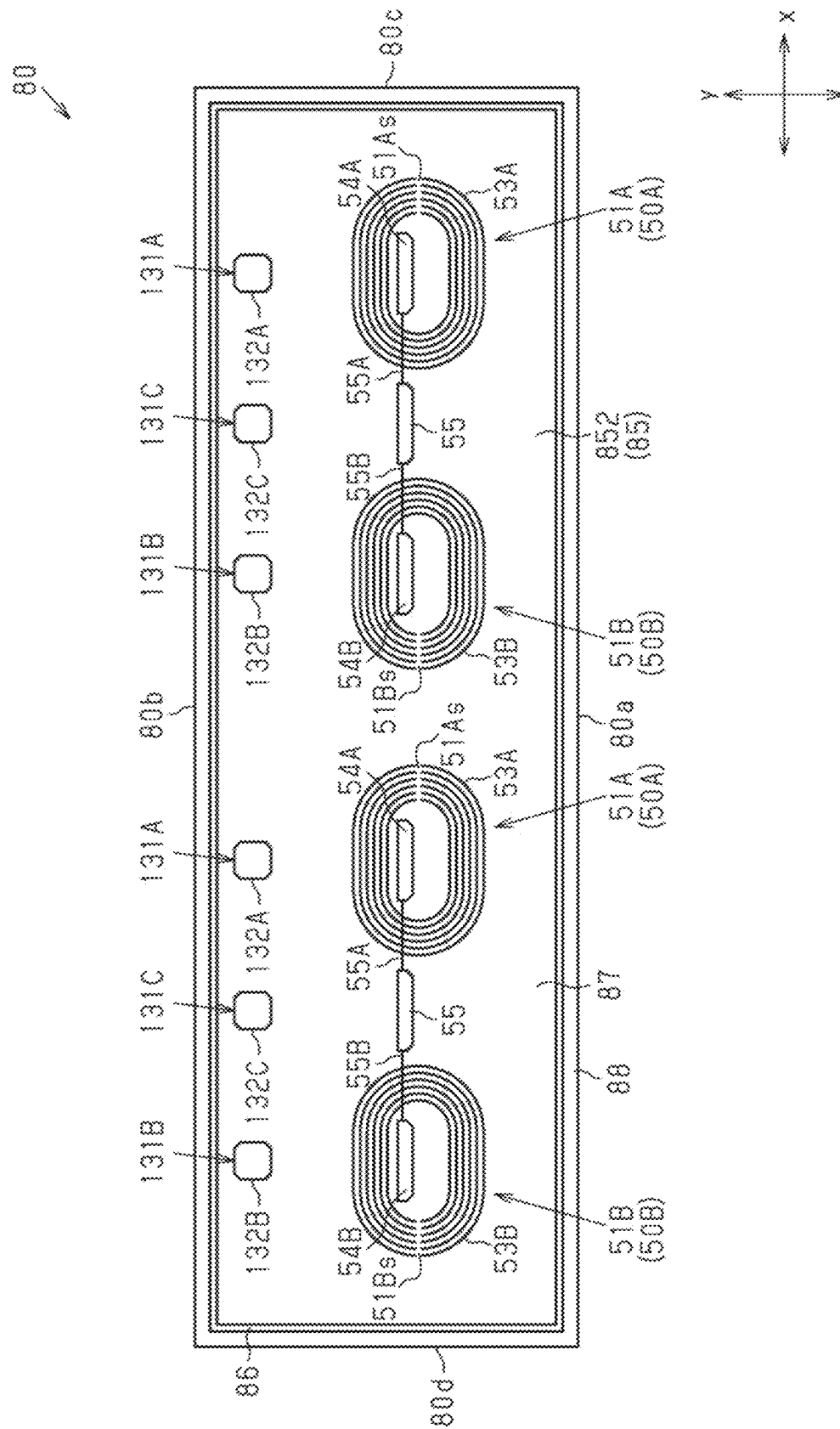


Fig.19

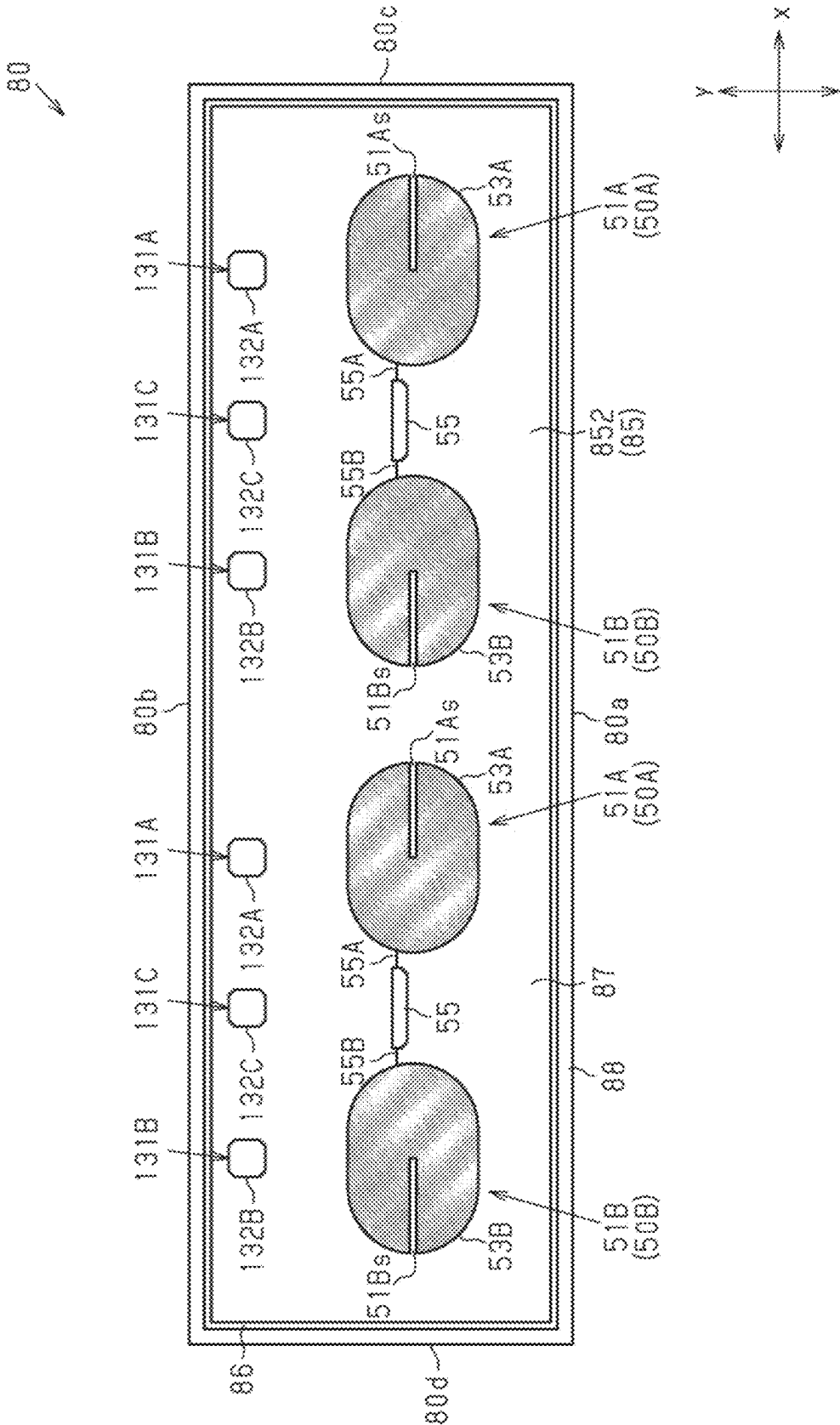


Fig.20

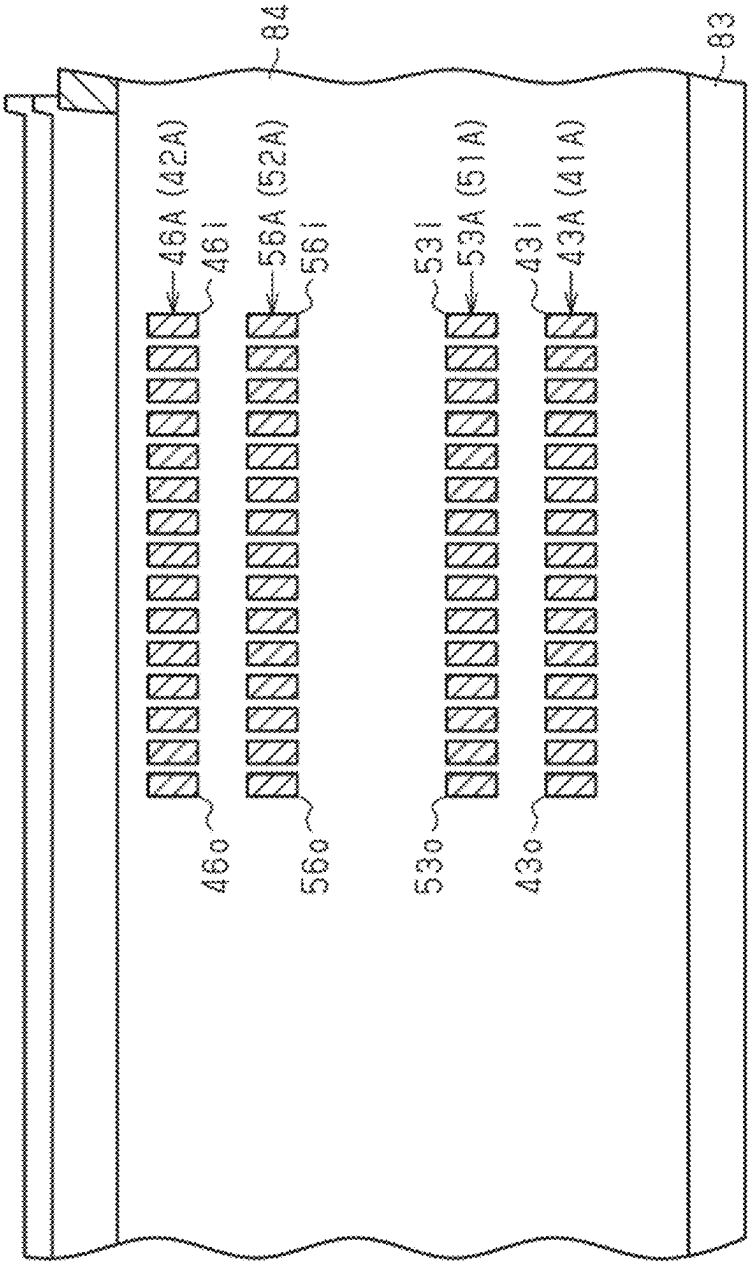


Fig.21

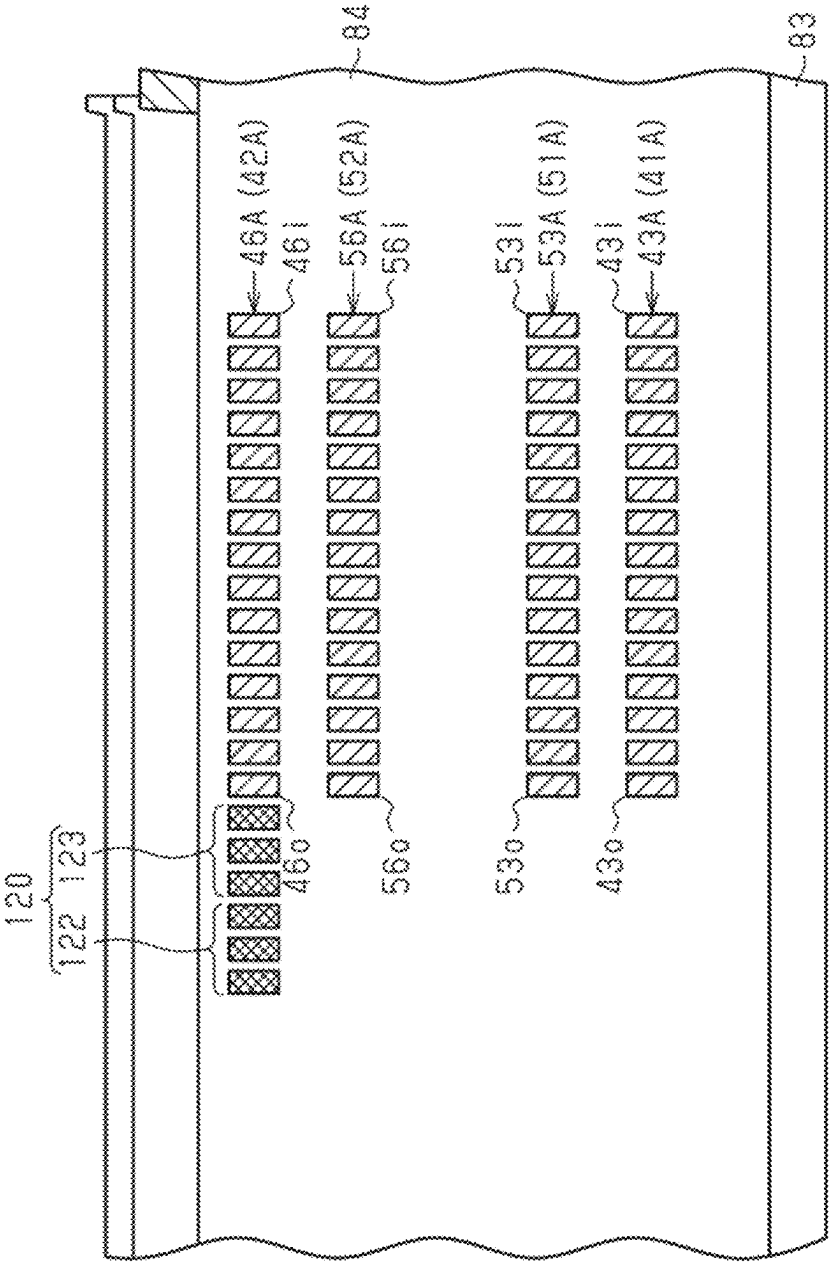


Fig.22

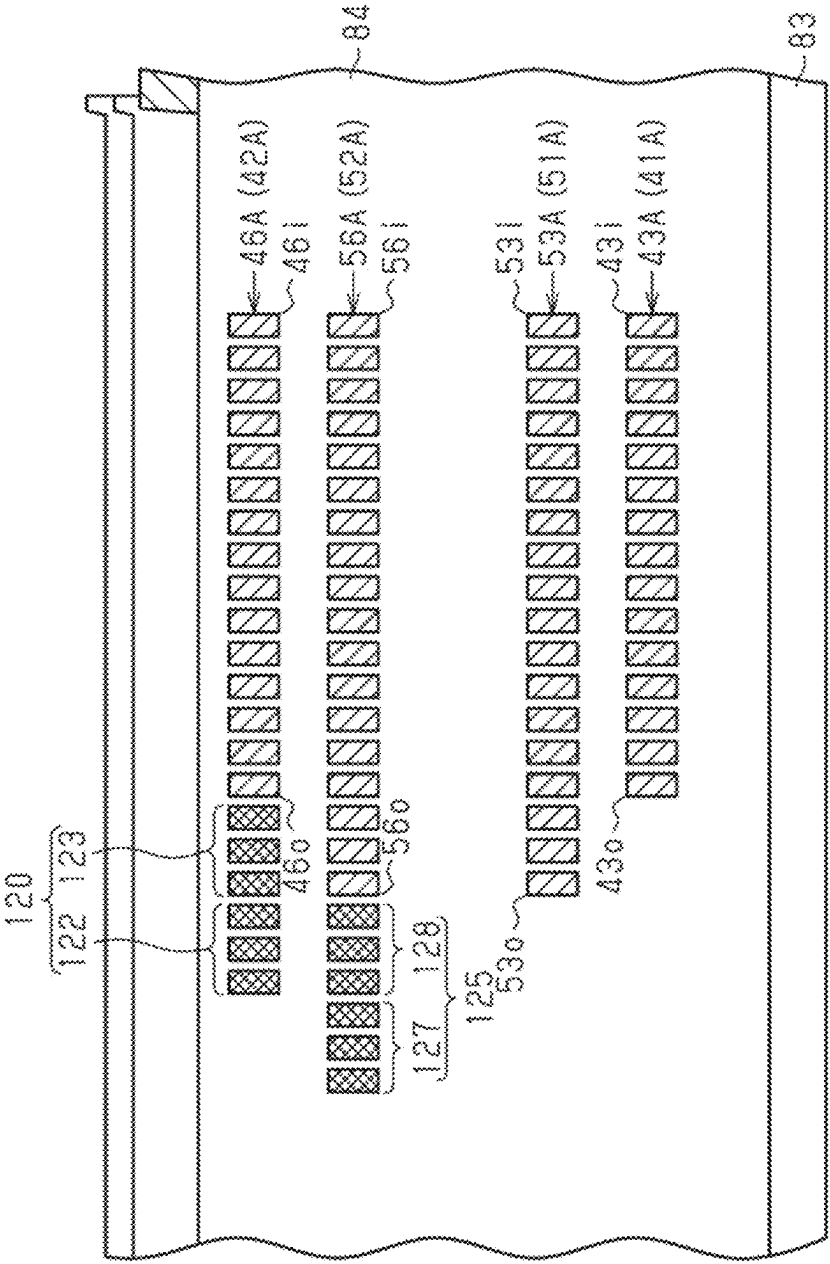


Fig.23

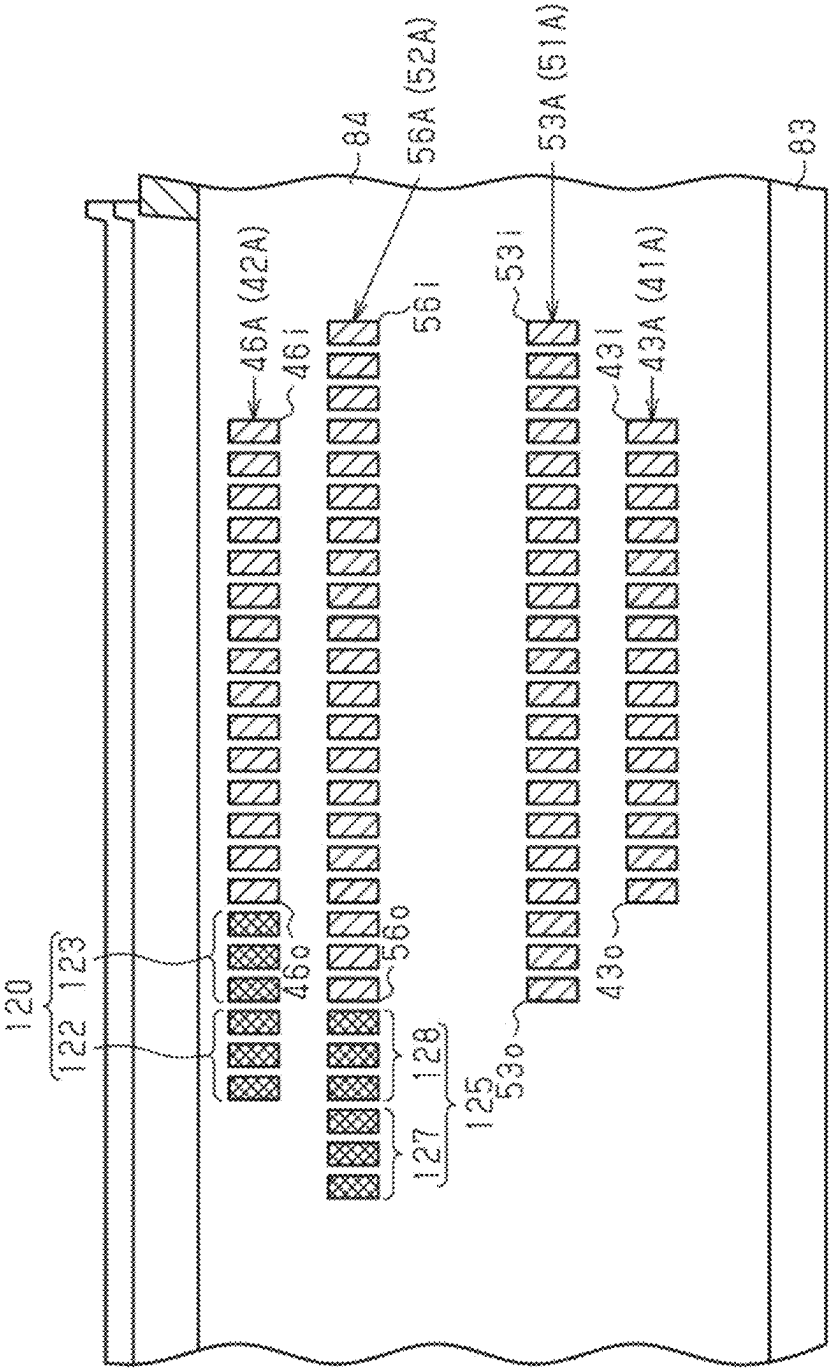


Fig.24

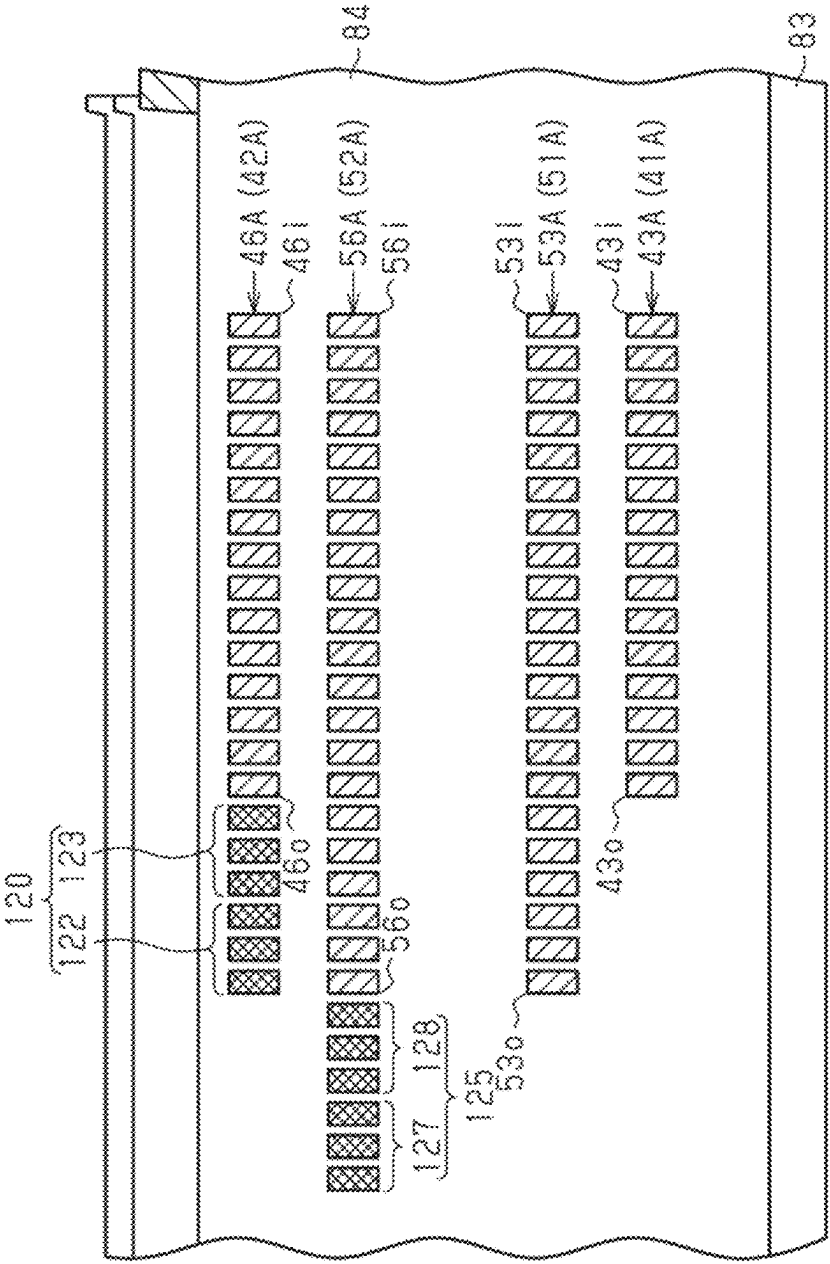


Fig.25

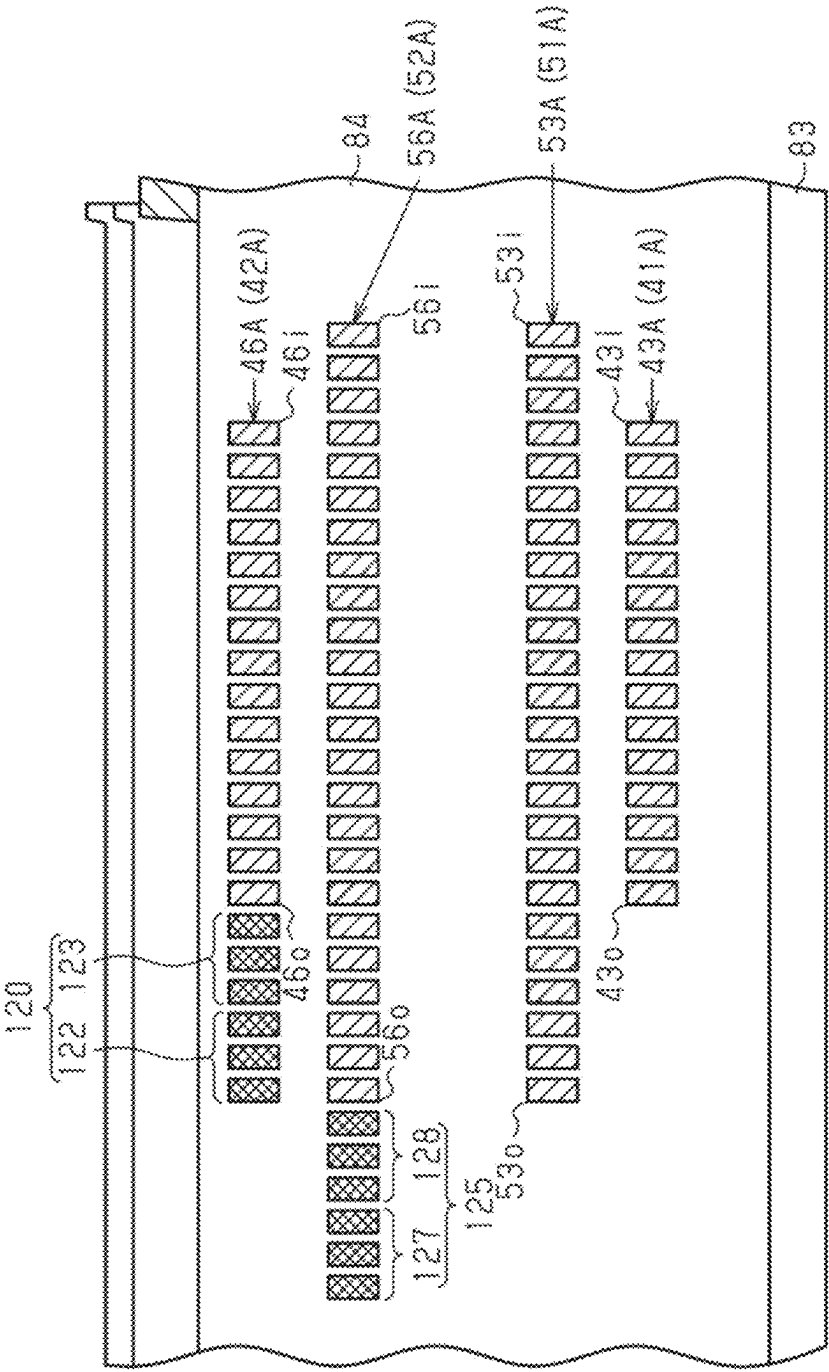


Fig.26

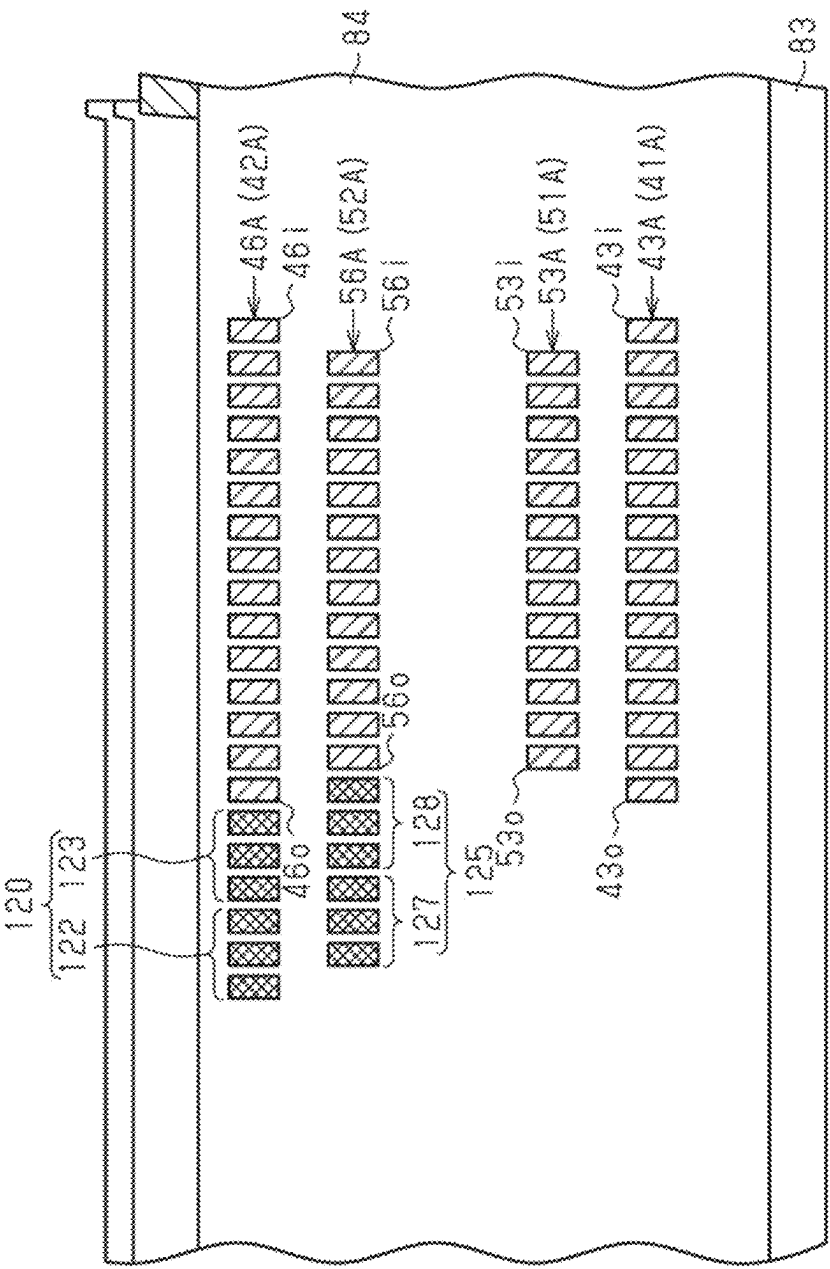


Fig.27

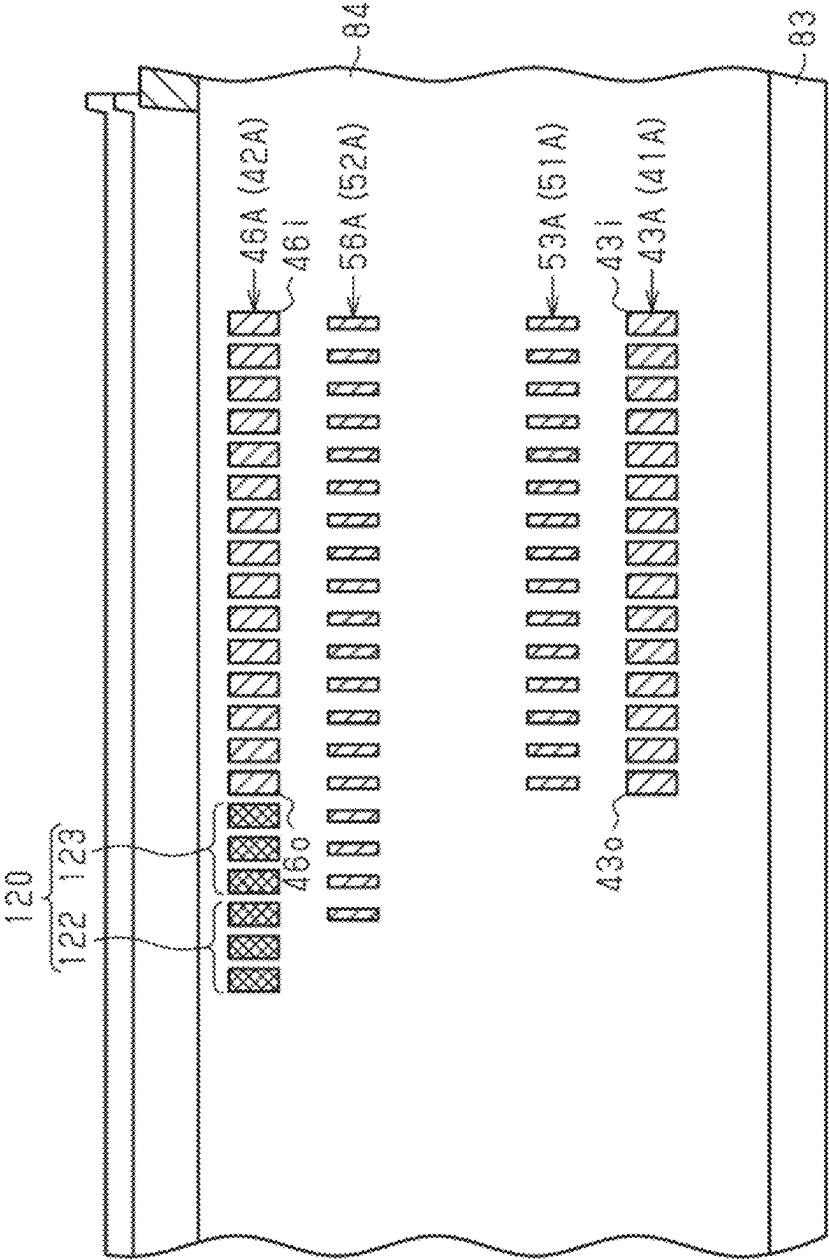
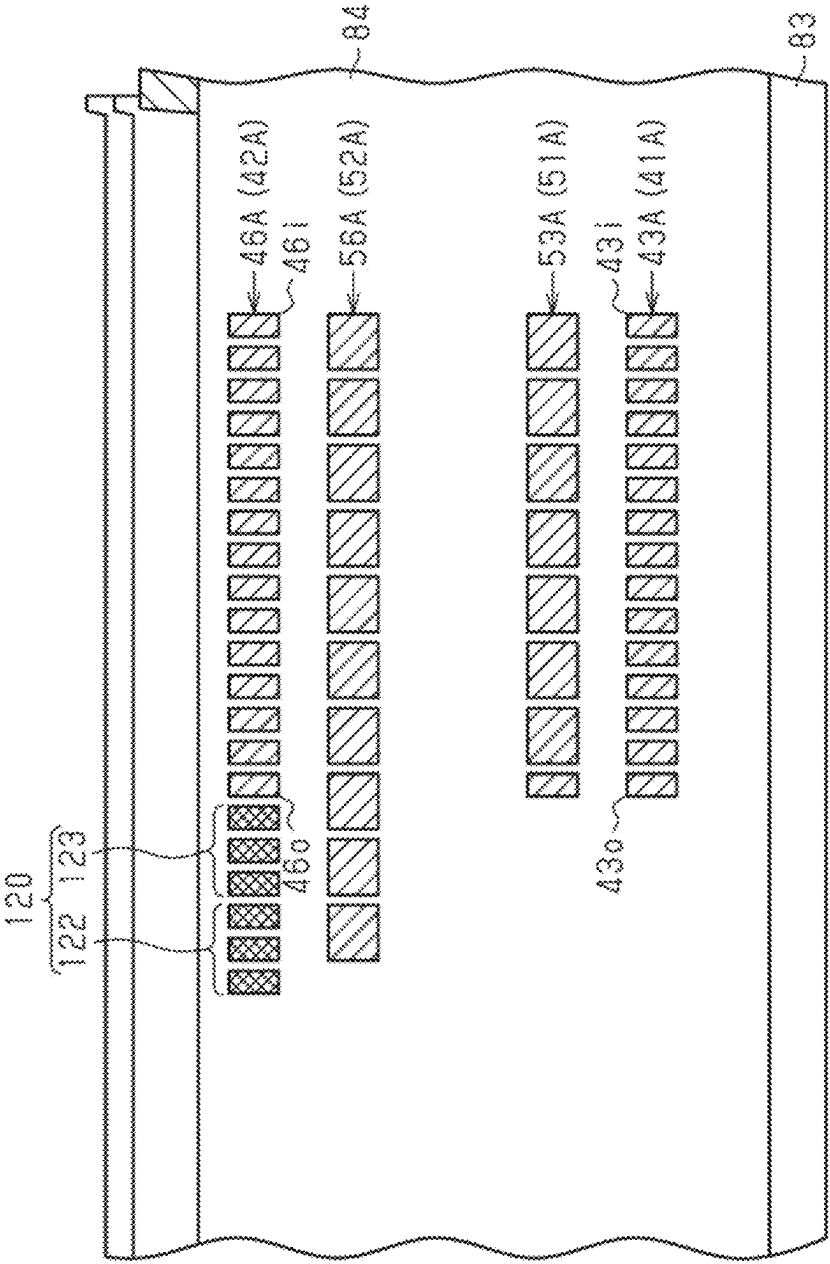


Fig.28



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ISOLATION TRANSFORMER

CROSS-REFERENCE TO RELATED APPLICATIONS

This present application is a bypass continuation of International Application No. PCT/JP2022/015019, filed Mar. 28, 2022, which claims priority from JP 2021-055721, filed Mar. 29, 2021, the entire contents of each are incorporated herein by reference.

BACKGROUND

The present disclosure relates to an isolation transformer. A known example of a gate driver that applies a gate voltage to the gate of a switching element such as a transistor is an insulated gate driver. JP 2018-78169 A describes an example of a semiconductor integrated circuit used as an insulated gate driver that includes a transformer. The transformer includes a first coil at the primary side and a second coil at the secondary side.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic circuit diagram showing a first embodiment of a gate driver.

FIG. 2 is a plan view showing an internal structure of the gate driver of the first embodiment.

FIG. 3 is a perspective view showing a transformer chip included in the gate driver of the first embodiment.

FIG. 4 is a plan view of the transformer chip shown in FIG. 3.

FIG. 5 is a schematic cross-sectional view showing a first coil of the transformer chip shown in FIG. 3.

FIG. 6 is a schematic cross-sectional view showing a second coil of the transformer chip shown in FIG. 3.

FIG. 7 is a schematic cross-sectional view showing a first capacitor electrode of the transformer chip shown in FIG. 3.

FIG. 8 is a schematic cross-sectional view showing a second capacitor electrode of the transformer chip shown in FIG. 3.

FIG. 9 is a cross-sectional view of the transformer chip mounted on a low-voltage die pad taken along line 9-9 in FIG. 4.

FIG. 10 is an enlarged view showing a portion of the transformer chip shown in FIG. 9.

FIG. 11 is an enlarged view showing a portion of the transformer chip shown in FIG. 9.

FIG. 12 is a circuit diagram showing an operation of a comparative example of a gate driver.

FIG. 13 is a circuit diagram showing an operation of the gate driver in the present embodiment.

FIG. 14 is a cross-sectional view of a transformer chip mounted on a low-voltage die pad in a second embodiment of a gate driver.

FIG. 15A is an enlarged partial view showing the transformer chip of the second embodiment.

FIG. 15B is an enlarged partial view showing the transformer chip of the second embodiment.

FIG. 16A is a cross-sectional view showing a process for forming a first capacitor electrode.

FIG. 16B is a cross-sectional view showing a process for forming the first capacitor electrode.

FIG. 17 is a schematic cross-sectional view showing a first capacitor electrode of a transformer chip in a modified example.

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FIG. 18 is a schematic cross-sectional view showing a first capacitor electrode of a transformer chip in a modified example.

FIG. 19 is a schematic cross-sectional view showing a first capacitor electrode of a transformer chip in a modified example.

FIG. 20 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 21 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 22 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 23 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 24 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 25 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 26 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 27 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

FIG. 28 is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

DETAILED DESCRIPTION

Embodiments of a gate driver will be described below with reference to the drawings.

The embodiments described below exemplify configurations and methods for embodying a technical concept and are not intended to limit the material, shape, structure, layout, dimensions, and the like of each component to those described below. Components in the drawings may be enlarged for simplicity and clarity. The dimensional proportion of a component may not be drawn to scale or may differ between drawings. In the cross-sectional views, hatching of components may be partially omitted to facilitate understanding.

First Embodiment

Gate Driver

A first embodiment of a gate driver 10 will be described with reference to FIGS. 1 to 11.

FIG. 1 schematically shows an example of a circuit configuration of the gate driver 10. The gate driver 10 is configured to apply a drive voltage signal to the gate of a switching element. In an example, the gate driver 10 is used

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in an inverter device **500** mounted on an electric vehicle or a hybrid vehicle. The inverter device **500** includes two switching elements **501** and **502** connected in series to each other, the gate driver **10**, and an electronic control unit **503** (ECU) that controls the gate driver **10**. In an example, the switching element **501** is a high-side switching element connected to a drive power supply. The switching element **502** is a low-side switching element. Examples of the switching elements **501** and **502** include transistors such as a Si metal-oxide-semiconductor field-effect transistor (Si MOSFET), a SiC MOSFET, and an insulated gate bipolar transistor (IGBT). The gate driver **10** of the present embodiment applies a drive voltage signal to the gate of the switching element **501**. In the description hereafter, SiC MOSFETs are used in the switching elements **501** and **502**.

The gate driver **10** is provided for each of the switching elements **501** and **502** and separately drives the switching elements **501** and **502**. In the present embodiment, the gate driver **10** that drives the switching element **501** will be described for the sake of convenience.

The gate driver **10** includes a low-voltage circuit **20** to which a first voltage V1 is applied, a high-voltage circuit **30** to which a second voltage V2 is applied, and transformers **40** arranged between the low-voltage circuit **20** and the high-voltage circuit **30**. The second voltage V2 is higher than the first voltage V1. More specifically, the low-voltage circuit **20** and the high-voltage circuit **30** are connected by the transformers **40**. The first voltage V1 and the second voltage V2 are direct current voltages.

The gate driver **10** of the present embodiment is configured, based on a control signal from the ECU **503**, to transmit a signal from the low-voltage circuit **20** to the high-voltage circuit **30** through the transformers **40** and to output a drive voltage signal from the high-voltage circuit **30**.

The signal transmitted from the low-voltage circuit **20** toward the high-voltage circuit **30**, that is, a signal output from the low-voltage circuit **20**, is, for example, for driving the switching element **501**. Examples of the signal include a set signal and a reset signal. The set signal transmits a rising edge of the control signal from the ECU **503**. The reset signal transmits a falling edge of the control signal from the ECU **503**. In other words, the set signal and the reset signal are signals for generating a drive voltage signal of the switching element **501**. The set signal and the reset signal correspond to a "first signal."

More specifically, the low-voltage circuit **20** is configured to be actuated by application of the first voltage V1. The low-voltage circuit **20** is electrically connected to the ECU **503** and generates a set signal and a reset signal based on a control signal received from the ECU **503**. In an example, the low-voltage circuit **20** generates the set signal in response to a rising edge of the control signal and generates the reset signal in response to a falling edge of the control signal. The low-voltage circuit **20** transmits the generated set signal and reset signal toward the high-voltage circuit **30**.

The high-voltage circuit **30** is configured to be actuated by application of the second voltage V2. The high-voltage circuit **30** is electrically connected to the gate of the switching element **501**. Based on the set signal and the reset signal received from the low-voltage circuit **20**, the high-voltage circuit **30** generates a drive voltage signal for driving the switching element **501** and applies the drive voltage signal to the gate of the switching element **501**. In other words, the high-voltage circuit **30** generates a drive voltage signal that is applied to the gate of the switching element **501** based on the first signal output from the low-voltage circuit **20**. More

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specifically, the high-voltage circuit **30** generates a drive voltage signal for activating the switching element **501** based on the set signal and applies the drive voltage signal to the gate of the switching element **501**. The high-voltage circuit **30** generates a drive voltage signal for deactivating the switching element **501** based on the reset signal and applies the drive voltage signal to the gate of the switching element **501**. Thus, the gate driver **10** controls the activation and deactivation of the switching element **501**.

The high-voltage circuit **30** includes, for example, an R-S flip-flop circuit, into which a set signal and a reset signal are input, and a driver unit. The driver unit generates a drive voltage signal based on an output signal of the R-S flip-flop circuit. However, the high-voltage circuit **30** may be changed to any specific circuit configuration.

In the gate driver **10** of the present embodiment, the low-voltage circuit **20** and the high-voltage circuit **30** are insulated from each other by the transformers **40**. More specifically, the transformers **40** restrict transmission of a direct current voltage between the low-voltage circuit **20** and the high-voltage circuit **30** while allowing transmission of various signals such as the set signal and the reset signal.

Thus, a state in which the low-voltage circuit **20** and the high-voltage circuit **30** are insulated from each other refers to a state in which transmission of a direct current voltage between the low-voltage circuit **20** and the high-voltage circuit **30** is interrupted, while transmission of a signal between the low-voltage circuit **20** and the high-voltage circuit **30** is allowed.

The insulation voltage of the gate driver **10** is, for example, in a range of 2500 Vrms to 7500 Vrms. In the present embodiment, the insulation voltage of the gate driver **10** is approximately 5000 Vrms. However, the insulation voltage of the gate driver **10** is not limited to these values and may be any specific numerical value.

In the present embodiment, ground GND1 of the low-voltage circuit **20** and ground GND2 of the high-voltage circuit **30** are arranged independently. In the description hereafter, the potential of the ground GND1 of the low-voltage circuit **20** is referred to as a first reference potential, and the potential of the ground GND2 of the high-voltage circuit **30** is referred to as a second reference potential. In this case, the first voltage V1 is a voltage from the first reference potential, and the second voltage V2 is a voltage from the second reference potential. The first voltage V1 is, for example, in a range of 4.5 V to 5.5 V. The second voltage V2 is, for example, in a range of 9 V to 24 V.

The transformers **40** will now be described in detail.

The gate driver **10** of the present embodiment includes two transformers **40** and two capacitors **50** corresponding to two signals transmitted from the low-voltage circuit **20** toward the high-voltage circuit **30**. More specifically, the gate driver **10** includes a transformer **40** and a capacitor **50** that are used to transmit a set signal (SET) and a transformer **40** and a capacitor **50** that are used to transmit a reset signal (RESET). Hereinafter, for the sake of brevity, the transformer **40** and the capacitor **50** used to transmit a set signal are referred to as a "transformer 40A" and a "capacitor 50A." The transformer **40** and the capacitor **50** used to transmit a reset signal are referred to as a "transformer 40B" and a "capacitor 50B."

The gate driver **10** includes a low-voltage signal line **21A**, which connects the low-voltage circuit **20** and the transformer **40A**, and a low-voltage signal line **21B**, which connects the low-voltage circuit **20** and the transformer **40B**. Thus, the low-voltage signal line **21A** transmits the set signal from the low-voltage circuit **20** to the transformer **40A**. The

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low-voltage signal line 21B transmits the reset signal from the low-voltage circuit 20 to the transformer 40B.

The gate driver 10 includes a high-voltage signal line 31A, which connects the transformer 40A and the high-voltage circuit 30, and a high-voltage signal line 31B, which connects the transformer 40B and the high-voltage circuit 30. Thus, the high-voltage signal line 31A transmits the set signal from the transformer 40A to the high-voltage circuit 30. The high-voltage signal line 31B transmits the reset signal from the transformer 40B to the high-voltage circuit 30.

The transformer 40A electrically insulates the low-voltage circuit 20 from the high-voltage circuit 30 while transmitting the set signal from the low-voltage circuit 20 to the high-voltage circuit 30.

The transformer 40A includes a first coil 41A and a second coil 42A. The first coil 41A and the second coil 42A are electrically insulated from each other and configured to be magnetically coupled to each other.

The first coil 41A is connected to the low-voltage circuit 20 by the low-voltage signal line 21A and is also connected to the ground GND1 of the low-voltage circuit 20. More specifically, the first coil 41A includes a first end electrically connected to the low-voltage circuit 20. The first coil 41A is configured to receive a low voltage through the first end of the first coil 41A. The first coil 41A includes a second end electrically connected to the ground GND1 of the low-voltage circuit 20. Thus, the potential of the second end of the first coil 41A equals the first reference potential. The first reference potential is, for example, 0V.

The second coil 42A is connected to the high-voltage circuit 30 by the high-voltage signal line 31A and is also connected to the ground GND2 of the high-voltage circuit 30. More specifically, the second coil 42A includes a first end electrically connected to the high-voltage circuit 30. The second coil 42A is configured to receive a high voltage through the first end of the second coil 42A. The second coil 42A includes a second end electrically connected to the ground GND2 of the high-voltage circuit 30. Thus, the potential of the second end of the second coil 42A equals the second reference potential. The ground GND2 of the high-voltage circuit 30 is connected to the source of the switching element 501. Hence, the second reference potential varies as the inverter device 500 is driven and may become, for example, greater than or equal to 600 V.

The transformer 40B electrically insulates the low-voltage circuit 20 from the high-voltage circuit 30 while transmitting the reset signal from the low-voltage circuit 20 to the high-voltage circuit 30. The transformer 40B includes a first coil 41B and a second coil 42B. The first coil 41B and the second coil 42B are electrically insulated from each other and configured to be magnetically coupled to each other. The connection configuration of the transformer 40B is the same as the connection configuration of the transformer 40A and thus will not be described in detail.

The capacitor 50A is connected to the transformer 40A. More specifically, the capacitor 50A is connected between the first coil 41A and the second coil 42A of the transformer 40A.

The capacitor 50A includes a first capacitor electrode 51A and a second capacitor electrode 52A. The first capacitor electrode 51A and the second capacitor electrode 52A are arranged between the first coil 41A and the second coil 42A of the transformer 40A. The first capacitor electrode 51A is connected to the second end of the first coil 41A. The second capacitor electrode 52A is connected to the second end of the second coil 42A. The second end of the first coil 41A is

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connected to the ground GND1 of the low-voltage circuit 20. Thus, the second end of the first coil 41A is a ground terminal. The first capacitor electrode 51A is connected to the ground terminal of the first coil 41A. The second end of the second coil 42A is connected to the ground GND2 of the high-voltage circuit 30. Thus, the second end of the second coil 42A is a ground terminal. Thus, the second capacitor electrode 52A is connected to the ground terminal of the second coil 42A.

The capacitor 50B is connected to the transformer 40B. More specifically, the capacitor 50B is connected between the first coil 41B and the second coil 42B of the transformer 40B.

The capacitor 50B includes a first capacitor electrode 51B and a second capacitor electrode 52B. The first capacitor electrode 51B and the second capacitor electrode 52B are arranged between the first coil 41B and the second coil 42B of the transformer 40B. The first capacitor electrode 51B is connected to a ground terminal of the first coil 41B. The second capacitor electrode 52B is connected to a ground terminal of the second coil 42B.

FIG. 2 shows an example of a plan view showing the internal structure of the gate driver 10. FIG. 1 shows a simplified circuit configuration of the gate driver 10. Hence, the number of external terminals of the gate driver 10 shown in FIG. 2 is greater than the number of external terminals of the gate driver 10 shown in FIG. 1. The number of external terminals of the gate driver 10 is the number of external electrodes configured to connect the gate driver 10 to electronic components arranged outside the gate driver 10, such as the ECU 503 or the switching element 501 (refer to FIG. 1). The number of signal lines (the number of wires W1 to W4 described later) that transmit a signal from the low-voltage circuit 20 to the high-voltage circuit 30 in the gate driver 10 shown in FIG. 2 is greater than the number of signal lines in the gate driver 10 shown in FIG. 1.

As shown in FIG. 2, the gate driver 10 is a semiconductor device including multiple semiconductor chips arranged in a single package and is, for example, mounted on a circuit substrate arranged in the inverter device 500. Each of the switching elements 501 and 502 is mounted on a mount substrate that differs from the circuit substrate. A cooling unit is attached to the mount substrate.

The package type of the gate driver 10 is small outline (SO) and, in the present embodiment, is a small outline package (SOP). The gate driver 10 includes a low-voltage circuit chip 60, a high-voltage circuit chip 70, and a transformer chip 80, which are semiconductor chips. The low-voltage circuit chip 60 is mounted on a low-voltage lead frame 90. The high-voltage circuit chip 70 is mounted on a high-voltage lead frame 100. The chips 60, 70, and 80 and a portion of the lead frames 90 and 100 are encapsulated by a mold resin 110. In the present embodiment, the transformer chip 80 corresponds to an "isolation transformer." The transformer chip 80 and the mold resin 110 correspond to an "insulation module" that insulates the low-voltage circuit 20 from the high-voltage circuit 30. In FIG. 2, the mold resin 110 is indicated by double-dashed lines to illustrate the internal structure of the gate driver 10. The package type of the gate driver 10 may be changed in any manner.

The mold resin 110 is formed from an electrical insulating material. An example of the resin includes a black epoxy resin. The mold resin 110 has the form of a rectangular plate having a thickness-wise direction conforming to the z-direction. The mold resin 110 includes four resin side surfaces 111 to 114. More specifically, the mold resin 110 includes

two end surfaces in the x-direction, namely, the resin side surfaces **111** and **112**, and two end surfaces in the y-direction, namely, the resin side surfaces **113** and **114**. The x-direction and the y-direction are orthogonal to the z-direction. The x-direction and the y-direction are orthogonal to each other. In the description hereafter, a plan view means a view in the z-direction.

Each of the low-voltage lead frame **90** and the high-voltage lead frame **100** is formed from an electrically conductive material. The low-voltage lead frame **90** and the high-voltage lead frame **100** are formed from copper (Cu), iron (Fe), or the like. The lead frames **90** and **100** extend from the inside to the outside of the mold resin **110**.

The low-voltage lead frame **90** includes a low-voltage die pad **91** arranged in the mold resin **110** and low-voltage leads **92** extending from the inside to the outside of the mold resin **110**. Each low-voltage lead **92** includes an external terminal configured to be electrically connected to an external electronic device such as the ECU **503** (refer to FIG. 1).

In the present embodiment, the low-voltage circuit chip **60** and the transformer chip **80** are mounted on the low-voltage die pad **91**. In plan view, the low-voltage die pad **91** is arranged so that the center of the low-voltage die pad **91** in the y-direction is located closer in the y-direction to the resin side surface **113** than the center of the mold resin **110** is. In the present embodiment, the low-voltage die pad **91** is not exposed from the mold resin **110**. In plan view, the low-voltage die pad **91** is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction.

The low-voltage leads **92** are spaced apart from each other in the x-direction. Among the low-voltage leads **92**, the low-voltage leads **92** located at opposite ends in the x-direction are integrated with the low-voltage die pad **91**. Each low-voltage lead **92** partially projects from the resin side surface **113** toward the outside of the mold resin **110**.

The high-voltage lead frame **100** includes a high-voltage die pad **101** arranged in the mold resin **110** and high-voltage leads **102** extending from the inside to the outside of the mold resin **110**. Each high-voltage lead **102** includes an external terminal configured to be electrically connected to an external electronic device such as the gate of the switching element **501** (refer to FIG. 1).

The high-voltage circuit chip **70** is mounted on the high-voltage die pad **101**. In plan view, the high-voltage die pad **101** is located closer in the y-direction to the resin side surface **114** than the low-voltage die pad **91** is. In the present embodiment, the high-voltage die pad **101** is not exposed from the mold resin **110**. In plan view, the high-voltage die pad **101** is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction.

The low-voltage die pad **91** and the high-voltage die pad **101** are spaced apart from each other in the y-direction. The y-direction may also be referred to as the arrangement direction of the two die pads **91** and **101**.

The dimension of the low-voltage die pad **91** and the high-voltage die pad **101** in the y-direction is set in accordance with the size and the number of semiconductor chips that are mounted. In the present embodiment, the low-voltage circuit chip **60** and the transformer chip **80** are mounted on the low-voltage die pad **91**, and the high-voltage circuit chip **70** is mounted on the high-voltage die pad **101**. Hence, the low-voltage die pad **91** is greater than the high-voltage die pad **101** in dimension in the y-direction.

The high-voltage leads **102** are spaced apart from each other in the x-direction. Among the high-voltage leads **102**, two of the high-voltage leads **102** are integrated with the

high-voltage die pad **101**. Each high-voltage lead **102** partially projects from the resin side surface **114** toward the outside of the mold resin **110**.

In the present embodiment, the number of the high-voltage leads **102** is the same as the number of the low-voltage leads **92**. As shown in FIG. 2, the low-voltage leads **92** and the high-voltage leads **102** are arranged in a direction (x-direction) orthogonal to the arrangement direction (y-direction) of the low-voltage die pad **91** and the high-voltage die pad **101**. The number of the high-voltage leads **102** and the number of the low-voltage leads **92** may be changed in any manner.

In the present embodiment, the low-voltage die pad **91** is supported by two low-voltage leads **92** that are integrated with the low-voltage die pad **91**. The high-voltage die pad **101** is supported by two high-voltage leads **102** that are integrated with the high-voltage die pad **101**. Hence, the die pads **91** and **101** are not provided with a suspension lead exposed from the resin side surfaces **111** and **112**. This allows for an increase in the insulation distance between the low-voltage lead frame **90** and the high-voltage lead frame **100**.

The low-voltage circuit chip **60**, the high-voltage circuit chip **70**, and the transformer chip **80** are spaced apart from each other in the y-direction. The low-voltage circuit chip **60**, the transformer chip **80**, and the high-voltage circuit chip **70** are arranged in this order in the y-direction from the low-voltage leads **92** toward the high-voltage leads **102**.

The low-voltage circuit chip **60** includes the low-voltage circuit **20** shown in FIG. 1. In plan view, the low-voltage circuit chip **60** is rectangular and has short sides and long sides. In plan view, the low-voltage circuit chip **60** is mounted on the low-voltage die pad **91** such that the long sides extend in the x-direction and the short sides extend in the y-direction. The low-voltage circuit chip **60** includes a chip main surface **60s** and a chip back surface (not shown) facing opposite directions in the z-direction. The chip back surface of the low-voltage circuit chip **60** is bonded to the low-voltage die pad **91** by a conductive bonding material such as solder or silver (Ag) paste.

First electrode pads **61**, second electrode pads **62**, and third electrode pads **63** are formed on the chip main surface **60s** of the low-voltage circuit chip **60**. The electrode pads **61** to **63** are electrically connected to the low-voltage circuit **20**.

The first electrode pads **61** are located on the chip main surface **60s** between the center of the chip main surface **60s** in the y-direction and the low-voltage leads **92**. The first electrode pads **61** are arranged in the x-direction. The second electrode pads **62** are arranged on one of the opposite ends of the chip main surface **60s** in the y-direction located closer to the transformer chip **80**. The second electrode pads **62** are arranged in the x-direction. The third electrode pads **63** are arranged on opposite ends of the chip main surface **60s** in the x-direction.

The high-voltage circuit chip **70** includes the high-voltage circuit **30** shown in FIG. 1. In plan view, the high-voltage circuit chip **70** is rectangular and has short sides and long sides. In plan view, the high-voltage circuit chip **70** is mounted on the high-voltage die pad **101** such that the long sides extend in the x-direction and the short sides extend in the y-direction. The high-voltage circuit chip **70** includes a chip main surface **70s** and a chip back surface (not shown) facing opposite directions in the z-direction. The chip back surface of the high-voltage circuit chip **70** is bonded to the high-voltage die pad **101** by a conductive bonding material.

First electrode pads **71**, second electrode pads **72**, and third electrode pads **73** are formed on the chip main surface

70s of the high-voltage circuit chip 70. The electrode pads 71 to 73 are electrically connected to the high-voltage circuit 30.

The first electrode pads 71 are arranged on one of the opposite ends of the chip main surface 70s in the y-direction located closer to the transformer chip 80. The first electrode pads 71 are arranged in the x-direction. The second electrode pads 72 are arranged on one of the opposite ends of the chip main surface 70s in the y-direction located farther away from the transformer chip 80. In other words, the second electrode pads 72 are arranged on one of the opposite ends of the chip main surface 70s in the y-direction located closer to the high-voltage leads 102. The second electrode pads 72 are arranged in the x-direction. The third electrode pads 73 are arranged on opposite ends of the chip main surface 70s in the x-direction.

The transformer chip 80 includes the transformers 40 (40A, 40B) and the capacitors 50 (50A, 50B), which are shown in FIG. 1. In plan view, the transformer chip 80 is rectangular and has short sides and long sides. In the present embodiment, in plan view, the transformer chip 80 is mounted on the low-voltage die pad 91 such that the long sides extend in the x-direction and the short sides extend in the y-direction.

The transformer chip 80 is arranged adjacent to the low-voltage circuit chip 60 in the y-direction. The transformer chip 80 is arranged closer to the high-voltage circuit chip 70 than the low-voltage circuit chip 60. That is, the transformer chip 80 is arranged between the low-voltage circuit chip 60 and the high-voltage circuit chip 70 in the y-direction.

The transformer chip 80 includes a chip main surface 80s and a chip back surface 80r (refer to FIG. 9) facing in opposite directions in the z-direction. The chip back surface 80r of the transformer chip 80 is bonded to the low-voltage die pad 91 by the conductive bonding material SD (refer to FIG. 9).

As shown in FIG. 2, first electrode pads 81 and second electrode pads 82 are formed on the chip main surface 80s of the transformer chip 80. In the present embodiment, each first electrode pad 81 corresponds to a "first electrode." Each second electrode pad 82 corresponds to a "second electrode."

The first electrode pads 81 are arranged, for example, on one of the opposite ends of the chip main surface 80s in the y-direction located closer to the low-voltage circuit chip 60. The first electrode pads 81 are arranged in the x-direction. The second electrode pads 82 are arranged, for example, near the center of the chip main surface 80s in the y-direction. The second electrode pads 82 are arranged in the x-direction.

As shown in FIG. 4, in plan view, the transformers 40A and 40B and the capacitors 50A and 50B are arranged near the center of the chip main surface 80s in the y-direction. In plan view, the second electrode pads 82 are arranged so as not to overlap with the transformers 40A and 40B and the capacitors 50A and 50B. The electrode pads 81 and 82 are electrically connected to the transformers 40A and 40B and the capacitors 50A and 50B.

As shown in FIG. 2, the lead frames 90 and 100 are located closest to each other at the low-voltage die pad 91 and the high-voltage die pad 101. Therefore, the low-voltage die pad 91 and the high-voltage die pad 101 need to be separated from each other so that the insulation voltage of the gate driver 10 is set to a predetermined insulation voltage. Hence, in plan view, the distance between the high-voltage circuit chip 70 and the transformer chip 80 is

greater than the distance between the low-voltage circuit chip 60 and the transformer chip 80.

The wires W1 to W4 are connected to each of the low-voltage circuit chip 60, the transformer chip 80, and the high-voltage circuit chip 70. Each of the wires W1 to W4 is a bonding wire formed by a wire bonder and is, for example, formed from a material including gold (Au), aluminum (Al), Cu, or the like.

The low-voltage circuit chip 60 is electrically connected to the low-voltage lead frame 90 by the wires W1. More specifically, the first electrode pads 61 and the third electrode pads 63 of the low-voltage circuit chip 60 are connected to the low-voltage leads 92 by the wires W1. The third electrode pads 63 of the low-voltage circuit chip 60 are connected by the wires W1 to the two low-voltage leads 92 integrated with the low-voltage die pad 91. Thus, the low-voltage circuit 20 is electrically connected to the low-voltage leads 92 (external electrodes of the gate driver 10 that are electrically connected to the ECU 503). In the present embodiment, the two low-voltage leads 92 integrated with the low-voltage die pad 91 include a ground terminal, and the low-voltage die pad 91 is electrically connected to the low-voltage circuit 20 by the wires W1. Thus, the low-voltage die pad 91 has the same potential as the ground GND1 of the low-voltage circuit 20.

The high-voltage circuit chip 70 and the high-voltage leads 102 of the high-voltage lead frame 100 are electrically connected by the wires W4. More specifically, the second electrode pads 72 and the third electrode pads 73 of the high-voltage circuit chip 70 are connected to the high-voltage leads 102 by the wires W4. Thus, the high-voltage circuit 30 is electrically connected to the high-voltage leads 102 (external electrodes of the gate driver 10 that are electrically connected to the switching element 501 or the like). In the present embodiment, the two high-voltage leads 102 integrated with the high-voltage die pad 101 include a ground terminal, and the high-voltage die pad 101 is electrically connected to the high-voltage circuit 30 by the wires W4. Thus, the high-voltage die pad 101 has the same potential as the ground GND2 of the high-voltage circuit 30.

The transformer chip 80 is connected to the low-voltage circuit chip 60 by the wires W2 and is also connected to the high-voltage circuit chip 70 by the wires W3. More specifically, the first electrode pads 81 of the transformer chip 80 and the second electrode pads 62 of the low-voltage circuit chip 60 are connected by the wires W2. More specifically, the second electrode pads 82 of the transformer chip 80 and the first electrode pads 71 of the high-voltage circuit chip 70 are connected by the wires W3.

The first coil 41A of the transformer 40A and the first coil 41B of the transformer 40B (refer to FIG. 1) are electrically connected to the ground GND1 of the low-voltage circuit 20 through the wires W2, the low-voltage circuit chip 60, and the like. The second coil 42A of the transformer 40A and the second coil 42B of the transformer 40B (refer to FIG. 1) is electrically connected to the ground GND2 of the high-voltage circuit 30 through the wires W3, the high-voltage circuit chip 70, and the like.

Structure of Transformer Chip

An example of the structure of the transformer chip 80 will now be described with reference to FIGS. 3 to 11.

In the following description, a direction from the chip back surface 80r of the transformer chip 80 toward the chip main surface 80s is referred to as an upward direction, and a direction from the chip main surface 80s toward the chip back surface 80r is referred to as a downward direction.

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FIG. 3 is a perspective view showing the outer appearance of the transformer chip 80.

FIG. 4 is a plan view of the transformer chip 80. The transformers 40A and 40B, the capacitors 50A and 50B, a shield electrode 86, and dummy patterns 120 and 125, which will be described later, are indicated by broken lines for the sake of illustration.

FIG. 5 is a cross-sectional view of the transformer chip 80 that is cut along an xy-plane through a position of the first coils 41A and 41B in the z-direction showing the connection relationship of the first coils 41A and 41B. FIG. 6 is a cross-sectional view of the transformer chip 80 that is cut along an xy-plane through a position of the second coils 42A and 42B in the z-direction showing the connection relationship of the second coils 42A and 42B. FIGS. 5 and 6 do not show hatching for the sake of convenience.

FIG. 7 is a cross-sectional view of the transformer chip 80 that is cut along an xy-plane through a position of the first capacitor electrodes 51A and 51B in the z-direction. FIG. 8 is a cross-sectional view of the transformer chip 80 that is cut along an xy-plane through a position of the second capacitor electrodes 52A and 52B in the z-direction. FIGS. 7 and 8 do not show hatching for the sake of convenience.

FIG. 9 is a cross-sectional view of the transformer chip 80 taken along line 9-9 in FIG. 4 showing a cross-sectional structure of the transformer 40A and the capacitor. FIG. 9 partially does not show hatching for simplicity and clarity. FIG. 10 is an enlarged partial view of FIG. 9 showing portions of the first coil 41A and the first capacitor electrode 51A. FIG. 11 is an enlarged partial view of FIG. 9 showing portions of the second coil 42A and the second capacitor electrode 52A. FIGS. 10 and 11 partially do not show hatching in the same manner as FIG. 9.

As shown in FIG. 4, the transformer chip 80 of the present embodiment includes two sets of the transformers 40A and 40B and the capacitors 50A and 50B. More specifically, the transformer chip 80 is a semiconductor chip that includes the transformers 40A and 40B and the capacitors 50A and 50B in the single chip. Thus, the transformer chip 80 is arranged separately from the low-voltage circuit chip 60 and the high-voltage circuit chip 70 (refer to FIG. 2).

The two sets of the transformers 40A and 40B and the capacitors 50A and 50B have the same structure. Furthermore, the transformer 40B and the transformer 40A have the same structure. The capacitor 50B and the capacitor 50A have the same structure. The structures of the transformer 40A and the capacitor 50A will be described in detail. The structures of the transformer 40B and the capacitor 50B will not be described in detail.

As shown in FIG. 4, the transformer chip 80 includes four chip side surfaces 80a, 80b, 80c, and 80d that are orthogonal to the chip main surface 80s and the chip back surface 80r. The chip side surfaces 80a to 80d are arranged between the chip main surface 80s and the chip back surface 80r in the z-direction. The chip side surfaces 80a and 80b define opposite surfaces of the transformer chip 80 in the y-direction. The chip side surfaces 80c and 80d define opposite surfaces of the transformer chip 80 in the x-direction. In plan view, the chip side surfaces 80a and 80b define long sides of the transformer chip 80. The chip side surfaces 80c and 80d define short sides of the transformer chip 80. In the present embodiment, the chip side surface 80a is located closer to the high-voltage circuit chip 70 (refer to FIG. 2) than the chip side surface 80b is. The chip side surface 80b is located closer to the low-voltage circuit chip 60 (refer to FIG. 2) than the chip side surface 80a is.

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As shown in FIGS. 4 and 9, the transformer chip 80 includes a substrate 83 and an insulation layer 84 formed on the substrate 83.

The substrate 83 is composed of, for example, a semiconductor substrate. In the present embodiment, the substrate 83 is formed from a material including silicon (Si). As the semiconductor substrate, a wide-bandgap semiconductor or a compound semiconductor may be used for the substrate 83. Alternatively, instead of the semiconductor substrate, an insulating substrate formed from a glass-containing material may be used for the substrate 83.

The wide-bandgap semiconductor is a semiconductor substrate having a band gap that is greater than or equal to 2.0 eV. The wide-bandgap semiconductor may be silicon carbide (SiC). The compound semiconductor may be a group III-V compound semiconductor. The compound semiconductor may include at least one of aluminum nitride (AlN), indium nitride (InN), gallium nitride (GaN), and gallium arsenide (GaAs).

The substrate 83 includes a substrate main surface 83s and a substrate back surface 83r facing opposite directions in the z-direction. The substrate back surface 83r defines the chip back surface 80r of the transformer chip 80.

As shown in FIG. 9, the insulation layer 84 of the present embodiment includes insulation films 85 stacked on the substrate main surface 83s of the substrate 83 in the z-direction. Thus, the z-direction may be referred to as a thickness-wise direction of the insulation layer 84. Also, the z-direction may be referred to as a thickness-wise direction of the insulation films 85. The insulation layer 84 is formed on the substrate main surface 83s of the substrate 83.

Each insulation film 85 includes a first insulation film 85A and a second insulation film 85B formed on the first insulation film 85A. Thus, the insulation layer 84 includes multiple first insulation films 85A and multiple second insulation films 85B. The first insulation films 85A and the second insulation films 85B are alternately stacked in the z-direction.

The first insulation film 85A is a thin film and is, for example, an etching stopper layer. The first insulation film 85A is formed from a material including silicon nitride (SiN), SiC, nitrogen-added silicon carbide (SiCN), or the like. In the present embodiment, the first insulation film 85A is formed from a material including SiN. The second insulation film 85B is, for example, an interlayer insulation film. The second insulation film 85B is formed from a material including silicon oxide (SiO₂). As shown in FIG. 9, the second insulation film 85B is greater in thickness than the first insulation film 85A. The thickness of the first insulation film 85A may be in a range of 100 nm to 1000 nm. The thickness of the second insulation film 85B may be in a range of 1000 nm to 3000 nm. In the present embodiment, the thickness of the first insulation film 85A is, for example, approximately 300 nm, and the thickness of the second insulation film 85B is, for example, approximately 2000 nm.

The second insulation films 85B include a lowermost insulation film 85L, which is in contact with the substrate main surface 83s of the substrate 83, and an uppermost insulation film 85U. Thus, each of the lowermost insulation film 85L and the uppermost insulation film 85U is smaller in thickness than other insulation films 85. The thickness of each of the lowermost insulation film 85L and the uppermost insulation film 85U is in a range of the thickness of the first insulation film 85A to the thickness of the second insulation film 85B.

The thickness of the lowermost insulation film 85L and the uppermost insulation film 85U may be changed in any

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manner. In an example, the thickness of each of the lowermost insulation film 85L and the uppermost insulation film 85U may be greater than the thickness of the second insulation film 85B or may be greater than or equal to the thickness of an insulation film 85 that includes the first insulation film 85A and the second insulation film 85B.

The transformer chip 80 includes the shield electrode 86 formed in the insulation layer 84. The shield electrode 86 limits entrance of moisture into the insulation layer 84 and formation of cracks in the insulation layer 84. In plan view, the shield electrode 86 is arranged on a peripheral portion of the insulation layer 84 (peripheral portion of the transformer chip 80). More specifically, as shown in FIGS. 4 to 8, the shield electrode 86 is spaced apart from the chip side surfaces 80a to 80d. In plan view, the shield electrode 86 is strip-shaped and extends along the chip side surfaces 80a to 80d. In the present embodiment, the shield electrode 86 has the form of a rectangular loop in plan view. The shield electrode 86 divides the insulation layer 84 into an inner region 87 and an outer region 88. In the present embodiment, as shown in FIG. 9, the uppermost insulation film 85U extends across the shield electrode 86 in plan view. That is, the uppermost insulation film 85U includes the outer region 88.

As shown in FIG. 4, the inner region 87 is a region of the insulation layer 84 protected by the shield electrode 86. In plan view, the inner region 87 is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction. The outer region 88 is a rectangular loop region surrounding the inner region 87 in plan view. The outer region 88 is located between the shield electrode 86 and each of the chip side surfaces 80a to 80d in plan view. Thus, the outer region 88 is a rectangular loop region including the chip side surfaces 80a to 80d.

As shown in FIG. 9, the shield electrode 86 extends through the insulation films 85 in the z-direction. More specifically, as viewed in a direction orthogonal to the z-direction, the shield electrode 86 is arranged to overlap the coils 41A, 41B, 42A, and 42B of the transformers 40A and 40B and the capacitor electrodes 51A, 51B, 52A, and 52B of the capacitors 50A and 50B. In the present embodiment, the shield electrode 86 extends in the z-direction from the insulation film 85 that is located immediately below the uppermost insulation film 85U to the insulation film 85 that is located immediately above the lowermost insulation film 85L. The shield electrode 86 is formed from a material including one or more selected from titanium (Ti), titanium nitride (TiN), Au, Ag, Cu, Al, and tungsten (W).

The lowermost insulation film 85L includes vias 89 extending through the lowermost insulation film 85L in the z-direction. The vias 89 are arranged to overlap the shield electrode 86 in plan view and connect the shield electrode 86 to the substrate 83. Thus, the shield electrode 86 is electrically connected to the substrate 83. In an example, the vias 89 and the shield electrode 86 may be formed from the same material.

As shown in FIG. 4, the transformers 40A and 40B and the capacitors 50A and 50B are embedded in the insulation layer 84. The transformers 40A and 40B and the capacitors 50A and 50B are arranged in the inner region 87. As viewed in the z-direction, the capacitor 50A overlaps the transformer 40A, and the capacitor 50B overlaps the transformer 40B. The transformers 40A and 40B and the capacitors 50A and 50B are aligned with each other in the y-direction and spaced apart from each other in the x-direction. In other words, the transformers 40A and 40B and the capacitors 50A

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and 50B are arranged in a direction orthogonal to a direction in which the chips 60, 70, and 80 are arranged.

As shown in FIG. 4, the transformer 40A and the capacitor 50A alternate with the transformer 40B and the capacitor 50B in the x-direction from the chip side surface 80c toward the chip side surface 80d.

As shown in FIG. 5, the first coil 41A of the transformer 40A includes a first coil wiring 43A, a first ground terminal 45 connected to one end of the first coil wiring 43A, and a first signal terminal 44A connected to the other end of the first coil wiring 43A. The first coil 41B of the transformer 40B includes a first coil wiring 43B, a first ground terminal 45 connected to one end of the first coil wiring 43B, and a first signal terminal 44B connected to the other end of the first coil wiring 43B. The first ground terminal 45 is configured to be a common terminal to the first coil 41A and the first coil 41B. Alternatively, the first coil 41A and the first coil 41B may be provided with separate first ground terminals.

The first coil wirings 43A and 43B are elliptical and spiral in plan view. The first signal terminals 44A and 44B are arranged at an inner side of the first coil wirings 43A and 43B. The first ground terminal 45 is arranged between the first coil 41A of the transformer 40A and the first coil 41B of the transformer 40B. The first coils 41A and 41B are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

The first signal terminal 44A is connected by an interconnect 131A to a first electrode pad 81A, shown in FIG. 4. The first signal terminal 44B is connected by an interconnect 131B to a first electrode pad 81B, shown in FIG. 4. The first ground terminal 45 is connected by an interconnect 131C to a first electrode pad 81C, shown in FIG. 4.

As shown in FIG. 6, the second coil 42A of the transformer 40A includes a second coil wiring 46A, a second ground terminal 48 connected to one end of the second coil wiring 46A, and a second signal terminal 47A connected to the other of the second coil wiring 46A. The second coil 42B of the transformer 40B includes a second coil wiring 46B, a second ground terminal 48 connected to one end of the second coil wiring 46B, and a second signal terminal 47B connected to the other end of the second coil wiring 46B. The second ground terminal 48 is configured to be a common terminal to the second coil 42A and the second coil 42B. Alternatively, the second coil 42A and the second coil 42B may be provided with separate second ground terminals.

The second coil wirings 46A and 46B are elliptical and spiral in plan view. The second signal terminals 47A and 47B are arranged at an inner side of the second coil wirings 46A and 46B. The second ground terminal 48 is arranged between the second coil 42A of the transformer 40A and the second coil 42B of the transformer 40B. The second coils 42A and 42B are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

The second signal terminal 47A is connected to a second electrode pad 82A, shown in FIG. 4. The second signal terminal 47B is connected to a second electrode pad 82B, shown in FIG. 4. The second ground terminal 48 is connected to a second electrode pad 82C, shown in FIG. 4.

In the present embodiment, the second coil wiring 46A is formed in the same winding direction as the first coil wiring 43A, shown in FIG. 5. The second coil wiring 46A and the first coil wiring 43A are wound the same number of times. In the present embodiment, the second coil wiring 46B is formed in the same winding direction as the first coil wiring

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43B, shown in FIG. 5. The second coil wiring 46B and the first coil wiring 43B are wound the same number of times.

As shown in FIG. 9, the first coil 41A and the second coil 42A of the transformer 40A are opposed to each other in the z-direction with an insulation film 85 interposed. In the present embodiment, the first coil 41A and the second coil 42A are opposed to each other in the z-direction with multiple insulation films 85 interposed.

As shown in FIG. 9, the first coil 41A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the first coil 41A is embedded in an insulation film 851. The insulation film 851 includes a coil groove 141 (first coil groove) extending through the second insulation film 85B in the z-direction. The conductive layer forming the first coil 41A is embedded in the coil groove 141 of the second insulation film 85B of the insulation film 851. The insulation film 851, in which the first coil 41A is embedded, is covered by insulation films 85 located next to the insulation film 851 in the z-direction. Thus, the first coil 41A is embedded in the insulation films 85.

As shown in FIG. 9, the second coil 42A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the second coil 42A is embedded in an insulation film 854. The insulation film 854 includes a coil groove 142 (second coil groove) extending through the second insulation film 85B in the z-direction. The conductive layer forming the second coil 42A is embedded in the coil groove 142 of the second insulation film 85B of the insulation film 854. The insulation film 854, in which the second coil 42A is embedded, is covered by insulation films 85 located next to the insulation film 854 in the z-direction. Thus, the second coil 42A is embedded in the insulation films 85.

In the z-direction, the second coil 42A is located farther away from the substrate 83 than the first coil 41A is. In other words, the second coil 42A is located above the first coil 41A. Also, in other words, the first coil 41A is located closer to the substrate 83 than the second coil 42A is. In the present embodiment, the distance between the first coil 41A and the second coil 42A in the z-direction is greater than the distance between the first coil 41A and the substrate main surface 83s of the substrate 83.

In plan view, the first capacitor electrode 51A of the capacitor 50A, shown in FIG. 7, overlaps the first coil 41A, shown in FIG. 5. The first capacitor electrode 51A is formed from a conductive material. More preferably, the first capacitor electrode 51A is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, tungsten titanium (TiW), tantalum (Ta), (tantalum nitride) TaN, chromium (Cr), chromium silicide (CrSi), Au, Ag, Cu, Al, and W. Alternatively, the first capacitor electrode 51A may be formed of a conductive material other than the materials described above. In the present embodiment, the first capacitor electrode 51A is formed from a material including Ti.

The first capacitor electrode 51A includes a first electrode wiring 53A, a first capacitor end 54A, and a first capacitor ground terminal 55. The first electrode wiring 53A is elliptical and spiral in the same manner as the first coil wiring 43A, shown in FIG. 5. The first electrode wiring 53A is identical in shape to the first coil wiring 43A of the first coil 41A, shown in FIG. 5. That is, the first electrode wiring 53A is set to be equal to the first coil wiring 43A in ratio of wiring width to wiring spacing.

The first electrode wiring 53A includes a first slit 51As extending from the center of the first electrode wiring 53A toward an outer side of the first electrode wiring 53A. The

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first electrode wiring 53A is annular and is open by the first slit 51As. The first slit 51As limits formation of a current loop in the first electrode wiring 53A.

The first capacitor end 54A overlaps the first signal terminal 44A of the first coil 41A, shown in FIG. 5. In plan view, the first capacitor end 54A is identical in shape to the first signal terminal 44A. The first capacitor end 54A is arranged at an inner side of the first electrode wiring 53A and is connected to the first electrode wiring 53A. The shape of the first capacitor end 54A may be changed in any manner. Moreover, the first capacitor end 54A may be omitted.

The first capacitor ground terminal 55 overlaps the first ground terminal 45 of the first coil 41A, shown in FIG. 5. In plan view, the first capacitor ground terminal 55 is identical in shape to the first ground terminal 45. The first capacitor ground terminal 55 is arranged between the first capacitor electrode 51A of the capacitor 50A and the first capacitor electrode 51B of the capacitor 50B. The first capacitor ground terminal 55 is electrically connected to each interconnect part of the first electrode wiring 53A by an interconnect 55A extending toward the center of the first electrode wiring 53A.

In plan view, the first capacitor electrode 51B of the capacitor 50B, shown in FIG. 7, overlaps the first coil 41B, shown in FIG. 5. The first capacitor electrode 51B is formed from a conductive material. More preferably, the first capacitor electrode 51B is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the first capacitor electrode 51B may be formed of a conductive material other than the materials described above.

The first capacitor electrode 51B includes a first electrode wiring 53B, a first capacitor end 54B, and a first ground terminal 55. Thus, the first capacitor electrodes 51A and 51B include the first ground terminal 55 as a common terminal in the same manner as the first coils 41A and 41B, shown in FIG. 5.

The first electrode wiring 53B is elliptical and spiral in the same manner as the first coil wiring 43B, shown in FIG. 5. The first electrode wiring 53B is identical in shape to the first coil wiring 43B of the first coil 41B, shown in FIG. 5. That is, the first electrode wiring 53B is set to be equal to the first coil wiring 43B in ratio of wiring width to wiring spacing. The first electrode wiring 53B includes a first slit 51Bs extending from the center of the first electrode wiring 53B toward an outer side of the first electrode wiring 53B. The first electrode wiring 53B is annular and is open by the first slit 51Bs. The first slit 51Bs limits formation of a current loop in the first electrode wiring 53B.

The first capacitor end 54B overlaps the first signal terminal 44B of the first coil 41B, shown in FIG. 5. In plan view, the first capacitor end 54B is identical in shape to the first signal terminal 44B. The first capacitor end 54B is arranged at an inner side of the first electrode wiring 53B and is connected to the first electrode wiring 53B. The shape of the first capacitor end 54B may be changed in any manner. Moreover, the first capacitor end 54B may be omitted.

The first capacitor ground terminal 55 is electrically connected to each interconnect part of the first electrode wiring 53B by an interconnect 55B extending toward the center of the first electrode wiring 53B.

In plan view, the second capacitor electrode 52A of the capacitor 50A, shown in FIG. 8, overlaps the second coil 42A, shown in FIG. 6. The second capacitor electrode 52A is formed from a conductive material. More preferably, the second capacitor electrode 52A is formed from a nonmag-

netic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the second capacitor electrode 52A may be formed of a conductive material other than the materials described above.

The second capacitor electrode 52A includes a second electrode wiring 56A, a second capacitor end 57A, and a second capacitor ground terminal 58. The second electrode wiring 56A is elliptical and spiral in the same manner as the second coil wiring 46A, shown in FIG. 6. The second electrode wiring 56A is identical in shape to the second coil wiring 46A of the second coil 42A, shown in FIG. 6. That is, the second electrode wiring 56A is set to be equal to the second coil wiring 46A in ratio of wiring width to wiring spacing.

The second electrode wiring 56A includes a second slit 52As extending from the center of the second electrode wiring 56A toward an outer side of the second electrode wiring 56A. The second electrode wiring 56A is annular and is open by the second slit 52As. The second slit 52As limits formation of a current loop in the second electrode wiring 56A.

The second capacitor end 57A overlaps the second signal terminal 47A of the second coil 42A, which is shown in FIG. 6. In plan view, the second capacitor end 57A is identical in shape to the second signal terminal 47A. The second capacitor end 57A is arranged at an inner side of the second electrode wiring 56A and is connected to the second electrode wiring 56A. The shape of the second capacitor end 57A may be changed in any manner. Moreover, the second capacitor end 57A may be omitted.

The second capacitor ground terminal 58 overlaps the second ground terminal 48 of the second coil 42A, which is shown in FIG. 6. In plan view, the second capacitor ground terminal 58 is identical in shape to the second ground terminal 48. The second capacitor ground terminal 58 is arranged between the second capacitor electrode 52A of the capacitor 50A and the second capacitor electrode 52B of the capacitor 50B. The second capacitor ground terminal 58 is electrically connected to each interconnect part of the second electrode wiring 56A by an interconnect 58A extending toward the center of the second electrode wiring 56A.

In plan view, the second capacitor electrode 52B of the capacitor 50B, shown in FIG. 8, overlaps the second coil 42B, shown in FIG. 6. The second capacitor electrode 52B is formed from a conductive material. More preferably, the second capacitor electrode 52B is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the second capacitor electrode 52B may be formed of a conductive material other than the materials described above.

The second capacitor electrode 52B includes a second electrode wiring 56B, a second capacitor end 57B, and a second capacitor ground terminal 58. Thus, the second capacitor electrodes 52A and 52B include the second capacitor ground terminal 58 as a common terminal in the same manner as the second coils 42A and 42B, shown in FIG. 6.

The second electrode wiring 56B is elliptical and spiral in the same manner as the second coil wiring 46B, shown in FIG. 6. The second electrode wiring 56B is identical in shape to the second coil wiring 46B of the second coil 42B, shown in FIG. 6. That is, the second electrode wiring 56B is set to be equal to the second coil wiring 46B in ratio of wiring width to wiring spacing. The second electrode wiring 56B includes a second slit 52Bs extending from the center of the second electrode wiring 56B toward an outer side of

the second electrode wiring 56B. The second electrode wiring 56B is annular and is open by the second slit 52Bs. The second slit 52Bs limits formation of a current loop in the second electrode wiring 56B.

The second capacitor end 57B overlaps the second signal terminal 47B of the second coil 42B, shown in FIG. 6. In plan view, the second capacitor end 57B is identical in shape to the second signal terminal 47B. The second capacitor end 57B is arranged at an inner side of the second electrode wiring 56B and is connected to the second electrode wiring 56B. The shape of the second capacitor end 57B may be changed in any manner. Moreover, the second capacitor end 57B may be omitted.

The second capacitor ground terminal 58 is electrically connected to each interconnect part of the second electrode wiring 56B by an interconnect 58B extending toward the center of the second electrode wiring 56B.

As shown in FIG. 9, the first capacitor electrode 51A and the second capacitor electrode 52A of the capacitor 50A are arranged between the first coil 41A and the second coil 42A of the transformer 40A. The first capacitor electrode 51A and the second capacitor electrode 52A are opposed to each other in the z-direction with multiple insulation films 85 interposed.

In the present embodiment, the first capacitor electrode 51A is formed in an insulation film 852 located next to the insulation film 851, in which the first coil 41A is formed. In the present embodiment, the second capacitor electrode 52A is formed in an insulation film 853 located next to the insulation film 854, in which the second coil 42A is formed.

The first capacitor electrode 51A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the insulation film 852, in which the first capacitor electrode 51A is embedded, includes a first insulation film 85A and a second insulation film 85B. The conductive layer forming the first capacitor electrode 51A is arranged between the first insulation film 85A and the second insulation film 85B of the insulation film 852. More specifically, one surface (lower surface) of the first insulation film 85A in the z-direction is in contact with the first coil 41A, and the other surface (upper surface) is in contact with the first capacitor electrode 51A. The first insulation film 85A of the insulation film 852 is arranged between the first capacitor electrode 51A and the first coil 41A. The first insulation film 85A of the insulation film 852 corresponds to a first thin film. The second insulation film 85B of the insulation film 852 corresponds to a first interlayer insulation film. Thus, the first capacitor electrode 51A is embedded in the insulation film 85.

The second capacitor electrode 52A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the insulation film 853, in which the second capacitor electrode 52A is embedded, includes a first insulation film 85A and a second insulation film 85B. The insulation film 854, in which the second coil 42A is embedded, includes a first insulation film 85A and a second insulation film 85B. The first insulation film 85A of the insulation film 854 is in contact with the second insulation film 85B of the insulation film 853. The conductive layer forming the second capacitor electrode 52A is arranged between the second insulation film 85B of the insulation film 853 and the first insulation film 85A of the insulation film 854 located next to the insulation film 853. More specifically, one surface (lower surface) of the first insulation film 85A in the z-direction is in contact with the second capacitor electrode 52A, and the other surface (upper surface) is in contact with the second coil 42A. The first insulation film

85A of the insulation film 854 is arranged between the second capacitor electrode 52A and the second coil 42A. The first insulation film 85A of the insulation film 854 corresponds to a second thin film. The second insulation film 85B of the insulation film 853 corresponds to a second interlayer insulation film. Thus, the second capacitor electrode 52A is embedded in the insulation films 85.

The distance between the first capacitor electrode 51A and the second capacitor electrode 52A is determined by the thickness of the insulation film 85 and the number of the insulation films 85 stacked between the first capacitor electrode 51A and the second capacitor electrode 52A. The distance is set in accordance with the insulation voltage or electric field strength of the transformer chip 80.

As shown in FIG. 9, the first capacitor electrode 51A is electrically connected to the first coil 41A.

The first coil 41A includes the first coil wiring 43A, the first signal terminal 44A, and the first ground terminal 45. The first capacitor electrode 51A includes the first electrode wiring 53A, the first capacitor end 54A, and the first capacitor ground terminal 55. The first electrode wiring 53A overlaps the first coil wiring 43A in the z-direction. The first capacitor end 54A overlaps the first signal terminal 44A in the z-direction. The first capacitor ground terminal 55 overlaps the first ground terminal 45 in the z-direction.

The first capacitor ground terminal 55 of the first capacitor electrode 51A is connected to the first ground terminal 45 of the first coil 41A. The first insulation film 85A of the insulation film 852, in which the first capacitor electrode 51A is embedded, includes a first opening 145 exposing the first ground terminal 45 of the first coil 41A. The first capacitor ground terminal 55 of the first capacitor electrode 51A is in contact with the first ground terminal 45 of the first coil 41A through the first opening 145. Thus, the first capacitor ground terminal 55 of the first capacitor electrode 51A is electrically connected to the first ground terminal 45 of the first coil 41A through the first opening 145.

As shown in FIG. 9, the second capacitor electrode 52A is electrically connected to the second coil 42A.

The second coil 42A includes the second coil wiring 46A, the second signal terminal 47A, and the second ground terminal 48. The second capacitor electrode 52A includes the second electrode wiring 56A, the second capacitor end 57A, and the second capacitor ground terminal 58. The second electrode wiring 56A overlaps the second coil wiring 46A in the z-direction. The second capacitor end 57A overlaps the second signal terminal 47A in the z-direction. The second capacitor ground terminal 58 overlaps the second ground terminal 48 in the z-direction.

The second capacitor ground terminal 58 of the second capacitor electrode 52A is connected to the second ground terminal 48 of the second coil 42A. The first insulation film 85A of the insulation film 854, in which the second coil 42A is embedded, includes a second opening 146 partially exposing the second capacitor ground terminal 58 of the second capacitor electrode 52A. The second ground terminal 48 of the second coil 42A is in contact with the second capacitor ground terminal 58 of the second capacitor electrode 52A through the second opening 146. Thus, the second ground terminal 48 of the second coil 42A is electrically connected to the second capacitor ground terminal 58 of the second capacitor electrode 52A through the second opening 146.

As shown in FIG. 4, in plan view, the first electrode pads 81 and the second electrode pads 82 are arranged in the inner region 87.

As shown in FIG. 9, the electrode pads 81 and 82 are formed on the uppermost insulation film 85U. In other

words, the electrode pads 81 and the electrode pads 82 are embedded in the insulation film 85. In the present embodiment, the electrode pad 81 and the electrode pad 82 are located farther away from the substrate 83 than the second coils 42A and 42B of the transformers 40A and 40B. In other words, the electrode pad 81 and the electrode pad 82 are located above the second coils 42A and 42B of the transformers 40A and 40B. In the present embodiment, the distance between the first coil 41A and the second coil 42A is greater than the distance between the second coil 42A and each of the electrode pads 81 and 82 in the z-direction.

As shown in FIG. 4, in plan view, each first electrode pad 81 is aligned with one of the two transformers 40A or one of the two transformers 40B in the x-direction or arranged between the transformer 40A and the transformer 40B in the x-direction. The first electrode pads 81 are arranged closer to the chip side surface 80b than the transformers 40A and 40B are in the y-direction. In other words, the first electrode pads 81 are arranged between the chip side surface 80b and the transformers 40A and 40B in the y-direction. In other words, in plan view, the first electrode pads 81 are arranged closer to the low-voltage leads 92 (refer to FIG. 2) than the transformers 40A and 40B are.

In the description hereafter, the first electrode pad 81 that is aligned with the transformer 40A in the x-direction is referred to as a first electrode pad 81A for the sake of brevity. The first electrode pad 81 that is aligned with the transformer 40B in the x-direction is referred to as a first electrode pad 81B for the sake of brevity. The first electrode pad 81 arranged between the transformer 40A and the transformer 40B in the x-direction is referred to as a first electrode pad 81C. The first electrode pads 81A to 81C will be referred to as the first electrode pads 81 when the description is common to the first electrode pads 81A to 81C.

The first electrode pads 81A are arranged to overlap the transformer 40A as viewed in the y-direction. The first electrode pads 81B are arranged to overlap the transformer 40B as viewed in the y-direction. The first electrode pads 81C are arranged to overlap a portion located between the transformer 40A and the transformer 40B in the x-direction as viewed in the y-direction. The first electrode pads 81A to 81C are aligned with each other in the y-direction and spaced apart from each other in the x-direction.

As shown in FIG. 4, in plan view, the first electrode pads 82 are arranged in the transformers 40A and 40B and between the transformer 40A and the transformer 40B in the x-direction. As viewed in the x-direction, the second electrode pads 82 are arranged to overlap the transformers 40A and 40B. In the description hereafter, the second electrode pad 82 arranged in the transformer 40A is referred to as a second electrode pad 82A, the second electrode pad 82 arranged in the transformer 40B is referred to as a second electrode pad 82B, and the second electrode pad arranged between the transformer 40A and the transformer 40B is referred to as the second electrode pad 82C. The second electrode pads 82A to 82C will be referred to as the second electrode pads 82 when the description is common to the second electrode pads 82A to 82C.

The second electrode pads 82A are arranged in an inner space of the transformer 40A defined by the elliptical, spiral second coil 42A. The second electrode pads 82B are arranged in an inner space of the transformer 40B defined by the elliptical, spiral second coil 42A. The second electrode pads 82C are arranged between the transformer 40A and the transformer 40B in the x-direction. Each of the second electrode pads 82A to 82C is formed of two electrode pads adjacent to each other in the x-direction. The second elec-

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trode pads **82A** to **82C** are aligned with each other in the y-direction and spaced apart from each other in the x-direction.

As shown in FIGS. **4**, **5**, and **9**, the first electrode pads **81A** are electrically connected to the first coil **41A** of the transformer **40A**. The first electrode pads **81C** are electrically connected to the first coil **41A** of the transformer **40A**. As shown in FIGS. **4** and **5**, the first electrode pads **81B** are electrically connected to the first coil **41B** of the transformer **40B**. The first electrode pads **81C** are electrically connected to the first coil **41B** of the transformer **40B**. That is, the first electrode pads **81C** are common to the transformer **40A** and the transformer **40B**.

As shown in FIGS. **4**, **6**, and **9**, each second electrode pad **82A** is separately electrically connected to the second coil **42A** of the transformer **40A**. The second electrode pads **82C** are electrically connected to the second coil **42A** of the transformer **40A**. The second electrode pads **82B**, shown in FIGS. **4** and **6**, are electrically connected to the second coil **42B** of the transformer **40B**. The second electrode pads **82C** are electrically connected to the second coil **42B** of the transformer **40B**. That is, the second electrode pads **82C** are common to the transformer **40A** and the transformer **40B**.

As shown in FIGS. **5** and **6**, the transformer chip **80** includes interconnects separately connecting the electrode pads **81A** to **81C** and **82A** to **82C** to the coils **41A**, **41B**, **42A**, and **42B** of the transformers **40A** and **40B**. In the present embodiment, the interconnects include the interconnects **131A**, **131B**, and **131C**, which connect the first electrode pads **81A** to **81C** to the first coils **41A** and **41B**. The interconnects **131A** to **131C** are arranged in the inner region **87**. The interconnects **131A** to **131C** are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

As shown in FIG. **5**, the interconnect **131A** connects the first electrode pad **81A** to the first signal terminal **44A** of the first coil **41A** of the transformer **40A**. The interconnect **131B** connects the first electrode pad **81B** to the first signal terminal **44B** of the first coil **41B** of the transformer **40B**. The interconnect **131C** connects the first electrode pad **81C** to the second end of the first coil **41A** of the transformer **40A** and the second end of the first coil **41B** of the transformer **40B**. The interconnect **131C** corresponds to a first voltage interconnect. The first electrode pad **81C** corresponds to a first ground electrode. The interconnects **131A** to **131C** have the same structure. Thus, in the description hereafter, the structure of the interconnect **131A** will be described. The interconnects **131B** and **131C** will not be described in detail.

As shown in FIG. **9**, the interconnect **131A** includes a first interconnect part **132A** extending through the insulation films **85** in the z-direction and a second interconnect part **133A** extending in the y-direction.

The first interconnect part **132A** is arranged to overlap the first electrode pad **81A** in plan view and is connected to the first electrode pad **81A**. The first interconnect part **132A** extends through the insulation films **85** from the insulation film **85** that is located immediately below the uppermost insulation film **85U** to the insulation film **85** that is located above the lowermost insulation film **85L** with one insulation film **85** interposed. The first interconnect part **132A** includes flat interconnect pieces and vias. The interconnect pieces are located at the same position as the insulation films **851** and **854**, in which the coils **41A** and **42A** are arranged. The vias are arranged between the two interconnect pieces in the z-direction, between the upper interconnect piece and the first electrode pad **81A**, and between the lower interconnect piece and the second interconnect part **133A**.

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The second interconnect part **133A** is arranged closer to the substrate **83** than the first interconnect part **132A** is. The second interconnect part **133A** is arranged closer to the substrate **83** than the first coil **41A** is. In the present embodiment, the second interconnect part **133A** is arranged in the insulation film **85** located immediately above the lowermost insulation film **85L** among the insulation films **85**. Of opposite ends of the second interconnect part **133A** in the x-direction, a first end is located closer to the chip side surface **80b** of the transformer chip **80** and overlaps the first interconnect part **132A** in plan view. The second interconnect part **133A** is connected to the first interconnect part **132A**. The second interconnect part **133A** includes a second end opposite to the first end. In plan view, the second end is arranged to overlap the first coil **41A** of the transformer **40A**. More specifically, in plan view, the second end overlaps the first signal terminal **44A** of the first coil **41A** of the transformer **40A**. The second interconnect part **133A** includes vias **134A** connecting the second interconnect part **133A** and the first signal terminal **44A**.

As shown in FIG. **9**, the interconnect **131C** includes a first interconnect part **132C** extending through the insulation films **85** in the z-direction and a second interconnect part **133C** extending in the y-direction.

The first interconnect part **132C** and the first interconnect part **132A** of the interconnect **131A** have the same structure.

The first interconnect part **132C** is arranged to overlap the first electrode pad **81C** in plan view and is connected to the first electrode pad **81C**. The first interconnect part **132C** extends through the insulation films **85** from the insulation film **85** that is located immediately below the uppermost insulation film **85U** to the insulation film **85** that is located above the lowermost insulation film **85L** with one insulation film **85** interposed. The first interconnect part **132C** includes flat interconnect pieces and vias. The interconnect pieces are located at the same position as the insulation films **851** and **854**, in which the coils **41A** and **42A** are arranged. The vias are arranged between the two interconnect pieces in the z-direction, between the upper interconnect piece and the first electrode pad **81A**, and between the lower interconnect piece and the second interconnect part **133C**.

The second interconnect part **133C** is arranged closer to the substrate **83** than the first interconnect part **132C** is. The second interconnect part **133C** is arranged closer to the substrate **83** than the first coil **41A** is. In the present embodiment, the second interconnect part **133C** is arranged in the insulation film **85** located immediately above the lowermost insulation film **85L** among the insulation films **85**. Of opposite ends of the second interconnect part **133C** in the x-direction, a first end is located closer to the chip side surface **80b** of the transformer chip **80** and overlaps the first interconnect part **132C** in plan view. The second interconnect part **133C** is connected to the first interconnect part **132C**. The second interconnect part **133C** includes a second end opposite to the first end. In plan view, the second end is arranged to overlap the first coil **41A** of the transformer **40A**. More specifically, in plan view, the second end overlaps the first ground terminal **45** of the first coil **41A** of the transformer **40A**. The second interconnect part **133C** includes vias **134C** connecting the second interconnect part **133C** and the first signal terminal **44A**. The second interconnect part **133C** of the interconnect **131C** is electrically connected to the substrate **83** by vias **136** that extend through the lowermost insulation film **85L**. The vias **136** may be omitted.

As shown in FIG. **9**, the second electrode pad **82A** is electrically connected to the second signal terminal **47A** of

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the second coil 42A by vias 135A extending through the uppermost insulation film 85U.

The second electrode pad 82C is electrically connected to the second ground terminal 48 of the second coil 42A by vias 135C extending through the uppermost insulation film 85U.

As shown in FIGS. 4 and 6, in the present embodiment, the transformer chip 80 includes a dummy pattern 120 arranged around the second coils 42A and 42B of the transformers 40A and 40B. The dummy pattern 120 is a dummy coil pattern.

The dummy pattern 120 is arranged in the inner region 87 and includes a first dummy pattern 121, a second dummy pattern 122, and a third dummy pattern 123. The dummy patterns 121 to 123 are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

In plan view, the first dummy pattern 121 is arranged in a region between the second coil 42A of the transformer 40A and the second coil 42B of the transformer 40B in the x-direction. The first dummy pattern 121 is patterned differently from the second coils 42A and 42B. The first dummy pattern 121 is electrically connected to the second ground terminal 48 of the second coil 42A. The first dummy pattern 121 may be electrically connected to the second ground terminal 48 of at least one of the two second coils 42A. Thus, the first dummy pattern 121 has the same potential as the second coils 42A and 42B. Therefore, as the second reference potential of the second coils 42A and 42B changes, the voltage of the first dummy pattern 121 may become higher than that of the first coil 41B in the same manner as the second coil 42B.

Although not shown, the first dummy pattern 121 is aligned with the second coils 42A and 42B in the z-direction. Thus, the first dummy pattern 121 is located farther away from the substrate 83 than the first coils 41A and 41B are. In other words, the dummy pattern 120 is arranged around coils of the transformers 40A and 40B located closer to the chip main surface 80s of the transformer chip 80.

The first dummy pattern 121 and the second coils 42A and 42B have the same voltage. This limits voltage drops between the first dummy pattern 121 and the second coils 42A and 42B. Thus, concentration of an electric field on the second coils 42A and 42B is limited.

As shown in FIG. 6, the third dummy pattern 123 is formed to surround the second coils 42A and 42B of the transformers 40A and 40B in plan view. The third dummy pattern 123 is electrically connected to the first dummy pattern 121. Thus, in the third dummy pattern 123, similar to the first dummy pattern 121, as the second reference potential of the second coil 42B changes, the voltage of the third dummy pattern 123 may become higher than that of the first coil 41B.

As shown in FIG. 9, the third dummy pattern 123 is aligned with the second coil 42A in the z-direction. Although not shown, the third dummy pattern 123 is aligned with the second coil 42B in the z-direction. Thus, the third dummy pattern 123 is located farther away from the substrate 83 than the first coils 41A and 41B are. As described above, the dummy patterns 121 to 123 are aligned with each other in the z-direction.

The third dummy pattern 123 and the second coils 42A and 42B have the same voltage. This limits voltage drops between the third dummy pattern 123 and the second coils 42A and 42B. Thus, concentration of an electric field on the second coils 42A and 42B is limited.

As shown in FIG. 6, the second dummy pattern 122 is formed to surround the third dummy pattern 123 in plan view. The second dummy pattern 122 is independent of the

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second coils 42A and 42B. In other words, the second dummy pattern 122 is not electrically connected to the second coils 42A and 42B.

As shown in FIG. 9, the second dummy pattern 122 is aligned with the second coil 42A in the z-direction. Although not shown, the second dummy pattern 122 is aligned with the second coil 42B in the z-direction. Thus, the second dummy pattern 122 is located farther away from the substrate 83 than the first coils 41A and 41B are. The second dummy pattern 122 limits an increase in the electric field strength around the second coils 42A and 42B and limits concentration of the electric field on the second electrode pads 82A to 82C.

As shown in FIG. 8, in the present embodiment, the transformer chip 80 includes a dummy pattern 125 arranged around the second capacitor electrodes 52A and 52B of the capacitors 50A and 50B. The dummy pattern 125 and the dummy pattern 120, described above, have the same structure. The dummy pattern 125 includes a slit extending from the inside of the dummy pattern 125 toward the outside of the dummy pattern 125. The slit limits formation of a current loop in the dummy pattern 125. The dummy pattern 125 is a dummy electrode pattern.

More specifically, the dummy pattern 125 is arranged in the inner region 87 and includes a first dummy pattern 126, a second dummy pattern 127, and a third dummy pattern 128. In an example, the dummy patterns 126 to 128 and the second capacitor electrode 52A are formed from the same material.

In plan view, the first dummy pattern 126 is arranged in a region between the second capacitor electrode 52A of the capacitor 50A and the second capacitor electrode 52B of the capacitor 50B in the x-direction. The first dummy pattern 126 is patterned differently from the second capacitor electrodes 52A and 52B. The first dummy pattern 126 is electrically connected to the second capacitor ground terminal 58 of the second capacitor electrode 52A. The first dummy pattern 126 may be electrically connected to the second capacitor ground terminal 58 of at least one of the two second capacitor electrodes 52A. Thus, the first dummy pattern 126 has the same potential as the second capacitor electrodes 52A and 52B. Therefore, as the second reference potential of the second capacitor electrodes 52A and 52B changes, the voltage of the first dummy pattern 126 may become higher than that of the first capacitor electrode 51B in the same manner as the second capacitor electrode 52B.

Although not shown, the first dummy pattern 126 is aligned with the second capacitor electrodes 52A and 52B in the z-direction. Thus, the first dummy pattern 126 is located farther away from the substrate 83 than the first capacitor electrodes 51A and 51B are. In other words, the dummy pattern 125 is arranged around coils of the capacitors 50A and 50B located closer to the chip main surface 80s of the transformer chip 80.

When the first dummy pattern 126 and the second capacitor electrodes 52A and 52B have the same voltage, a voltage drop between the first dummy pattern 126 and the second capacitor electrodes 52A and 52B is limited. This limits concentration of an electric field on the second capacitor electrodes 52A and 52B.

As shown in FIG. 8, the third dummy pattern 128 is formed to surround the second capacitor electrodes 52A and 52B of the capacitors 50A and 50B in plan view. The third dummy pattern 128 is electrically connected to the first dummy pattern 126. Thus, in the third dummy pattern 128, similar to the first dummy pattern 126, as the second reference potential of the second capacitor electrode 52B

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changes, the voltage of the third dummy pattern 128 may become higher than the voltage of the first capacitor electrode 51B.

As shown in FIG. 9, the third dummy pattern 128 is aligned with the second capacitor electrode 52A in the z-direction. Although not shown, the third dummy pattern 128 is also aligned with the second capacitor electrode 52B in the z-direction. Thus, the third dummy pattern 128 is located farther away from the substrate 83 than the first capacitor electrodes 51A and 51B are. As described above, the dummy patterns 126 to 128 are aligned with each other in the z-direction.

When the third dummy pattern 128 and the second capacitor electrodes 52A and 52B have the same voltage, a voltage drop between the third dummy pattern 128 and the second capacitor electrodes 52A and 52B is limited. This limits concentration of an electric field on the second capacitor electrodes 52A and 52B.

As shown in FIG. 8, the second dummy pattern 127 is formed to surround the third dummy pattern 128 in plan view. The second dummy pattern 127 is independent of the second capacitor electrodes 52A and 52B. In other words, the second dummy pattern 127 is not electrically connected to the second capacitor electrodes 52A and 52B.

As shown in FIG. 9, the second dummy pattern 127 is aligned with the second capacitor electrode 52A in the z-direction. Although not shown, the second dummy pattern 127 is also aligned with the second capacitor electrode 52B in the z-direction. Thus, the second dummy pattern 127 is located farther away from the substrate 83 than the first capacitor electrodes 51A and 51B are. The second dummy pattern 127 limits an increase in the electric field strength around the second capacitor electrodes 52A and 52B and limits concentration of the electric field on the second electrode pads 82A to 82C.

As shown in FIG. 9, the transformer chip 80 includes a protection film 150 and a passivation film 160. The protection film 150 is formed on a surface 84s of the insulation layer 84. The protection film 150 protects the insulation layer 84. The protection film 150 is a film formed from a material including, for example, silicon oxide.

The passivation film 160 is a surface protection film for the transformer chip 80. The passivation film 160 is formed from a material including, for example, silicon nitride. The material including silicon nitride includes, for example, SiN and SiCN. In the present embodiment, the passivation film 160 is formed from a material including SiN. The passivation film 160 defines the chip main surface 80s of the transformer chip 80.

The first electrode pads 81 and the second electrode pads 82 are covered by the protection film 150 and the passivation film 160. The protection film 150 and the passivation film 160 each include openings partially exposing the first electrode pads 81 and the second electrode pads 82. Thus, each first electrode pad 81 includes an exposed surface configured to be connected to the wire W2. Also, each second electrode pad 82 includes an exposed surface configured to be connected to the wire W3.

As shown in FIGS. 3 and 4, the transformer chip 80 includes a resin layer 180 formed on the passivation film 160. The resin layer 180 is formed from a material including, for example, polyimide (PI). The resin layer 180 is separated by a separation groove 183 into an inner resin layer 181 and an outer resin layer 182. As shown in FIG. 4, in plan view, the separation groove 183 surrounds the transformers 40A and 40B. The resin layer 180 includes a first resin opening

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184 exposing the first electrode pad 81 and a second resin opening 185 exposing the second electrode pad 82.

FIG. 10 is an enlarged partial view of FIG. 9 showing an example of the structure of the first coil 41A and the first capacitor electrode 51A. In the example of the structure, the first coil 41A is formed from Cu.

As shown in FIG. 10, the coil groove 141 is elliptical and spiral and is formed in the insulation film 851. The coil groove 141 extends through the second insulation film 85B. Therefore, the upper end and the lower end of the coil groove 141 are respectively open to the first insulation film 85A of the upper insulation film 851 and the first insulation film 85A of the insulation film 852, in which the first coil 41A is embedded.

A barrier film 143 is formed on the wall surfaces (side wall surface and bottom wall surface) of the coil groove 141. The barrier film 143 is formed on the side wall surface and the bottom wall surface so that the coil groove 141 has a cavity that is open upward. In the embodiment, the barrier film 143 is formed from a material including Ta, TaN, Ti, TiN, or the like. A body layer 144 is embedded in the coil groove 141 at an inner side of the barrier film 143. This forms the second coil 42A, which is an example of an embedded coil formed of the barrier film 143 and the body layer 144. The body layer 144 is formed from a material including one or more selected from Cu, Al, and W.

The first coil 41A includes an upper surface that is flush with the upper surface of the insulation film 851. As a result, the side surface, the upper surface, and the lower surface of the first coil 41A are in contact with different insulation films 85. More specifically, in the insulation film 851, in which the first coil 41A is embedded, the first insulation film 85A and the second insulation film 85B are in contact with the side surface of the second coil 42A. In the insulation film 852 formed above the insulation film 851, only the first insulation film 85A, which is the lower layer, is in contact with the upper surface of the second coil 42A. In the insulation film 85 located below, only the second insulation film 85B, which is the upper layer, is in contact with the lower surface of the second coil 42A.

The first capacitor electrode 51A is in contact with the upper surface of the first insulation film 85A forming the insulation film 852. The upper surface and the side surface of the first capacitor electrode 51A are in contact with the second insulation film 85B of the insulation film 852. One surface (lower surface) of the first insulation film 85A of the insulation film 852 in the z-direction is in contact with the second insulation film 85B (interlayer insulation film). The other surface (upper surface) of the first insulation film 85A of the insulation film 852 in the z-direction is in contact with the first capacitor electrode 51A. The first insulation film 85A of the insulation film 852 corresponds to a first thin film. The second insulation film 85B of the insulation film 852 corresponds to a first interlayer insulation film.

The first insulation film 85A of the insulation film 852 covers the second insulation film 85B of the insulation film 851, which is located below the insulation film 852, and the first coil 41A, which is embedded in the insulation film 851. The first insulation film 85A includes the first opening 145, which partially exposes the upper surface of the first ground terminal 45 of the first coil 41A. Then, a film of a conductive material is formed on the first insulation film 85A to form the first capacitor electrode 51A. The conductive material film is partially removed to form the first capacitor electrode 51A. The second insulation film 85B is formed to cover the first

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capacitor electrode 51A and the first insulation film 85A. This forms the first capacitor electrode 51A embedded in the insulation film 852.

FIG. 11 is an enlarged partial view of FIG. 9 showing an example of the structure of the second coil 42A and the second capacitor electrode 52A. In the example of the structure, the second coil 42A is formed from Cu.

As shown in FIG. 11, the coil groove 142 is elliptical and spiral and formed in the second insulation film 85B of the insulation film 854. The coil groove 142 extends through the second insulation film 85B. Therefore, the upper end and the lower end of the coil groove 142 are respectively open to the first insulation film 85A of the upper insulation film 85 and the first insulation film 85A of the insulation film 854, in which the second coil 42A is embedded.

The barrier film 143 is formed on the wall surface (side wall surface and bottom wall surface) of the coil groove 142. The barrier film 143 is formed on the side wall surface and the bottom wall surface so that the coil groove 142 has a cavity that is open upward. In the embodiment, the barrier film 143 is formed from a material including Ta, TaN, Ti, TiN, or the like. A body layer 144 is embedded in the coil groove 142 at an inner side of the barrier film 143. This forms the second coil 42A, which is an example of an embedded coil formed of is formed of the barrier film 143 and the body layer 144. The body layer 144 is formed from a material including one or more selected from Cu, Al, and W.

The second coil 42A includes an upper surface that is flush with the upper surface of the insulation film 854. As a result, the side surface, the upper surface, and the lower surface of the second coil 42A are in contact with different insulation films. More specifically, in the insulation film 85 in which the second coil 42A is embedded, the second insulation film 85B is in contact with the side surface of the second coil 42A. In the insulation film 85 formed above the insulation film 85 in which the second coil 42A is embedded, only the lower first insulation film 85A is in contact with the upper surface of the second coil 42A. In the insulation film 854, in which the second coil 42A is embedded, the first insulation film 85A is in contact with the lower surface of the second coil 42A.

The second capacitor electrode 52A is in contact with the lower surface of the first insulation film 85A of the insulation film 854, in which the second coil 42A is embedded. The lower surface and the side surface of the second capacitor electrode 52A are in contact with the second insulation film 85B of the insulation film 853. One surface (lower surface) of the first insulation film 85A of the insulation film 853 in the z-direction is in contact with the second capacitor electrode 52A. The other surface (upper surface) is in contact with the second insulation film 85B (interlayer insulation film). The first insulation film 85A of the insulation film 854 corresponds to a second thin film. The second insulation film 85B of the insulation film 853 corresponds to a second interlayer insulation film.

As shown in FIG. 11 with broken lines, the second insulation film 85B of the insulation film 853 includes two layers of insulation film 85B1 and 85B2. First, the insulation film 85B1 is formed on the upper surface of the first insulation film 85A of the insulation film 853. Then, a film of a conductive material is formed on the upper surface of the insulation films 85B1 to form the second capacitor electrode 52A. The conductive material film is partially removed to form the second capacitor electrode 52A. The second capacitor electrode 52A is further embedded in the insulation film 85B2 to form the second insulation film 85B.

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The first insulation film 85A of the insulation film 854 is formed to cover the second insulation film 85B and the second capacitor electrode 52A. This forms the second capacitor electrode 52A embedded in the insulation film 85.

Although not described and shown, the dummy patterns 120 and 125, shown in FIG. 9, are formed in the same manner as the second coil 42A and the second capacitor electrode 52A. The interconnects 131A and 131C, shown in FIG. 9, are formed in the same manner as the first coil 41A and the second coil 42A by embedding a barrier film and a body layer in a groove extending through the insulation film 85.

Operation

The operation of the gate driver 10 according to the present embodiment will now be described.

Comparative Example

A comparative example of the gate driver 10 of the present embodiment will now be described.

FIG. 12 shows a comparative example of a gate driver 10R. The gate driver 10R of the comparative example does not include the capacitors 50 (capacitors 50A and 50B), which are shown in FIG. 1. In the gate driver 10R, a current i_{C1} flowing to parasitic capacitance C1 between the first coil 41A and the second coil 42A of the transformer 40A may produce noise on a set signal transmitted from the low-voltage circuit 20 to the high-voltage circuit 30. This may result in an erroneous operation. Also, a current I_{C2} flowing to parasitic capacitance C2 between the first coil 41B and the second coil 42B of the transformer 40B may produce noise on a reset signal transmitted from the low-voltage circuit 20 to the high-voltage circuit 30. This may result in an erroneous operation.

To mitigate an erroneous operation caused by the currents i_{C1} and i_{C2} , the high-voltage circuit 30 includes a circuit that masks noise. In an example, after receiving a reset signal (RESET), the mask circuit masks reception of a signal for a fixed period. This mitigates an erroneous operation caused by the currents i_{C1} and i_{C2} flowing to the second coils 42A and 42B due to the parasitic capacitances C1 and C2.

The capacitance values of the parasitic capacitances C1 and C2 may vary depending on each gate driver 10R and operational state. That is, the capacitance values of the parasitic capacitances C1 and C2 are inconstant. Therefore, the position of noise superimposed on the set signal and the reset signal may change. In this regard, the high-voltage circuit 30 needs to set a long mask period in accordance with positions where noise may be produced. Since a signal cannot be transmitted during the mask period, fast transmission of a signal from the low-voltage circuit 20 to the high-voltage circuit 30 is hindered. The same issue arises when a signal is transmitted from the high-voltage circuit 30 to the low-voltage circuit 20. Hence, the low-voltage circuit 20 also needs a mask circuit in the same manner as the high-voltage circuit 30.

FIG. 13 shows an operation of the gate driver 10 of the present embodiment.

As described above, the gate driver 10 of the present embodiment includes the transformer 40A including the first coil 41A and the second coil 42A and the transformer 40B including the first coil 41B and the second coil 42B. The gate driver 10 of the present embodiment further includes the capacitor 50A connected between the ground terminal of the first coil 41A and the ground terminal of the second coil 42A.

and the capacitor 50B connected between the ground terminal of the first coil 41B and the ground terminal of the second coil 42B.

The first capacitor electrode 51A of the capacitor 50A and the first coil 41A are electrically connected and have the same potential. The second capacitor electrode 52A of the capacitor 50A and the second coil 42A are electrically connected and have the same potential. The first capacitor electrode 51B of the capacitor 50B and the first coil 41B are electrically connected and have the same potential. The second capacitor electrode 52B of the capacitor 50B and the second coil 42B are electrically connected and have the same potential.

The transformer 40A and the transformer 40B are used to, for example, transmit a set signal (SET) and a reset signal (RESET) from the low-voltage circuit 20 toward the high-voltage circuit 30.

At this time, the set signal output from the low-voltage circuit 20 cause a current i_{1A} to flow to the first coil 41A of the transformer 40A. A current i_{2A} flows to the second coil 42A, which is magnetically coupled to the first coil 41A. The current i_{2A} causes the high-voltage circuit 30 to generate a pulse signal, that is, receive a set signal.

The set signal causes a current i_{CA} to flow between the first capacitor electrode 51A and the second capacitor electrode 52A of the capacitor 50A. The current i_{CA} flows from the second capacitor electrode 52A to the ground terminal of the second coil 42A, namely, the ground GND2. This reduces the effect of the current i_{CA} flowing to the capacitor 50A on the current i_{2A} flowing to the second coil 42A of the transformer 40A. Thus, the effect on signal transmission between the low-voltage circuit 20 and the high-voltage circuit 30 is reduced.

The reset signal output from the low-voltage circuit 20 causes a current i_{1B} to flow to the first coil 41B of the transformer 40B. A current i_{2B} flows to the second coil 42B, which is magnetically coupled to the first coil 41B. The current i_{2B} causes the high-voltage circuit 30 to generate a pulse signal, that is, receive a reset signal.

The reset signal causes a current i_{CB} to flow between the first capacitor electrode 51B and the second capacitor electrode 52B of the capacitor 50B. The current i_{CB} flows from the second capacitor electrode 52B to the ground terminal of the second coil 42B, namely, the ground GND2. This reduces the effect of the current i_{2B} flowing to the second coil 42B of the transformer 40B on the current i_{CB} flowing to the capacitor 50B. Thus, the effect on signal transmission between the low-voltage circuit 20 and the high-voltage circuit 30 is reduced.

The phase of current i_{CA} is retarded from the phase of current i of the set signal in accordance with the impedance value of the transformer 40A and the capacitance value of the capacitor 50A. The phase of current i_{CB} is retarded from the phase of the current i of the reset signal in accordance with the impedance value of the transformer 40B and the capacitance value of the capacitor 50B. With this configuration, even when the currents i_{CA} and i_{CB} affect signal reception of the high-voltage circuit 30, a mask period may be set in accordance with timings of the currents i_{CA} and i_{CB} . The mask period is shorter than that for the case of the parasitic capacitance C1. This reduces the effect on the fast signal transmission.

Advantages

The gate driver 10 of the present embodiment obtains the following advantages.

(1-1) The transformer chip 80 includes the insulation layer 84, the transformer 40A, and the capacitor 50A. The transformer 40A includes the first coil 41A and the second coil 42A embedded in the insulation layer 84 and separated from each other in the thickness-wise direction of the insulation layer 84. The capacitor 50A includes the first capacitor electrode 51A and the second capacitor electrode 52A arranged between the first coil 41A and the second coil 42A of the transformer 40A. The first capacitor electrode 51A is connected to the first ground terminal 45 of the first coil 41A. The second capacitor electrode 52A is connected to the second ground terminal 48 of the second coil 42A. The insulation layer 84 includes the insulation films 85 (851 to 856). The first capacitor electrode 51A is formed between the second insulation film 85B of the insulation film 851 and the first insulation film 85A of the insulation film 852. The second capacitor electrode 52A is formed between the second insulation film 85B of the insulation film 853 and the first insulation film 85A of the insulation film 854.

With this structure, the transformer 40A is used to, for example, transmit a set signal (SET) from the low-voltage circuit 20 toward the high-voltage circuit 30. The set signal causes the current i_{CA} flowing through the capacitor 50A to flow from the second capacitor electrode 52A to the ground terminal of the second coil 42A, namely, the ground GND2. This reduces the effect of the current i_{CA} flowing to the capacitor 50A on the current i_{2A} flowing to the second coil 42A of the transformer 40A. Thus, the effect on signal transmission between the low-voltage circuit 20 and the high-voltage circuit 30 is reduced.

The transformer chip 80 includes the insulation layer 84, the transformer 40B, and the capacitor 50B. The transformer 40B includes the first coil 41B and the second coil 42B embedded in the insulation layer 84 and separated from each other in the thickness-wise direction of the insulation layer 84. The capacitor 50B includes the first capacitor electrode 51B and the second capacitor electrode 52B arranged between the first coil 41B and the second coil 42B of the transformer 40B. The first capacitor electrode 51B is connected to the first ground terminal 45 of the first coil 41B. The second capacitor electrode 52B is connected to the second ground terminal 48 of the second coil 42B.

The transformer 40B is used to, for example, transmit a reset signal (RESET) from the low-voltage circuit 20 toward the high-voltage circuit 30. The reset signal causes the current i_{CB} flowing through the capacitor 50B to flow from the second capacitor electrode 52B to the ground terminal of the second coil 42B, namely, the ground GND2. This reduces the effect of the current i_{2B} flowing to the second coil 42B of the transformer 40B on the current i_{CB} flowing to the capacitor 50B. Thus, the effect on signal transmission between the low-voltage circuit 20 and the high-voltage circuit 30 is reduced.

(1-2) The phase of current i_{CA} flowing through the capacitor 50A is retarded from the phase of current i of the set signal in accordance with the impedance value of the transformer 40A and the capacitance value of the capacitor 50A. The phase of current i_{CB} flowing through the capacitor 50B is retarded from the phase of current i of the reset signal in accordance with the impedance value of the transformer 40B and the capacitance value of the capacitor 50B. With this configuration, even when the currents i_{CA} and i_{CB} affect signal reception of the high-voltage circuit 30, a mask period may be set in accordance with phases of the currents i_{CA} and i_{CB} . This simplifies the design of the high-voltage circuit 30.

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The mask period is shorter than that for the case of the parasitic capacitance C1. This reduces the effect on the fast signal transmission.

(1-3) In plan view, the dummy pattern 120 is arranged around the second coils 42A and 42B. This structure limits concentration of an electric field on the second coils 42A and 42B.

(1-4) In plan view, the dummy pattern 125 is arranged around the second capacitor electrodes 52A and 52B. This structure limits concentration of an electric field on the second capacitor electrodes 52A and 52B.

(1-5) The gate driver 10 includes the low-voltage circuit 20, the high-voltage circuit 30, and the transformer chip 80. The low-voltage circuit 20 and the high-voltage circuit 30 are connected by the transformer chip 80 and configured to transmit a signal through the transformer chip 80. The transformer chip 80 includes the transformers 40A and 40B and the capacitors 50A and 50B. The capacitor 50A includes the first capacitor electrode 51A and the second capacitor electrode 52A arranged between the first coil 41A and the second coil 42A of the transformer 40A. The capacitor 50B includes the first capacitor electrode 51B and the second capacitor electrode 52B arranged between the first coil 41B and the second coil 42B of the transformer 40B. The first capacitor electrodes 51A and 51B are connected to the first ground terminals 45 of the first coils 41A and 41B. The second capacitor electrodes 52A and 52B are connected to the second ground terminals 48 of the second coils 42A and 42B. This structure obtains the advantage (1) described above. Thus, the effect of the gate driver 10 on a transmitting signal is reduced.

(1-6) In an example of a structure in which the gate driver 10 includes a transformer 40 and a capacitor 50, the gate driver 10 may include a low-voltage circuit chip including the low-voltage circuit 20, the transformer 40, and the capacitor 50. Alternatively, the gate driver 10 may include a high-voltage circuit chip including the high-voltage circuit 30, the transformer 40, and the capacitor 50. In such structures, when the circuit configuration of the low-voltage circuit 20 or the high-voltage circuit 30 is changed, the corresponding chip entirely needs to be changed. This increases costs when different types of gate drivers are manufactured.

In this regard, in the present embodiment, the transformer 40 and the capacitor 50 are included in the transformer chip 80, which is independent of the low-voltage circuit chip 60 and the high-voltage circuit chip 70. That is, the chip is dedicated to the transformer 40. This allows the common transformer chip 80 to be used for the low-voltage circuit 20 and the high-voltage circuit 30, which differ from each other. As a result, costs are reduced when manufacturing different types of gate drivers 10 that differ in at least one of the low-voltage circuit 20 and the high-voltage circuit 30.

(1-7) The first coil 41A and the first capacitor electrode 51A of the present embodiment are located at opposite sides of the first insulation film 85A of the insulation film 852. The first coil 41A and the first capacitor electrode 51A are formed from a conductive metal. The first insulation film 85A is formed from, for example, SiN. In other words, the first coil 41A, the first insulation film 85A, and the first capacitor electrode 51A may have a metal-insulator-metal (MIM) structure. Also, the second coil 42A, the second capacitor electrode 52A, and the first insulation film 85A of the insulation film 854 may have a metal-insulator-metal (MIM) structure. Thus, a capacitor having the MIM structure is readily formed on the transformer chip 80.

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(1-8) The first capacitor electrode 51A and the second capacitor electrode 52A are formed from a nonmagnetic material. When, for example, TiN, CrSi, or the like is selected as a magnetic material, a resistive element may be readily formed from the nonmagnetic material on the transformer chip 80.

Second Embodiment

A second embodiment of a gate driver 10 will be described with reference to FIGS. 14, 15A, and 15B. The gate driver 10 of the present embodiment differs from the gate driver 10 of the first embodiment mainly in the positions where the first capacitor electrode 51A of the capacitor 50A and the second capacitor electrode 52A are formed. In the following description, the differences from the gate driver 10 of the first embodiment will be described in detail. The same reference characters are given to those components that are the same as the corresponding components of the gate driver 10 of the first embodiment. Such components will not be described in detail.

As shown in FIG. 14, the first capacitor electrode 51A and the second capacitor electrode 52A of the capacitor 50A are arranged between the first coil 41A and the second coil 42A of the transformer 40A. The first capacitor electrode 51A and the second capacitor electrode 52A are opposed to each other in the z-direction with multiple insulation films 85 interposed.

At least one insulation film 85 is interposed between the first capacitor electrode 51A and the first coil 41A. In the present embodiment, one insulation film 85 is interposed. The second capacitor electrode 52A is formed in the insulation film 853 located next to the insulation film 854, in which the second coil 42A is formed.

As shown in FIGS. 14 and 15A, the first capacitor electrode 51A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the insulation film 852, in which the first capacitor electrode 51A is embedded, includes a first insulation film 85A and a second insulation film 85B. An insulation film 855 is formed between the insulation film 851, including the first coil 41A, and the insulation film 852 and includes a first insulation film 85A and a second insulation film 85B.

The first capacitor electrode 51A is formed on the upper surface of the insulation film 855, more specifically, the upper surface of the second insulation film 85B of the insulation film 855. The first insulation film 85A of the insulation film 852 is formed to cover the first capacitor electrode 51A. Thus, the upper surface and the side surface of the first capacitor electrode 51A are in contact with the first insulation film 85A of the insulation film 852. The lower surface of the first capacitor electrode 51A is flush with the lower surface of the first insulation film 85A of the insulation film 852. The lower surface of the first capacitor electrode 51A is in contact with the second insulation film 85B of the insulation film 855. Thus, the first capacitor electrode 51A is embedded in the insulation film 85. The second insulation film 85B of the insulation film 855 is arranged between the first capacitor electrode 51A and the first coil 41A. The first insulation film 85A of the insulation film 852 corresponds to a first thin film. The second insulation film 85B of the insulation film 855 corresponds to a first interlayer insulation film.

As shown in FIGS. 14 and 15B, the second capacitor electrode 52A is configured to be a conductive layer embedded in one insulation film 85. More specifically, the insulation film 853, in which the second capacitor electrode 52A

is embedded, includes a first insulation film **85A** and a second insulation film **85B**. An insulation film **856** is located next to the insulation film **853** in a direction toward the first capacitor electrode **51A** and includes a first insulation film **85A** and a second insulation film **85B**. The second capacitor electrode **52A** is formed on the upper surface of the insulation film **856**, more specifically, the upper surface of the second insulation film **85B** of the insulation film **856**. The first insulation film **85A** of the insulation film **853** is formed to cover the second capacitor electrode **52A**. Thus, the upper surface and the side surface of the second capacitor electrode **52A** are in contact with the first insulation film **85A** of the insulation film **853**. The lower surface of the second capacitor electrode **52A** is flush with the lower surface of the first insulation film **85A** of the insulation film **853**. The lower surface of the second capacitor electrode **52A** is in contact with the second insulation film **85B** of the insulation film **856**. Thus, the second capacitor electrode **52A** is embedded in the insulation films **85**. In other words, the second insulation film **85B** of the insulation film **856** and the first coil **41A** are located at opposite sides of the first capacitor electrode **51A**. The second insulation film **85B** of the insulation film **853** is arranged between the first capacitor electrode **51A** and the first coil **41A**. The first insulation film **85A** of the insulation film **853** corresponds to a second thin film. The second insulation film **85B** of the insulation film **856** corresponds to a second interlayer insulation film.

The distance between the first capacitor electrode **51A** and the second capacitor electrode **52A** is determined by the thickness of the insulation film **85** and the number of the insulation films **85** stacked between the first capacitor electrode **51A** and the second capacitor electrode **52A**. The distance is set in accordance with the insulation voltage or electric field strength of the transformer chip **80**.

The first capacitor electrode **51A** is formed from a material including, for example, Ta, TaN, Ti, or TiN. The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. The first capacitor electrode **51A** and the barrier film **143**, which includes vias embedded in the insulation film **855** located next to the insulation film **852** including the first capacitor electrode **51A**, are formed at the same time.

As shown in FIG. **14**, the first capacitor electrode **51A** is electrically connected to the first coil **41A**.

The first coil **41A** includes the first coil wiring **43A**, the first signal terminal **44A**, and the first ground terminal **45**. The first capacitor electrode **51A** includes the first electrode wiring **53A**, the first capacitor end **54A**, and the first capacitor ground terminal **55**. The first electrode wiring **53A** overlaps the first coil wiring **43A** in the z-direction. The first capacitor end **54A** overlaps the first signal terminal **44A** in the z-direction. The first capacitor ground terminal **55** overlaps the first ground terminal **45** in the z-direction.

The first capacitor ground terminal **55** of the first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**. The insulation film **855**, which is arranged between the first capacitor electrode **51A** and the first coil **41A**, includes vias **137** extending through the insulation film **855** in the z-direction. The vias **137** connect the first capacitor ground terminal **55** of the first capacitor electrode **51A** and the first ground terminal **45** of the first coil **41A**. Thus, the first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**.

As shown in FIG. **14**, the second capacitor electrode **52A** is electrically connected to the second coil **42A**.

The second coil **42A** includes the second coil wiring **46A**, the second signal terminal **47A**, and the second ground terminal **48**. The second capacitor electrode **52A** includes the second electrode wiring **56A**, the second capacitor end **57A**, and the second capacitor ground terminal **58**. The second electrode wiring **56A** overlaps the second coil wiring **46A** in the z-direction. The second capacitor end **57A** overlaps the second signal terminal **47A** in the z-direction. The second capacitor ground terminal **58** overlaps the second ground terminal **48** in the z-direction.

The second capacitor ground terminal **58** of the second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**. The insulation film **853**, which is arranged between the second capacitor electrode **52A** and the second coil **42A**, includes vias **138** extending through the insulation film **853** in the z-direction. The vias **138** further extend through the first insulation film **85A** of the insulation film **854**, in which the second coil **42A** is embedded. The vias **138** connect the second capacitor ground terminal **58** of the second capacitor electrode **52A** and the second ground terminal **48** of the second coil **42A**. The second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**.

FIGS. **16A** and **16B** show some of manufacturing processes of the transformer chip **80** including processes of forming vias **131V** embedded in the insulation film **855** and forming the first capacitor electrode **51A**. The vias **131V** are included in the interconnect **131C**, shown in FIG. **14**.

As shown in FIG. **16A**, via holes **151** are formed in the insulation film **855** and extends through the insulation film **855** in the thickness-wise direction. A barrier film **152** is formed on the surface of the insulation film **855**, more specifically, the upper surface of the second insulation film **85B**, and the side and bottom wall surfaces of the via holes **151**. The via holes **151** are filled with the body layer **144** at an inner side of the barrier film **152**. A resist film **153** is formed to cover predetermined portions of the upper surface of the barrier film **152**. The predetermined portions covered by the resist film **153** correspond to where the first capacitor electrode **51A** (refer to FIG. **14**) is formed. The barrier film **152** exposed from the resist film **153** (refer to FIG. **16B**) is removed by, for example, etching, and then the resist film **153** is removed. This obtains the first capacitor electrode **51A** shown in FIG. **16B**.

Although not shown and described, the second capacitor electrode **52A** of the present embodiment may be formed through the same processes as the first capacitor electrode. Advantages

The gate driver **10** of the present embodiment obtains the following advantages in addition to the advantages of the first embodiment.

(2-1) The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. This allows the first capacitor electrode **51A** to be formed at the same time as, for example, the barrier film **143** forming the vias **131V** included in the interconnect **131C**. Thus, the number of processes for the first capacitor electrode **51A** is not increased. The second capacitor electrode **52A** may also be formed through the same processes as the first capacitor electrode **51A**. Thus, the number of processes is not increased.

(2-2) The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. Thus, the barrier film **143** may be used to readily form other elements such as a resistive element on the transformer chip **80**.

The above embodiments exemplify, without any intention to limit, applicable forms of an insulation module and a gate driver according to the present disclosure. The insulation module and the gate driver according to the present disclosure can be applicable to forms differing from the above embodiments. In an example of such a form, the structure of the above embodiment is partially replaced, changed, or omitted, or a further structure is added to the above embodiment. The modified examples described below may be combined with one another as long as there is no technical inconsistency. In the modified examples, the same reference characters are given to those components that are the same as the corresponding components of the above embodiment. Such components will not be described in detail.

In the embodiments, the shape of the capacitor 50 in plan view may be changed.

FIG. 17 shows a modified example of first capacitor electrodes 51A and 51B. The first capacitor electrode 51A is elliptical and annular and overlaps the first coil wiring 43A of the first coil 41A shown in FIG. 5. The first capacitor electrode 51A of this modified example includes a first electrode wiring 53A formed of annular wires arranged from an inner end 43i of the first coil wiring 43A to an outer end 43o of the first coil wiring 43A, which are shown in FIG. 5. The first capacitor electrode 51A includes a slit 51As extending outward from the center. The first capacitor electrode 51A is annular and open by the slit 51As.

The first capacitor electrode 51B is elliptical and annular and overlaps the first coil wiring 43A of the first coil 41B shown in FIG. 5. The first capacitor electrode 51B of this modified example is formed of annular wires arranged from the inner end 43i of the first coil wiring 43B to the outer end 43o of the first coil wiring 43B, which are shown in FIG. 5. The first capacitor electrode 51B includes a first slit 51Bs extending outward from the center. The first capacitor electrode 51B is annular and open by the first slit 51Bs.

FIG. 18 shows a modified example of first capacitor electrodes 51A and 51B. The first capacitor electrode 51A is elliptical and annular and overlaps the first coil wiring 43A of the first coil 41A shown in FIG. 5. The first capacitor electrode 51A of this modified example includes a first electrode wiring 53A that is plate-shaped and continuous from the inner end 43i of the first coil wiring 43A to the outer end 43o of the first coil wiring 43B, which are shown in FIG. 5. The first capacitor electrode 51A includes a first slit 51As extending outward from the center. The first capacitor electrode 51A is annular and open by the first slit 51As.

The first capacitor electrode 51B is elliptical and annular and overlaps the first coil wiring 43A of the first coil 41B shown in FIG. 5. The first capacitor electrode 51B of this modified example includes a first electrode wiring 53B that is plate-shaped and continuous from the inner end 43i of the first coil wiring 43A to the outer end 43o of the first coil wiring 43B, which are shown in FIG. 5. The first capacitor electrode 51B includes a first slit 51Bs extending outward from the center. The first capacitor electrode 51B is annular and open by the first slit 51Bs.

FIG. 19 shows a modified example of first capacitor electrodes 51A and 51B. The first capacitor electrode 51A is elliptical-plate-shaped and overlaps the first coil wiring 43A and the first signal terminal 44A of the first coil 41A, shown in FIG. 5. The first capacitor electrode 51A of this modified example includes a first electrode wiring 53A that is plate-shaped and continuous from the center of the first coil 41A to the outer end 43o of the first coil wiring 43B, shown in

FIG. 5. Thus, the first electrode wiring 53A of this modified example includes the first signal terminal 44A, shown in FIG. 5. The first capacitor electrode 51A includes a first slit 51As extending outward from the center.

The first capacitor electrode 51B is elliptical-plate-shaped and overlaps the first coil wiring 43A and the first signal terminal 44B of the first coil 41B, shown in FIG. 5. The first capacitor electrode 51B of this modified example is plate-shaped and continuous from the center of the first coil 41B to the outer end 43o of the first coil wiring 43B, shown in FIG. 5. Thus, the first electrode wiring 53B of this modified example includes the first signal terminal 44B, shown in FIG. 5. The first capacitor electrode 51B includes a first slit 51Bs extending outward from the center.

In the embodiments, the cross section of the first coil 41A and the second coil 42A, the first capacitor electrode 51A and the second capacitor electrode 52A, and the dummy patterns 120 and 125 in the z-direction may be changed.

FIGS. 20 to 28 schematically show portions of the transformer chip 80 in a cross section in the z-direction.

As shown in FIG. 20, the dummy pattern 120 corresponding to the second coil 42A and the dummy pattern 125 (refer to FIG. 9) corresponding to the second capacitor electrode 52A may be omitted.

As shown in FIG. 21, the dummy pattern 125 (refer to FIG. 9) corresponding to the second capacitor electrode 52A may be omitted so that only the dummy pattern 120 corresponding to the second coil wiring 46A of the second coil 42A is included. Alternatively, the dummy pattern 120 corresponding to the second coil wiring 46A of the second coil 42A may be omitted so that only the dummy pattern 125 (refer to FIG. 9) corresponding to the second capacitor electrode 52A is included.

As shown in FIG. 22, the first electrode wiring 53A of the first capacitor electrode 51A includes an outer end 53o located at an outer side of the outer end 43o of the first coil wiring 43A of the first coil 41A. The first electrode wiring 53A of the first capacitor electrode 51A includes an inner end 53i located at the same position as the inner end 43i of the first coil wiring 43A of the first coil 41A. The second electrode wiring 56A of the second capacitor electrode 52A includes an outer end 56o located at an outer side of an outer end 46o of the second coil wiring 46A of the second coil 42A. The second electrode wiring 56A of the second capacitor electrode 52A includes an inner end 56i located at the same position as an inner end 46i of the second coil wiring 46A of the second coil 42A.

As shown in FIG. 23, the outer end 53o of the first electrode wiring 53A of the first capacitor electrode 51A is located at an outer side of the outer end 43o of the first coil wiring 43A of the first coil 41A. The inner end 53i of the first electrode wiring 53A of the first capacitor electrode 51A is located at an inner side of the inner end 43i of the first coil wiring 43A of the first coil 41A. The outer end 56o of the second electrode wiring 56A of the second capacitor electrode 52A is located at an outer side of the outer end 46o of the second coil wiring 46A of the second coil 42A. The inner end 56i of the second electrode wiring 56A of the second capacitor electrode 52A is located at an inner side of the inner end 46i of the second coil wiring 46A of the second coil 42A.

As shown in FIG. 24, the first electrode wiring 53A of the first capacitor electrode 51A and the second electrode wiring 56A of the second capacitor electrode 52A overlap the second coil wiring 46A of the second coil 42A and the dummy pattern 120. More specifically, the outer end 53o of the first electrode wiring 53A of the first capacitor electrode

51A is located at the same position as the outer end of the dummy pattern 120 formed at an outer side of the second coil wiring 46A of the second coil 42A. In the same manner, the outer end 56o of the second electrode wiring 56A of the second capacitor electrode 52A is located at the same position as the outer end of the dummy pattern 120 formed at an outer side of the second coil wiring 46A of the second coil 42A. The first electrode wiring 53A of the first capacitor electrode 51A includes an inner end 53i located at the same position as the inner end 43i of the first coil wiring 43A of the first coil 41A. The second electrode wiring 56A of the second capacitor electrode 52A includes an inner end 56i located at the same position as an inner end 46i of the second coil wiring 46A of the second coil 42A.

As shown in FIG. 25, the first electrode wiring 53A of the first capacitor electrode 51A and the second electrode wiring 56A of the second capacitor electrode 52A overlap the second coil wiring 46A of the second coil 42A and the dummy pattern 120. More specifically, the outer end 53o of the first electrode wiring 53A of the first capacitor electrode 51A is located at the same position as the outer end of the dummy pattern 120 formed at an outer side of the second coil wiring 46A of the second coil 42A. In the same manner, the outer end 56o of the second electrode wiring 56A of the second capacitor electrode 52A is located at the same position as the outer end of the dummy pattern 120 formed at an outer side of the second coil wiring 46A of the second coil 42A. The inner end 53i of the first electrode wiring 53A of the first capacitor electrode 51A is located at an inner side of the inner end 43i of the first coil wiring 43A of the first coil 41A. The inner end 56i of the second electrode wiring 56A of the second capacitor electrode 52A is located at an inner side of the inner end 46i of the second coil wiring 46A of the second coil 42A.

As shown in FIG. 26, the outer end 53o of the first electrode wiring 53A of the first capacitor electrode 51A is located at an inner side of the outer end 43o of the first coil wiring 43A of the first coil 41A. The inner end 53i of the first electrode wiring 53A of the first capacitor electrode 51A is located at an outer side of the inner end 43i of the first coil wiring 43A of the first coil 41A. The outer end 56o of the second electrode wiring 56A of the second capacitor electrode 52A is located at an inner side of the outer end 46o of the second coil wiring 46A of the second coil 42A. The inner end 56i of the second electrode wiring 56A of the second capacitor electrode 52A is located at an outer side of the inner end 46i of the second coil wiring 46A of the second coil 42A.

As shown in FIG. 27, the width of the first electrode wiring 53A of the first capacitor electrode 51A is set to be less than the width of the first coil wiring 43A of the first coil 41A. That is, the ratio of wiring width to wiring spacing of the first electrode wiring 53A is set to be less than the ratio of wiring width to wiring spacing of the first coil wiring 43A. The width of the second electrode wiring 56A of the second capacitor electrode 52A is set to be less than the width of the second coil wiring 46A of the second coil 42A. That is, the ratio of wiring width to wiring spacing of the second electrode wiring 56A is set to be less than the ratio of wiring width to wiring spacing of the second coil wiring 46A.

As shown in FIG. 28, the width of the first electrode wiring 53A of the first capacitor electrode 51A is set to be greater than the width of the first coil wiring 43A of the first coil 41A. That is, the ratio of wiring width to wiring spacing of the first electrode wiring 53A is set to be greater than the ratio of wiring width to wiring spacing of the first coil wiring

43A. The width of the second electrode wiring 56A of the second capacitor electrode 52A is set to be greater than the width of the second coil wiring 46A of the second coil 42A. That is, the ratio of wiring width to wiring spacing of the second electrode wiring 56A is set to be greater than the ratio of wiring width to wiring spacing of the second coil wiring 46A.

As described above, the shape of the first coil wiring 43A (first coil 41A) and the second coil wiring 46A (second coil 42A) may be set in any manner. The shape of the first electrode wiring 53A (first capacitor electrode 51A) and the second electrode wiring 56A (second capacitor electrode 52A) may be set in any manner. In an example, in the z-direction, the thickness of the first coil wiring 43A and the second coil wiring 46A may differ from the thickness of the first electrode wiring 53A and the second electrode wiring 56A.

In the embodiments, the passivation film 160 is not limited to a material including silicon nitride and may be any layer that protects the insulation layer 84.

In the embodiments, the arrangement of the transformers 40A and 40B and the capacitors 50A and 50B may be changed in any manner. In an example, the transformer 40A, the transformer 40B, and the transformer 40B may be arranged in this order from the chip side surface 80c toward the chip side surface 80d of the transformer chip 80. The capacitors 50A and 50B are arranged in accordance with the positions of the transformers 40A and 40B.

In the embodiments, the first dummy pattern 121 of the dummy pattern 120 is electrically connected to the second coil 42B. However, there is no limit to such a configuration. In an example, the first dummy pattern 121 may be arranged independently of the second coils 42A and 42B. In other words, the first dummy pattern 121 may be configured not to be electrically connected to the second coils 42A and 42B. In the embodiments, the third dummy pattern 123 is electrically connected to the first dummy pattern 121. However, there is no limit to such a configuration. In an example, the third dummy pattern 123 may be configured not to be electrically connected to the first dummy pattern 121.

In the embodiments, the first dummy pattern 126 of the dummy pattern 125 is electrically connected to the first capacitor electrode 51A. However, there is no limit to such a configuration. In an example, the first dummy pattern 126 may be arranged independently of the first capacitor electrodes 51A and 51B. That is, the first dummy pattern 126 may be configured not to be electrically connected to the first capacitor electrodes 51A and 51B. In the embodiments, the third dummy pattern 128 is electrically connected to the first dummy pattern 126. However, there is no limit to such a configuration. In an example, the third dummy pattern 128 may be configured not to be electrically connected to the first dummy pattern 126.

In the embodiments, the structure of the dummy pattern 120 corresponding to the second coils 42A and 42B may be changed in any manner. In an example, one or two of the first dummy pattern 121, the second dummy pattern 122, and the third dummy pattern 123 may be omitted from the dummy pattern 120. The dummy pattern 120 may be omitted from the transformer chip 80.

In the embodiments, the structure of the dummy pattern 125 corresponding to the second capacitor electrodes 52A and 52B may be changed in any manner. In an example, one or two of the first dummy pattern 126, the second dummy pattern 127, and the third dummy pattern 128 may be

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omitted from the dummy pattern 125. The dummy pattern 125 may be omitted from the transformer chip 80.

In the embodiments, the low-voltage circuit 20 and the transformer 40 are mounted on separate chips. However, there is no limit to such a configuration. The transformer 40 and the low-voltage circuit 20 may be mounted on a single chip. In an example, the low-voltage circuit 20 may be formed on the substrate 83 of the transformer chip 80. The transformer chip 80 is covered by the mold resin 110.

In the embodiments, the high-voltage circuit 30 and the transformer 40 are mounted on separate chips. However, there is no limit to such a configuration. The transformer 40 and the high-voltage circuit 30 may be mounted on a single chip. In an example, the high-voltage circuit 30 may be formed on the substrate 83 of the transformer chip 80. In this case, the transformer chip 80 is mounted on the high-voltage die pad 101. The transformer chip 80 is covered by the mold resin 110.

In the embodiments, the gate driver 10 may include an insulation module in which the transformers 40 are accommodated in a single package. The insulation module includes the transformer chip 80 and the mold resin 110 encapsulating the transformer chip 80. The insulation module may further include a die pad on which the transformer chip 80 is mounted, leads, and wires connecting the leads to the transformer chip 80. The mold resin 110 encapsulates the transformer chip 80, the die pad, and the wires. The leads are configured to be electrically connected to both the low-voltage circuit 20 and the high-voltage circuit 30.

In the embodiments, the gate driver 10 may include a low-voltage circuit unit in which the low-voltage circuit 20 and the transformer 40 are accommodated in a single package. The low-voltage circuit unit may include the low-voltage circuit chip 60, the transformer chip 80, and the mold resin 110 encapsulating the low-voltage circuit chip 60 and the transformer chip 80. The low-voltage circuit unit may further include a die pad, first leads, first wires connecting the first leads to the low-voltage circuit chip 60, second leads, and second wires connecting the second leads to the transformer chip 80. The mold resin 110 encapsulates at least the low-voltage circuit chip 60, the transformer chip 80, the die pad, and the wires. In an example, the first leads are configured to be electrically connected to the ECU 503, and the second leads are configured to be electrically connected to the high-voltage circuit 30.

In the embodiments, the gate driver 10 may include a high-voltage circuit unit in which the high-voltage circuit 30 and the transformer 40 are accommodated in a single package. The high-voltage circuit unit may include the high-voltage circuit chip 70, the transformer chip 80, and the mold resin 110 encapsulating the high-voltage circuit chip 70 and the transformer chip 80. The high-voltage circuit unit may further include a die pad, first leads, first wires connecting the first leads to the high-voltage circuit chip 70, second leads, and second wires connecting the second leads to the transformer chip 80. The mold resin 110 encapsulates at least the high-voltage circuit chip 70, the transformer chip 80, the die pad, and the wires. In an example, the first leads are configured to be electrically connected to the source of the switching element 501, and the second leads are configured to be electrically connected to the low-voltage circuit 20.

In the embodiments, the high-voltage circuit 30 may be configured to transmit a signal to the low-voltage circuit 20 through the transformer 40 and the capacitor 50. Alternatively, the low-voltage circuit 20 and the high-voltage circuit

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30 may be configured to bidirectionally transmit signals to each other through the transformer 40 and the capacitor 50.

In the embodiments, the first coil 41A and the second coil 42A may have different number of windings, and the first coil 41B and the second coil 42B may have different number of windings. The first coil 41A and the second coil 42A may be wound in different directions, and the first coil 41B and the second coil 42B may be wound in different directions.

In the embodiments, the positions of the slits 51As and 51Bs shown in FIG. 7 may be changed in any manner. In FIG. 7, the slits 51As and 51Bs extend in the x-direction. Instead, the slits 51As and 51Bs may extend in the y-direction. The slits 51As and Bs may extend in the same direction, for example, a direction toward the chip side surface 80c. Also, the positions of the slits 52As and 52Bs shown in FIG. 8 may be changed. The slit 51As shown in FIG. 7 and the slit 52As shown in FIG. 8 may extend in different directions. Also, the slit 51Bs shown in FIG. 7 and the slit 52Bs shown in FIG. 8 may extend in different directions.

In the present disclosure, the term “on” includes the meaning of “above” in addition to the meaning of “on” unless otherwise clearly indicated in the context. Thus, the phrase “A is formed on B” is intended to mean that A may be disposed directly on B in contact with B in the present embodiment and also that A may be disposed above B without contacting B in a modified example. In other words, the term “on” does not exclude a structure in which another member is formed between A and B.

The z-direction as referred to in the present disclosure does not necessarily have to be the vertical direction and does not necessarily have to fully conform to the vertical direction. In the structures according to the present disclosure, “upward” and “downward” in the z-direction as referred to in the present description are not limited to “upward” and “downward” in the vertical direction. In an example, the x-direction may conform to the vertical direction. In another example, the y-direction may conform to the vertical direction.

In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

REFERENCE SIGNS LIST

10) gate driver
 10R) gate driver
 20) low-voltage circuit
 21A, 21B) low-voltage signal line
 30) high-voltage circuit
 31A, 31B) high-voltage signal line
 40) transformer
 40A, 40B) transformer
 41A, 41B) first coil
 42A, 42B) second coil
 43A, 43B) first coil wiring
 43i) inner end
 43o) outer end
 44A, 44B) first signal terminal
 45) first ground terminal
 46A, 46B) second coil wiring
 46i) inner end
 46o) outer end
 47A, 47B) second signal terminal
 48) second ground terminal
 50) capacitor
 50A, 50B) capacitor
 51A, 51B) first capacitor electrode
 51As, 51Bs) first slit

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52A, 52B) second capacitor electrode
 52As, 52Bs) second slit
 53A, 53B) first electrode wiring
 53i) inner end
 53o) outer end
 54A, 54B) first capacitor end
 55) first capacitor ground terminal
 55A, 55B) interconnect
 56A, 56B) second electrode wiring
 56i) inner end
 56o) outer end
 57A, 57B) second capacitor end
 58) second capacitor ground terminal
 58A, 58B) interconnect
 60) low-voltage circuit chip
 60s) chip main surface
 61) first electrode pad
 62) second electrode pad
 63) third electrode pad
 70) high-voltage circuit chip
 70s) chip main surface
 71) first electrode pad
 72) second electrode pad
 73) third electrode pad
 80) transformer chip
 80a to 80d) chip side surface
 80r) chip back surface
 80s) chip main surface
 81) first electrode pad
 81A to 81C) first electrode pad
 82) second electrode pad
 82A to 82C) second electrode pad
 83) substrate
 83r) substrate back surface
 83s) substrate main surface
 84) insulation layer
 84s) surface
 85) insulation film
 85A) first insulation film
 85B) second insulation film
 85B1) insulation film
 85B2) insulation film
 85L) insulation film
 85U) insulation film
 86) shield electrode
 87) inner region
 88) outer region
 89) via
 90) low-voltage lead frame
 91) low-voltage die pad
 92) low-voltage lead
 100) high-voltage lead frame
 101) high-voltage die pad
 102) high-voltage lead
 110) mold resin
 111 to 114) resin side surface
 120) dummy pattern
 121) first dummy pattern
 122) second dummy pattern
 123) third dummy pattern
 125) dummy pattern
 126) first dummy pattern
 127) second dummy pattern
 128) third dummy pattern
 131A to 131C) interconnect
 131V) via
 132A, 132C) first interconnect part

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133A, 133C) second interconnect part
 134A, 134C) via
 135A, 135C) via
 136 to 138) via
 5 141, 142) coil groove
 143) barrier film
 144) body layer
 145) first opening
 145) opening
 10 146) second opening
 150) protection film
 151) via hole
 152) barrier film
 153) resist film
 15 160) passivation film
 180) resin layer
 181) inner resin layer
 182) outer resin layer
 183) separation groove
 20 184) first resin opening
 185) second resin opening
 500) inverter device
 501, 502) switching element
 503) ECU
 25 851 to 856) insulation film
 GND1, GND2) ground
 i) current
 i_{1A}, i_{1B}) current
 i_{2A}, i_{2B}) current
 30 i_{CA}, i_{CB}) current
 SD) conductive bonding material
 V1) first voltage
 V2) second voltage;
 W1 to W4) wire
 35 The invention claimed is:
 1. An isolation transformer, comprising:
 an insulation layer;
 a transformer embedded in the insulation layer and
 including a first coil and a second coil separated from
 each other in a thickness-wise direction of the insula-
 40 tion layer, the first coil including a first signal terminal
 and a first ground terminal and being configured to
 receive a low voltage through the first signal terminal,
 and the second coil including a second signal terminal
 45 and a second ground terminal and being configured to
 receive a high voltage through the second signal ter-
 minal; and
 a capacitor including a first capacitor electrode and a
 second capacitor electrode, the first capacitor electrode
 50 being arranged between the first coil and the second
 coil and being connected to the first ground terminal,
 and the second capacitor electrode being arranged
 between the first capacitor electrode and the second coil
 and connected to the second ground terminal, wherein
 55 the insulation layer includes thin films and interlayer
 insulation films alternately stacked in the thickness-
 wise direction,
 the thin films include a first thin film and a second thin
 film separated from each other in the thickness-wise
 60 direction,
 the interlayer insulation films include a first interlayer
 insulation film located next to the first thin film in the
 thickness-wise direction and a second interlayer insu-
 lation film located next to the second thin film in the
 65 thickness-wise direction,
 the first capacitor electrode is formed between the first
 thin film and the first interlayer insulation film, and

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the second capacitor electrode is formed between the second thin film and the second interlayer insulation film.

2. The isolation transformer according to claim 1, wherein the first thin film is arranged between the first capacitor electrode and the first coil, and

the second thin film is arranged between the second capacitor electrode and the second coil.

3. The isolation transformer according to claim 1, wherein the first interlayer insulation film is arranged between the first capacitor electrode and the first coil, the second interlayer insulation film and the second coil are located at opposite sides of the second capacitor electrode, and

at least one of the interlayer insulation films is arranged between the second thin film and the second coil.

4. The isolation transformer according to claim 1, further comprising:

- a chip main surface and a chip back surface facing opposite directions in the thickness-wise direction, wherein the first coil is located closer to the chip back surface than the second coil is;
- a first ground electrode arranged on the chip main surface; and
- a first voltage interconnect connecting the first ground electrode to the first coil and the first capacitor electrode, wherein

the first voltage interconnect includes a via including a barrier film arranged on a wall surface of a via hole formed in the insulation layer, and

the barrier film of the via and at least one of the first capacitor electrode and the second capacitor electrode are formed from a same material.

5. The isolation transformer according to claim 1, wherein at least one of the first capacitor electrode and the second capacitor electrode includes a nonmagnetic material.

6. The isolation transformer according to claim 1, wherein as viewed in the thickness-wise direction, the first capacitor electrode is annular and open and includes a first slit extending in a direction from a center of the first coil toward an outer side of the first coil, and

as viewed in the thickness-wise direction, the second capacitor electrode is annular and open and includes a second slit extending in a direction from a center of the second coil toward an outer side of the second coil.

7. The isolation transformer according to claim 1, wherein the first coil includes a spiral first coil wiring, the first ground terminal connected to one end of the first coil wiring, and the first signal terminal connected to another end of the first coil wiring, and

the second coil includes a spiral second coil wiring, the second ground terminal connected to one end of the second coil, and the second signal terminal connected to another end of the second coil wiring.

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8. The isolation transformer according to claim 7, wherein as viewed in the thickness-wise direction, the first capacitor electrode includes a first electrode wiring overlapping the first coil wiring, a first capacitor end overlapping the first signal terminal, and a first capacitor ground terminal overlapping the first ground terminal, and

as viewed in the thickness-wise direction, the second capacitor electrode includes a second electrode wiring overlapping the second coil wiring, a second capacitor end overlapping the second signal terminal, and a second capacitor ground terminal overlapping the second ground terminal.

9. The isolation transformer according to claim 8, wherein the first electrode wiring is set to be equal to the first coil wiring in ratio of wiring width to wiring spacing, and the second electrode wiring is set to be equal to the second coil wiring in ratio of wiring width to wiring spacing.

10. The isolation transformer according to claim 7, wherein

- the first capacitor electrode is plate-shaped and continuous from an inner end of the first coil wiring to an outer end of the first coil wiring as viewed in the thickness-wise direction, and
- the second capacitor electrode is plate-shaped and continuous from an inner end of the second coil wiring to an outer end of the second coil wiring as viewed in the thickness-wise direction.

11. The isolation transformer according to claim 8, wherein as viewed in the thickness-wise direction, an outer end of the second electrode wiring is located at either an outer side of an outer end of the second coil wiring or a same position as an outer end of the second coil wiring.

12. The isolation transformer according to claim 11, wherein as viewed in the thickness-wise direction, an inner end of the second electrode wiring is located at either a same position as an inner end of the second coil wiring or an inner side of an inner end of the second coil wiring.

13. The isolation transformer according to claim 11, wherein as viewed in the thickness-wise direction, an outer end of the first electrode wiring is located at either an outer side of an outer end of the first coil wiring or a same position as an outer end of the first coil wiring.

14. The isolation transformer according to claim 13, wherein as viewed in the thickness-wise direction, an inner end of the first electrode wiring is located at a same position as an inner end of the first coil wiring.

15. The isolation transformer according to claim 13, wherein as viewed in the thickness-wise direction, an inner end of the first electrode wiring is located at an inner side of an inner end of the first coil wiring.

16. The isolation transformer according to claim 4, further comprising:

- a dummy coil pattern arranged around the second coil and connected to the second coil.

* * * * *